

Consecutive Powerful Numbers and Pell Equations: Triples, Pairs, and Further Observations

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Abstract

A positive integer is *powerful* if every prime that divides it divides it at least twice. Pairs of consecutive powerful numbers exist in abundance, but no three consecutive powerful numbers are known, and Erdős asked whether they exist. This volume collects three parts. Part I turns a hypothetical triple into an address on a Pell equation, proves a lower bound for its middle in terms of its kernel, and reduces the existence of a triple to primes that divide a Pell number exactly once. Using Reinhart's complete list of the kernels m up to $1.5 \cdot 10^{12}$ that divide $U_1(m)$, where $T_1 + U_1\sqrt{m}$ is the smallest solution of $x^2 - my^2 = 1$ with $y > 0$, it shows by computation that no triple has a middle below $1.83 \cdot 10^{18}$. Independently of this bound, it asks whether the 961 building blocks attached to the lattice points it examines can be powerful: 954 are shown not to be, by witnesses proved prime, and 7 remain open. Part II establishes the explicit lower bound $\liminf N_1(x)/\log x \geq 1.6822$ for the number $N_1(x)$ of pairs $n, n+1$ with $n+1 \leq x$, a logarithmic lower bound at distance two, and shows that the number of pairs at distance two up to x differs by $O(x^{2/9})$ from the number of kernels m with $m \mid U_1(m)$, $T_1(m)$ even and $T_1(m) \leq x$. Part III collects the gap spectrum of powerful numbers, powerful values of cyclotomic polynomials, and tests of the independence assumptions behind the heuristic, each with the power of the test. Every computed number is generated from deposited data or from a cited constant, every numbered statement is labeled by the kind of evidence behind it, and the open questions are collected as Problems A to I.

About this series. This volume collects three parts that share one source, one bibliography, and one numbering. A statement numbered I.4.5 carries that number in its part and in the collected volume. Every number that we computed is generated from the result files of the data deposit or from a cited constant; numbers from the literature carry a citation. The ranges and parameters of a computation and the constants of an argument are written in the text. The open questions of all parts are collected as Problems A to I in the framework introduction below and in the file `OPEN_PROBLEMS.md` of the data deposit; both are generated from the parts. This volume is a preprint. The author corrects any error that becomes visible after the release in a new version of the record; earlier versions remain available under their own version numbers.

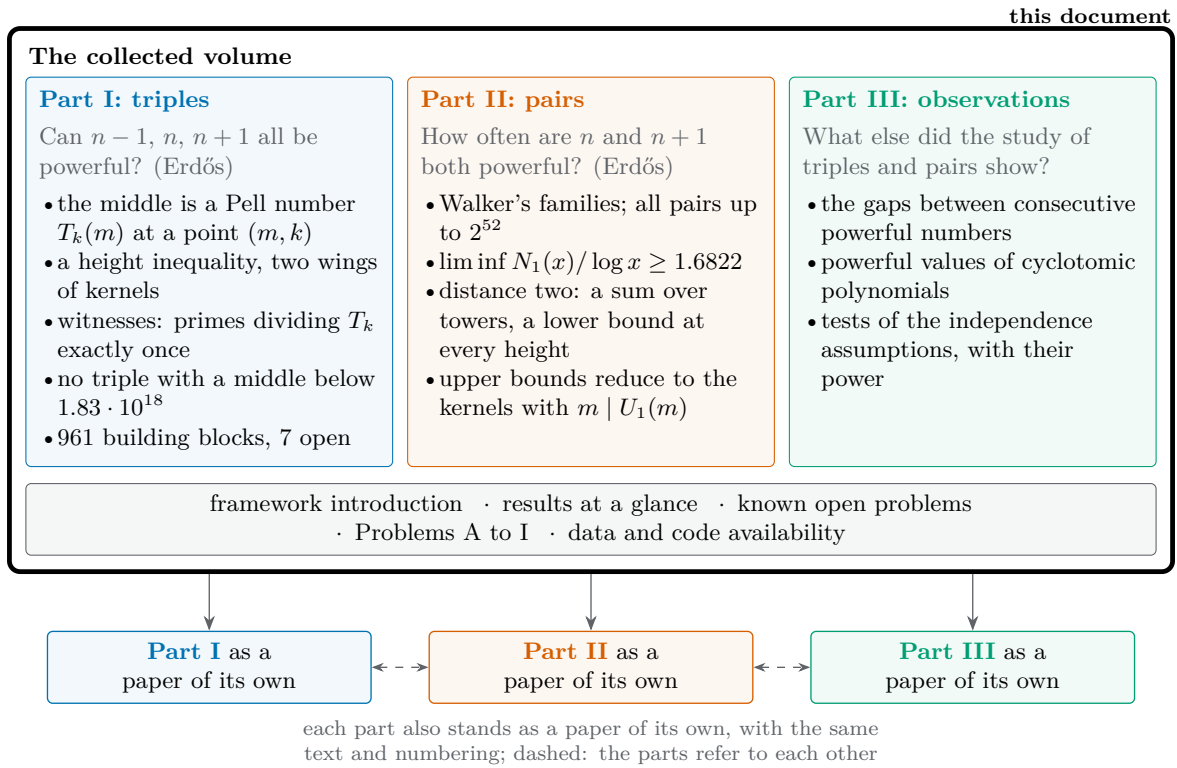


Figure 1: **What this research did.** It followed two questions of Erdős on consecutive powerful numbers, on triples and on pairs, through Pell equations, proved or computed what it could reach, and recorded the rest as observations and open problems. The collected volume contains the three parts and the framework that ties them together; each part also stands as a paper of its own, with the same text and numbering, and the parts refer to each other. The heavier frame marks the document in hand.

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Framework introduction

A positive integer is *powerful* if every prime that divides it divides it at least twice. Two consecutive integers can both be powerful, as 8, 9 shows, and there are infinitely many such pairs. No three consecutive powerful numbers are known. We expect that there are none; the introduction of Part I gives the reasons, and none of them is a proof. This volume collects three parts on these two situations.

Part I concerns triples $n-1, n, n+1$. It translates the middle n into an address (m, k) on a Pell equation, bounds n from below in terms of the kernel m , and profiles a counterexample: where it would have to live, and which kernels are the weak point of the bound. *Part II* concerns pairs $n, n+1$. It enumerates the pairs by families, gives an explicit lower bound for their count, compares the count with a heuristic yardstick, and localizes the count of pairs at distance two in the kernels that Part I singles out, together with their counterparts of the other residue class. *Part III* collects further observations: the gaps between consecutive powerful numbers, powerful values of cyclotomic polynomials, and two tests of the independence assumptions behind the heuristic of Part I, each with the power of the test.

How to read the volume. The parts follow one arc. Part I states the question and its address system, shows what can be excluded and why the rest depends on witnesses, and names the step at which the methods stop; a certified witness is found for 954 of the 961 building blocks of its finite record. Part II turns the two halves of a triple into pairs and asks how many there are: towers, families, an explicit lower bound, and why local models mislead. Part III collects what we saw around the problem: the gap spectrum, cyclotomic values, measured independence, and negative results with their reach. The open problems at the end of this introduction are what remains.

The parts share one source and one numbering: a statement numbered I.4.5 carries that number here and in Part I. Every numbered statement carries a label that says what kind of evidence stands behind it (proved, literature, computed, measured, heuristic, open); the table at the end of each part lists them. A statement that combines several kinds carries the label of the part that needs the most trust: a proof that uses our own computation is labeled computed, and a proof that uses a published result is labeled proved. Every number that we computed is generated from the deposited data files or from a cited constant; the build refuses a number that has no source.

The author is an independent researcher who likes numbers and science, not a number theorist or mathematician by training. The series therefore keeps every claim as narrow as its proof or computation allows and says plainly where, to our knowledge, a question remains open. It grew out of about two months of research. Specialists are welcome to extend, correct, or cite this work; the author intends it to help this field progress and hopes that a number theorist will take the next steps. The question whether three consecutive powerful numbers exist can be settled in only two ways: a triple is found, or a general statement is found that holds at all of the infinitely many addresses at once. Such a statement would have to agree with all results in this field, the formulas and laws of this volume and of the literature, and those still to be found. It need not be a single law: it could also be one exclusion that covers all addresses, or many exclusions that together cover them, in the way that the exclusions for cube middles [19, 71, 69] each cover a part. We do not know yet whether such a statement exists.

Results at a glance

The table lists every statement of our own in each part, grouped by the type of evidence behind it, with the main statements in bold. Statements taken from the literature are not listed here; the complete list of the statements of a part, with the type of each, stands at the end of that part. The open questions are listed completely in the section on the open problems below.

part	type	statements
Part I (triples)	proved	I.2.1 height inequality I.2.4 two wings I.4.6 reduction to a witness statement I.4.8 the divisor count of the Wieferich conditions I.6.10 Wieferich primes of the unit and the primitive parts <i>also:</i> I.2.2 the constant in the height bound; I.2.3 kernel bound; I.3.2 weaker bounds with less work; I.4.3 valuation identity; I.4.4 exponent-3 condition; I.4.7 the T_1 -part of a lattice point; I.4.10 what the witness condition does not follow from; I.4.11 where a witness in an outer number can sit; I.4.12 two elementary constraints, and the size of the gap; I.4.13 the cyclotomic axis
	computed	I.3.1 reduction I.4.5 the b-box I.5.1 the strip I.5.3 smooth kernels I.5.4 square-type middles I.6.2 no triple below $1.83 \cdot 10^{18}$, and the pairs at distance two with middle divisible by four I.6.8 the primitive part over an explicit finite set I.6.13 a second primitive prime factor I.7.1 profile <i>also:</i> I.5.5 the share of the candidates covered; I.6.3 ladders for the known kernels dividing U_1 ; I.6.4 the smallest witnesses; I.6.5 the condition modulo 36 and the witnesses; I.6.6 how many conditions there are, and how often they are met; I.6.7 middles that are powers of two; I.6.11 the Wieferich events of the units; I.6.12 Lucas–Wieferich primes; I.8.2 Yokoi’s representation; I.8.3 small units
	measured open	I.6.1 exponents at the rank <i>finiteness of the kernels of wing (ii) (Problem A); the primitive part of $T_d(m)$ (Problem B); a middle that is a power of 2 (Problem C); Reinhart’s condition (C) (Problem D)</i>

part	type	statements
Part II (pairs)	proved	II.6.2 the count is a sum over towers II.6.4 localization of the pair count II.6.6 where Tao's argument is weakest <i>also:</i> II.2.1 powerful numbers; II.6.5 what the localization says, and what it does not say; II.7.1 pairs on one Pell equation
	computed	II.3.1 the known pairs II.4.1 explicit lower bound II.5.2 the families with a square overtake the comparison function II.6.1 the middles that are not divisible by four II.6.3 lower bound at every height II.7.2 the third number <i>also:</i> II.3.2 completeness inside the families
	heuristic	II.5.1 comparison function
	open	<i>convergence of the rates</i> (Problem E); <i>growth of the tower rate</i> (Problem F); <i>the kernels that divide U_1</i> (Problem G)
Part III (further observations)	proved	III.2.2 two explicit chains of powerful values of Φ_3 and Φ_6 III.2.5 the simplest case of Φ_8
	computed	III.1.2 the head of the gap spectrum III.2.3 powerful values up to a bound <i>also:</i> III.1.3 a ceiling for common divisors; III.2.1 the values $x^2 + 1$; III.2.4 a class without squares
	measured	III.1.4 a counting model, measured III.3.1 no joint event along the divisors, and the power of the test III.4.1 no deviation from equidistribution, memory or coupling <i>also:</i> III.1.5 common factors of pairs
	heuristic	III.5.3 a conjecture on the most frequent gap
	open	<i>the most frequent gap</i> (Problem H); <i>powerful values of $x^4 + 1$</i> (Problem I)

In words. A proved statement stands on a proof in the text, which may use cited results. A computed statement stands on a computation whose data are in the deposit; where it depends on a published search, the statement says so. A measured statement describes a finite range and claims nothing beyond it, and a heuristic is a yardstick, not a claim.

Where this work meets known open problems

The questions of this volume touch several known open problems. The table names each one, says what this volume reaches toward it, and says what stays open. None of these problems is solved here.

problem	what this volume reaches	what stays open
Three consecutive powerful numbers (Erdős; in a form about Pell numbers, Mollin and Walsh [51, p. 111])	no triple has a middle below $1.83 \cdot 10^{18}$ (Proposition I.6.2(d); Table 3 compares it with the bounds in the literature, among them a machine-checked proof below 10^{14} [49]); every lattice point with kernel up to 10^8 and middle below 10^{2000} is excluded (Theorem I.3.1); the profile of a counterexample (Theorem I.7.1)	the question itself; Problems A to D
Wieferich primes of Pell units (for primes that have a rank, the Lucas–Wieferich primes of McIntosh and Roettger [47])	if the unit of a kernel has only finitely many Wieferich primes of odd rank, then all but finitely many of its building blocks are not powerful (Proposition I.6.10); below $3 \cdot 10^6$ the 79 units of Proposition I.6.8 carry 54 Wieferich events, 18 of them of odd rank $d \geq 5$ (Remark I.6.11)	whether there are only finitely many; Problem B
Kernels m with $m \mid U_1(m)$ (Walsh’s question [86, p. 97])	they are the weak point of the height bound (Corollary I.2.4); Reinhart’s list of them is complete up to $1.5 \cdot 10^{12}$ [61, Remark 5.5], and for the two known kernels $m \equiv 7 \pmod{8}$ with T_1 even a ladder shows how large a middle would have to be (Proposition I.6.3); the count of pairs at distance two is localized in them (Proposition II.6.4)	finiteness, Problem A; a bound for E_3 , Problem G
The <i>abc</i> conjecture	an <i>abc</i> inequality with exponent below $4/3$ would exclude all large triples; the exponents derived from the explicit form of the conjecture are larger, and that form gives only a conditional bound on the middle (Remark I.4.12)	an unconditional route
Primitive parts of Lucas sequences (after the primitive divisor theorems of Carmichael [17] and of Bilu, Hanrot and Voutier [9])	every divisor $d \geq 5$ of an odd index gives a prime of rank d , and every such prime satisfies $p \equiv \pm 1 \pmod{4d}$ (Corollary I.4.8); on 26 of the 79 kernels every building block has at least two distinct prime factors (Remark I.6.13)	whether a building block can be powerful; Problem B
How many pairs there are (upper bounds at distance two by Chan [18], Blomer and Schöbel [10], and Tao [79])	an explicit lower bound $\liminf N_1(x)/\log x \geq 1.6822$ at distance one (Proposition II.4.1); at distance two a bound N^θ for the number of kernels with $m \mid U_1(m)$ and $T_1(m)$ even gives the bound $N^{\max(2/9, \theta)}$ for all pairs, since the two counts differ by $O(N^{2/9})$ (Proposition II.6.4), and under a squarefreeness assumption the case in which Tao’s argument is weakest lies among those kernels (Remark II.6.6)	the true growth; Problems E, F and G

problem	what this volume reaches	what stays open
Powerful values of polynomials (De Koninck, Doyon and Luca [26])	two explicit infinite chains, each giving powerful values of both $x^2 + x + 1$ and $x^2 - x + 1$ (Proposition III.2.2); one of them contains the prime x reported by Lavi [40]; $x^4 + 1 = q^3 a^2$ has no solution for primes $q < 2 \cdot 10^{11}$ (Proposition III.2.5, by a theorem of Cohn [24] and the published verification of the Ankeny–Artin–Chowla conjecture [81, 82])	whether $x^4 + 1$ is ever powerful; Problem I
Katz’s conjecture on Wieferich fractions [36]	on 25 units and the primes below $3 \cdot 10^6$ the Wieferich fractions show no deviation from equidistribution in our measurement, which states its power (Remark III.4.1)	the conjecture itself

The open problems

The parts end with questions. We collect them here with letters, in the order of the parts, in the wording of the parts. The discussion of each question, that is, what is known and what an answer would give, stands behind the question in its part. The same list, generated from the same source, is the file `OPEN_PROBLEMS.md` of the data deposit. Erdős’s own questions, whether a triple exists and whether $N_1(x)$ is bounded by a power of $\log x$, are not repeated; they are the questions the parts start from.

Notation used in the problems.

- A positive integer is *powerful* if every prime that divides it divides it at least twice.
- For a squarefree $m \geq 2$, $\varepsilon_m = T_1 + U_1\sqrt{m}$ is the fundamental solution of $x^2 - my^2 = 1$, and $\varepsilon_m^k = T_k + U_k\sqrt{m}$; $T_k(m)$, $U_k(m)$ name the kernel m . m' is the product of the primes of m that do not divide U_1 .
- The *rank* $\alpha_m(p)$ of a prime $p \nmid m$ is the smallest k with $p \mid T_k(m)$; M_d is the part of $T_d(m)$ supported on the primes of rank exactly d (the *primitive part*).
- *Wing (ii)* of Part I: the kernels $m \equiv 7 \pmod{8}$ with $m \mid U_1(m)$; they are the weak point of the lower bound for a triple.
- In Part II, the *families* are Walker’s Pell families of pairs $n, n+1$, F_f is the growth factor from one pair of the family f to the next, $w(m) = 1/(2m' \log_{10} \varepsilon_m)$ is the rate of the tower of the kernel m , $P_2(N)$ counts the pairs $n-1, n+1$ of powerful numbers with $n \leq N$, and $E_3(N)$ counts the odd squarefree m with $m \mid U_1(m)$, $T_1(m)$ even and $T_1(m) \leq N$.
- Reinhart’s condition (C) is Definition 1.4 of [62].

Problem A (finiteness of the kernels of wing (ii); Question I.9.1 in Part I). Are there only finitely many squarefree $m \equiv 7 \pmod{8}$ with $m \mid U_1(m)$ and $T_1(m)$ even?

Problem B (the primitive part of $T_d(m)$; Question I.9.2 in Part I). Let $m \equiv 7 \pmod{8}$ be squarefree with $T_1(m)$ even, let $d \geq 5$ be odd, and let M_d be the part of $T_d(m)$ supported on the primes of rank exactly d . Can M_d be powerful?

Problem C (a middle that is a power of 2; Question I.9.3 in Part I). Can $2^a - 1$ and $2^a + 1$ both be powerful for some $a \geq 2$?

Problem D (Reinhart's condition (C); Question I.9.4 in Part I). Does Reinhart's condition (C) [62, Definition 1.4] hold for some squarefree d , that is, is there an index k with $T_k(d)$ powerful and $d \mid U_k(d)$? For $d \mid U_1(d)$ the second condition holds for every k .

Problem E (convergence of the rates; Question II.5.3 in Part II). Does $\sum_f 1/\log F_f$, taken over all families, converge?

Problem F (growth of the tower rate; Question II.6.7 in Part II). Does $c_M = \sum_{m \leq M} w(m)$, the sum over odd squarefree m with $T_1(m)$ even of $w(m) = 1/(2m' \log_{10} \varepsilon_m)$, stay bounded as $M \rightarrow \infty$?

Problem G (the kernels that divide U_1 ; Question II.8.1 in Part II). Is there a $\theta < 61/180$ with $E_3(N) \ll N^\theta$? Since $E_3(N) \leq P_2(N)$, Blomer and Schöbel give $E_3(N) \ll N^\gamma$ for every $\gamma > 61/180$ [10, Theorem 3]; the question is whether the squarefree odd m with $m \mid U_1(m)$, $T_1(m)$ even and $T_1(m) \leq N$ are rarer than that.

Problem H (the most frequent gap; Question III.5.1 in Part III). Let $g(x)$ be the most frequent gap between consecutive powerful numbers up to x . Is $g(x)$ powerful for all large x , and does the share of powerful values among the N most frequent gaps up to x tend to 1 when N is fixed and $x \rightarrow \infty$?

Problem I (powerful values of $x^4 + 1$; Question III.5.2 in Part III). Is $x^4 + 1$ powerful for some integer $x \geq 1$?

One gap, seen from two sides. Problems A and G concern two nested sets of kernels. Part I bounds the middle of a triple from below in two wings (Corollary I.2.4); wing (ii) consists of the kernels m with $m \mid U_1(m)$, and it is the weaker wing. Part II localizes the count $P_2(N)$ of pairs at distance two in the kernels with $m \mid U_1(m)$ and $T_1(m)$ even, of both residue classes, of which those with $m \equiv 7 \pmod{8}$ form wing (ii): Proposition II.6.4 says that the count differs from $E_3(N)$ by at most $51N^{2/9}$, where $E_3(N)$ counts the odd squarefree m with $m \mid U_1(m)$ and $T_1(m)$ even whose first middle is at most N . Problem A asks whether these kernels, restricted to $m \equiv 7 \pmod{8}$, are finite in number; Problem G asks how fast $E_3(N)$ grows. What is known is small: Reinhart's list of these kernels is complete up to $1.5 \cdot 10^{12}$ [61, Remark 5.5] and, if his extended search missed nothing, up to $5.325 \cdot 10^{13}$ [62] (Remark I.2.5); the number of its odd kernels with $T_1(m)$ even is 1 below the first bound and 3 below the second (Section 8 of Part II). A bound $E_3(N) \ll N^\theta$ with $\theta < 61/180$ would improve the best exponent we know for pairs at distance two; a proof that the kernels of wing (ii) are finite in number would confine that wing to finitely many kernels, each of which still carries infinitely many admissible indices k , and for the known ones Problem D is open.

One step behind several problems. Problems B and C, and the witness statement of Part I (Theorem I.4.6), stop at the same step, and so does every route we tried. Whether a prime divides a term $T_k(m)$ is decided modulo p by the rank of the prime, and this controls all indices at once (Lemma I.4.1, Corollary I.4.8). Whether a prime of rank d divides $T_d(m)$

twice is a condition modulo p^2 of Wieferich type, and the structure modulo p does not decide it. Corollary I.4.8 turns a counterexample into one such condition at every divisor $d \geq 5$ of the index, and Remark I.4.11 shows that in the outer numbers of a triple no prime can carry exponent one at all. This describes the reach of the methods used here and of the tools we know; it says nothing about whether a triple exists. As a reference point from outside our methods, Granville [33] showed that if there are no three consecutive powerful numbers, then infinitely many primes p satisfy $p^2 \nmid 2^p - 2$ (Theorem I.4.9). To our knowledge this last statement is known only under further assumptions such as the *abc* conjecture, so a proof of nonexistence would have to include it. We take this as a sign that the step is hard, not as evidence for either answer. Problems A and G concern the different gap described in the preceding paragraph.

In words. The questions are printed in one place so that a reader can see them together and check them off. Two of them, Problems A and G, are two questions about nested sets of kernels: how many kernels m divide their own $U_1(m)$ and have an even first term. Part I needs to know them, so that its lower bound for a triple has no weak wing; Part II needs them to be few, so that its count of pairs at distance two is small. Below $1.5 \cdot 10^{12}$ the number of such kernels is exactly 1, and whether there are finitely many is open; Walsh asked it for all kernels dividing U_1 .

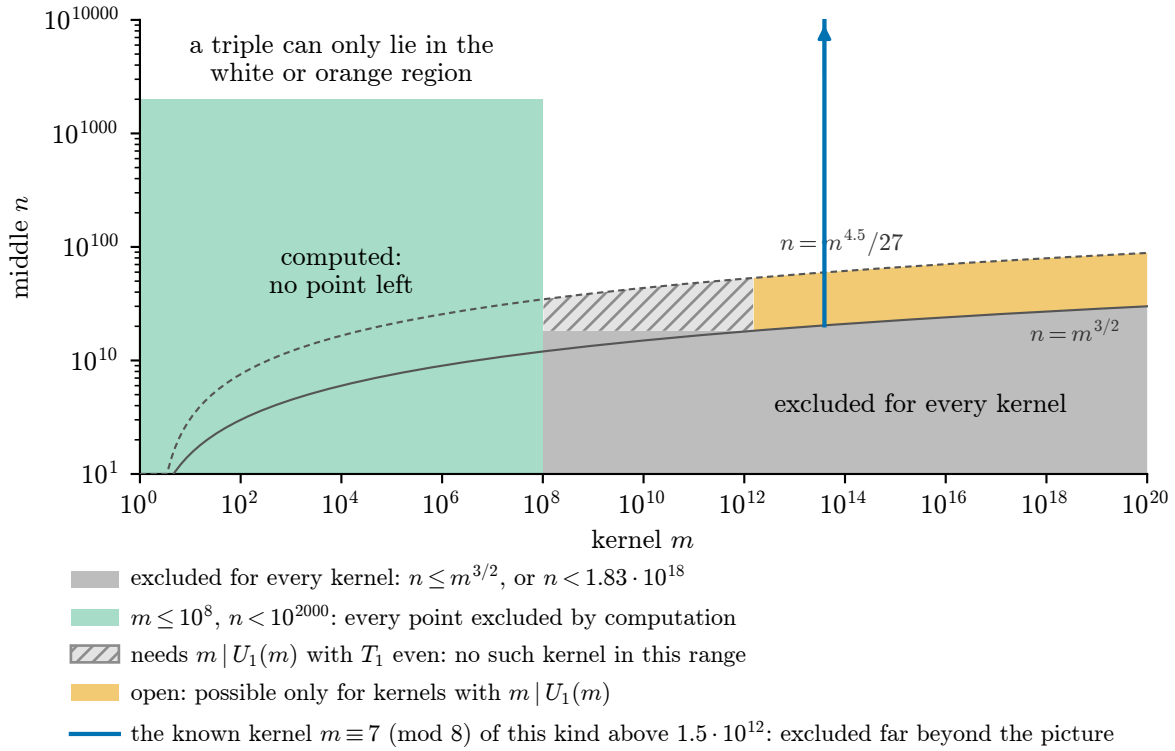


Figure 2: **Where a triple could still lie.** The kernel m on a logarithmic scale, the middle n on a doubly logarithmic one. Gray: excluded for every kernel, by the height bound of Corollary I.2.3 or by Proposition I.6.2(d). Hatched: here a middle needs $m \mid U_1(m)$ with $T_1(m)$ even, and Reinhart’s complete list has no such kernel between 10^8 and $1.5 \cdot 10^{12}$ (Remark I.2.5; the parity of T_1 is computed in Proposition I.6.3). Green: the box of Theorem I.3.1, where every lattice point is excluded by computation. Orange: the weak point of the bound, kernels with $m \mid U_1(m)$ above $1.5 \cdot 10^{12}$ (Question I.9.1); the one known kernel of this kind above $1.5 \cdot 10^{12}$ in class 7 with T_1 even, drawn as a line, admits no middle below the height given in Table 5, and so does the other known one, which lies in the box. The picture does not draw the classes of kernels that Section 5 excludes, so a triple could lie only in parts of the white and orange regions.

Part I: triples

1 Introduction and the address system

A positive integer is *powerful* if every prime that divides it divides it at least twice. Erdős asked whether three consecutive integers can all be powerful, and conjectured that they cannot [27, 11]; on the attribution among Erdős’s papers see [3, Section 7]. Mollin and Walsh conjectured it in a form about Pell numbers [51, p. 111] (Section 4). Two consecutive powerful numbers exist in abundance, since $x^2 - 8y^2 = 1$ has infinitely many solutions, and

four consecutive ones are impossible, since one of four consecutive integers is 2 modulo 4. No triple is known. The *abc* conjecture implies that there are only finitely many, and its explicit forms give finite bounds (Remark I.4.12); unconditionally the question is open. Bloom records that no triple has a middle below $7.38 \cdot 10^{28}$ by OEIS A076445 [11, 54], a statement that rests on the listed terms of that sequence (Section 6).

We expect that there is no triple, and we give the reasons as evidence, not as proof. First, the *abc* conjecture implies that there are at most finitely many (Remark I.4.12). Second, none has been found: none has a middle below $1.83 \cdot 10^{18}$ (Proposition I.6.2(d)), and none lies below 10^{22} by an enumeration of pairs that is described as exhaustive but is not certified [55, 49]. Mian and Siddique prove, with machine-checked theorems in Lean 4, that no triple lies below 10^{14} [49]; Table 3 lists these bounds with what each rests on. Third, a triple needs a powerful Pell number $T_k(m)$, and this means that for each divisor $d \geq 5$ of k with $k < d(4d - 1)$ every prime of rank d , and there is at least one, must divide $T_d(m)$ at least twice (Corollary I.4.8); at our 445 lattice points the median number of such conditions is 3 (Proposition I.6.6). In the data, a prime p divides the first Pell term that it divides at least twice about as often as the model with probability $1/p$ predicts (Proposition I.6.1); under that model each condition is one more rare event. The third reason is a heuristic, and the model is not proved.

no triple below	what the bound rests on	source
10^{14}	machine-checked theorems in Lean 4	[49]
$1.83 \cdot 10^{18}$	computation in this part, every witness proved prime, together with Reinhart's computer proof that his list of the kernels dividing U_1 is complete up to $1.5 \cdot 10^{12}$	Proposition I.6.2(d)
$3.88 \cdot 10^{20}$	the same, if Reinhart's extended search missed nothing	Proposition I.6.2(d)
10^{22}	an enumeration of the pairs, described as exhaustive, not certified	[55, 49]
$7.38 \cdot 10^{28}$	the listed terms of OEIS A076445; the entry does not state that the list is complete	[11, 54]

Table 3: **Bounds below which no triple of consecutive powerful numbers is known, and what each rests on.** Our two rows and the last one bound the middle of a triple; the rows for 10^{14} and 10^{22} are stated for the triple, whose members differ from its middle by at most one.

The problem has an exact address system, due to Mollin and Walsh, Walker and Sentance (Lemmas I.1.1 to I.1.4): the middle n of a triple is a term $T_k(m)$ of the Pell sequence of a squarefree kernel $m \equiv 7 \pmod{8}$, at an odd index k that is a multiple of the part m' of m prime to U_1 . So a triple is a lattice point (m, k) at which $T_k(m)$ is even and powerful, and the question becomes a question about the prime factors of Pell numbers. Two facts make it hard. Granville showed that the nonexistence of triples implies that there are infinitely many primes p with $p^2 \nmid 2^p - 2$ (Theorem I.4.9), a statement that is not known unconditionally; so a proof would also settle an open problem about Wieferich primes. And an *abc* inequality with exponent below $4/3$ would exclude all large triples, while the exponents derived from the explicit form of the conjecture, itself unproved, are 1.7 and more (Remark I.4.12).

This part does three things. It proves a height inequality that bounds the middle of a

triple from below in terms of its kernel, and splits the kernels into two wings by whether m divides $U_1(m)$. It recasts the reduction of Mollin and Walsh as a statement about witnesses, primes that divide a Pell number exactly once, and organizes the witnesses by their rank. And it computes: every lattice point with kernel up to 10^8 and $T_k < 10^{2000}$ is excluded by a certified witness or by parity, and of the 961 building blocks M_d that occur there, 954 are shown not to be powerful by a certified witness, a prime that divides the block exactly once and is proved prime; 7 remain open. The weak point of the bound is wing (ii), the kernels with $m \mid U_1$: for them only $n > m^{3/2}$ is available, all such kernels up to $1.5 \cdot 10^{12}$ are known [61, Remark 5.5], up to $5.325 \cdot 10^{13}$ with a caveat of the author [62], and beyond that we know no lower bound for their middles other than $n > m^{3/2}$ (Section 2). Part II meets the kernels with $m \mid U_1(m)$ and $T_1(m)$ even, of both residue classes, from the side of pairs, and the collected volume lists the open problems of both parts together.

Main results.

- *The height inequality* (Theorem I.2.1): at every lattice point,

$$\log_{10} T_k(m) \geq m'(1.5 \log_{10} m - \log_{10} m') - \log_{10} 2.$$

With Remark I.2.2, a triple with middle below 10^H has its kernel in one of two wings (Corollary I.2.4): $m \nmid U_1$ and $m < (27 \cdot 10^H)^{2/9}$, or $m \mid U_1$ and $m < 10^{2H/3}$.

- *The reduction theorem* (Theorem I.3.1): among the 10 132 126 kernels up to 10^8 , exactly 445 lattice points have $T_k < 10^{2000}$; 74 of them have T_k odd, and each of the other 371 has a certified witness below $2 \cdot 10^5$. So no triple has a middle $n < 10^{2000}$ with $\text{sqfree}(n^2 - 1) \leq 10^8$, and no triple at all has a middle below $1.83 \cdot 10^{18}$ (Proposition I.6.2(d)).
- *The witness statement* (Theorem I.4.6): there is no triple if and only if every lattice point with $T_1(m)$ even has a witness, and a witness can then be found among the primes whose rank divides k with odd cofactor; a powerful T_k would also have to satisfy the exponent-3 condition of Proposition I.4.4. The b -box (Theorem I.4.5) excludes every middle, of any size, with kernel up to 10^6 and squarefree part up to 10^4 ; the strip (Theorem I.5.1) shows that a triple with kernel up to 2000 has a middle of more than $8.4 \cdot 10^7$ digits; the smooth kernels (Theorem I.5.3) exclude a further range below 10^{2000} .
- *The building blocks* (Proposition I.6.8): of the 961 primitive parts M_d on the 79 kernels of the lattice points, 954 have a certified witness; 7 are open, each with at least 173 digits. Theorem I.7.1 collects the conditions on a counterexample that this part proves or computes.

How to read this part. Every statement is followed by a paragraph *In words* that says what it means with as few formulas as possible; a reader who wants the results can read the statements and these paragraphs. A reader who wants to check can read the proofs, the paragraphs *The computation*, and the tables; every number we computed is generated from the data files of the deposit or from a cited constant, and every numbered statement carries a label that says whether it is proved, taken from the literature, computed, measured, a heuristic, or open (see the table at the end). A statement that combines several kinds carries

the label of the part that needs the most trust: a proof that uses our own computation is labeled computed, and a proof that uses a published result is labeled proved. Table 4 collects the notation.

Plan. The rest of this section sets up the notation and the address system. Section 2 proves the height inequality and the two wings. Section 3 states the reduction theorem and describes the computation behind it. Section 4 proves the exponent condition and the reduction to a witness statement, computes the b -box, and records Granville’s theorem. Section 5 adds the strip, the smooth kernels and the square-type middles. Section 6 reports the data: the exponents at the rank, the list of pairs at distance two, the ladders for the known kernels dividing U_1 , the witnesses, and the building blocks. Section 7 collects the profile of a counterexample, Section 8 records the routes that do not close the problem, and Section 9 states the open questions.

1.1 Notation

A positive integer n is *powerful* if p^2 divides n for every prime p that divides n . For a squarefree integer $m \geq 2$ let $\varepsilon_m = T_1 + U_1\sqrt{m}$ be the fundamental solution of $x^2 - my^2 = 1$, that is, the solution in positive integers with the smallest x . For $k \geq 1$ write

$$\varepsilon_m^k = T_k + U_k\sqrt{m},$$

and $T_k(m)$, $U_k(m)$ when the kernel m has to be named. Put

$$m' = \prod_{p|m, p \nmid U_1} p,$$

the product of the primes of m that do not divide U_1 . For an integer $N \geq 1$ let $\text{sqfree}(N)$ be the product of the primes that divide N to an odd power.

In words. A number is powerful when every prime in it appears at least twice; for example $72 = 2^3 \cdot 3^2$ is powerful and $12 = 2^2 \cdot 3$ is not. The equation $x^2 - my^2 = 1$ is Pell’s equation. Its smallest solution ε_m produces all other solutions as powers, and T_k and U_k are the two parts of the k -th power. The number m' collects the primes of m that do not divide U_1 .

1.2 The address system

Lemma I.1.1 (Mollin and Walsh [51, Theorem, p. 110]). [literature] *There are three consecutive powerful numbers if and only if there are a squarefree $m \equiv 7 \pmod{8}$ and an odd k such that $T_k(m)$ is an even powerful number and m divides $U_k(m)$. In that case the triple is $(T_k - 1, T_k, T_k + 1)$, and $m = \text{sqfree}(T_k^2 - 1)$.*

Mollin and Walsh state the criterion with the fundamental unit of $\mathbb{Q}(\sqrt{m})$. For $m \equiv 7 \pmod{8}$ this unit is ε_m : the ring of integers is $\mathbb{Z}[\sqrt{m}]$, and m has a prime factor $\equiv 3 \pmod{4}$, so no unit of norm -1 exists.

In words. The middle n of a triple of consecutive powerful numbers is always even, and it is a Pell number T_k for exactly one kernel m . So the search for triples becomes a search over pairs (m, k) .

Lemma I.1.2 (Walker [85, Theorems 2.2(3) and 2.5]; Mollin and Walsh [51, p. 111]). [literature] *Let m be squarefree and let p be a prime dividing m . Then $p \mid U_k$ if and only if $p \mid kU_1$. Hence $m \mid U_k$ if and only if $m' \mid k$.*

Proof. By the binomial theorem, $U_k = \sum_{i \text{ odd}} \binom{k}{i} T_1^{k-i} U_1^i m^{(i-1)/2}$. For $p \mid m$ every term with $i \geq 3$ is divisible by p , so $U_k \equiv kT_1^{k-1}U_1 \pmod{p}$. From $T_1^2 - mU_1^2 = 1$ we get $T_1^2 \equiv 1 \pmod{p}$, so T_1 is invertible modulo p , and the first claim follows. A prime $p \mid m$ with $p \mid U_1$ divides every U_k ; a prime $p \mid m$ with $p \nmid U_1$ divides U_k exactly when $p \mid k$. Since m is squarefree, this gives the second claim. $\square$

In words. Whether m divides U_k depends only on whether k is a multiple of m' . So for each kernel the useful indices k are exactly the multiples of m' .

Lemma I.1.3 (Sentance [70, Theorem and Converse, p. 272]). [literature] *Let $n \equiv 0 \pmod{4}$. Then $n - 1$ and $n + 1$ are both powerful if and only if $n = T_k(m)$ for a squarefree $m \equiv 7 \pmod{8}$ and an odd k with $m' \mid k$.*

Proof. ($\Leftarrow$) We have $n^2 - 1 = mU_k^2$ and $m \mid U_k$ by Lemma I.1.2, so $n^2 - 1 = m^3w^2$ for an integer w . As n is even, $n - 1$ and $n + 1$ are coprime, so for every prime p the full power of p in m^3w^2 lies in one of them. Hence each of $n \pm 1$ has the form a^3s^2 with $a \mid m$ squarefree, and such a number is powerful. ($\Rightarrow$) If $n \pm 1$ are powerful and coprime, then $N = n^2 - 1$ is powerful. Write $N = a^2b^3$ with b squarefree; then $N = b(ab)^2$ with $b \mid ab$. So (n, ab) solves $x^2 - by^2 = 1$, and $(n, ab) = (T_k, U_k)$ for $m = b$ and some $k \geq 1$. From $n \equiv 0 \pmod{4}$ we get $n^2 - 1 \equiv 7 \pmod{8}$; as U_k is odd, $m \equiv 7 \pmod{8}$. An even T_k forces k odd, since $T_{2j} = 2T_j^2 - 1$ is odd. Finally $m \mid U_k$ gives $m' \mid k$ by Lemma I.1.2. $\square$

Lemma I.1.4 (Sentance [70, p. 272]; Mollin and Walsh [50, Lemma 1.1]). [literature] *(General even middle.) Let $n \geq 2$ be even. Then $n - 1$ and $n + 1$ are both powerful if and only if $n = T_k(m)$ for a squarefree odd $m \geq 3$ and an odd k with $m' \mid k$ and $T_1(m)$ even. The pair (m, k) is unique: $m = \text{sqfree}(n^2 - 1)$, and k is determined because T_k increases with k . Moreover $m \equiv 7 \pmod{8}$ if and only if $n \equiv 0 \pmod{4}$, and $m \equiv 3 \pmod{8}$ if and only if $n \equiv 2 \pmod{4}$; kernels $m \equiv 1, 5 \pmod{8}$ never occur. An odd n never occurs.*

Proof. As for Lemma I.1.3; only the residues change. For $n \equiv 2 \pmod{4}$ we get $n^2 - 1 \equiv 3 \pmod{8}$, and with U_k odd, $m \equiv 3 \pmod{8}$. For odd k we have $T_k \equiv T_1 \pmod{2}$: if T_1 is even, every term of $T_k = \sum_j \binom{k}{2j} T_1^{k-2j} (mU_1^2)^j$ contains a factor T_1 ; if T_1 is odd, every term with $j \geq 1$ contains the even factor $mU_1^2 = T_1^2 - 1$. So T_k is even exactly when T_1 is. If n were odd, $n \pm 1$ would be even powerful numbers, hence both divisible by 4, which is impossible at distance two. $\square$

The residue split and the uniqueness statement are added here; the rest is Sentance's theorem.

We call a pair (m, k) with $m \equiv 7 \pmod{8}$ squarefree and k an odd multiple of m' a *lattice point*. By Lemma I.1.3, the lattice points with T_k even are exactly the pairs of powerful numbers at distance two whose middle is $\equiv 0 \pmod{4}$. By Lemma I.1.1, a triple needs one more condition: T_k itself must be powerful.

In words. Two powerful numbers that differ by 2 always sit around a Pell number T_k . So every such pair has an address (m, k) , and each address carries at most one pair. A triple is

a pair whose middle is powerful as well. The kernel also tells the remainder: $m \equiv 7 \pmod{8}$ exactly when the middle is a multiple of 4, and $m \equiv 3 \pmod{8}$ when it is 2 more than a multiple of 4.

1.3 Notation and terms at a glance

Table 4 collects the symbols and words used in this part, with the place where each is defined.

symbol or term	meaning	defined in
powerful	every prime factor occurs at least twice	§1
$v_p(N)$	the exponent of the prime p in N	standard
$\text{sqfree}(N)$	the product of the primes that divide N to an odd power	§1
kernel m	a squarefree $m \geq 2$; for a middle n it is $m = \text{sqfree}(n^2 - 1)$	Lemma I.1.4
ε_m, T_k, U_k	the smallest solution $T_1 + U_1\sqrt{m}$ of $x^2 - my^2 = 1$, and its k -th power $T_k + U_k\sqrt{m}$	§1
m'	the product of the primes of m that do not divide U_1	§1
address (m, k)	the pair with $n = T_k(m)$ for a middle n	Lemma I.1.3
lattice point	a pair (m, k) with $m \equiv 7 \pmod{8}$ squarefree and k an odd multiple of m'	§1
wings (i), (ii)	kernels with $m \nmid U_1$, and with $m \mid U_1$	Corollary I.2.4
rank $\alpha_m(p)$	the smallest $k \geq 1$ with $p \mid T_k(m)$	Lemma I.4.1
witness	a prime that divides a number exactly once, which shows that the number is not powerful; the number is mostly $T_k(m)$ or M_d	Theorem I.4.6
class 7, class 3	kernels $m \equiv 7$, resp. $3 \pmod{8}$; they belong to middles $n \equiv 0$, resp. $2 \pmod{4}$	Lemma I.1.4
$\tau(k)$	the number of divisors of k	standard
$\text{rad}(N)$	the product of the distinct primes dividing N	standard
primitive part M_d , building block	the part of $T_d(m)$ on the primes of rank exactly d	Proposition I.6.8
Φ_n	the n -th cyclotomic polynomial	standard
Wieferich prime (to base 2)	a prime p with $p^2 \mid 2^{p-1} - 1$	standard
Wieferich prime for ε_m	an odd prime $p \nmid m$ with $p^2 \mid T_{\alpha_m(p)}(m)$	Proposition I.6.10
abc conjecture	$c < C_\epsilon \text{rad}(abc)^{1+\epsilon}$ for coprime $a + b = c$ and every $\epsilon > 0$	Remark I.4.12
certified prime	a prime proved by a deterministic test or by a certificate that a second program checks	Proposition I.6.8
ψ_{12}	the smallest number that passes the strong test to the first twelve prime bases without being prime, 318 665 857 834 031 151 167 461 [75]	Proposition I.6.8
V_n	the Lucas sequence $V_n(2T_1, 1) = 2T_n$	Proposition I.6.8
B_1	the first-stage bound of the elliptic curve method	Table 7

Table 4: Symbols and terms of Part I.

2 The height inequality

Theorem I.2.1 (height inequality). [proved] *Let $m \equiv 7 \pmod{8}$ be squarefree and let k be an odd multiple of m' , so that $m \mid U_k$. Then*

$$\log_{10} T_k(m) \geq m'(1.5 \log_{10} m - \log_{10} m') - \log_{10} 2.$$

In particular this bound holds for the middle $n = T_k \equiv 0 \pmod{4}$ of every pair of powerful numbers at distance two, and so for the middle of every triple of consecutive powerful numbers.

Proof. Put $d = m/m'$. By definition d divides U_1 , so $U_1 \geq d$. Hence $T_1 = \sqrt{mU_1^2 + 1} > \sqrt{m}U_1 \geq m^{3/2}/m'$. Since $T_k = (\varepsilon_m^k + \varepsilon_m^{-k})/2 \geq \varepsilon_m^k/2 \geq T_1^k/2$ and $k \geq m'$, we get $T_k \geq T_1^{m'}/2 > (m^{3/2}/m')^{m'}/2$. Take logarithms. $\square$

In words. The larger m' is, the larger the middle must be: its number of digits exceeds $m' \log_{10}(m^{1.5}/m') - 1$. This already forces the middle of any triple with a large m' to be enormous.

We did not find this inequality in the literature. Aktaş and Murty [1, proof of Theorem 1.3, p. 251] bound the index using only the trivial lower bound for the fundamental solution, and Chan [18, Section 2] counts solutions of Thue equations; neither uses that m' divides the index. Mollin and Walsh [51, p. 111] and Stephens and Williams [76, p. 619] note that k must be large.

Remark I.2.2 (the constant in the height bound). [proved] All terms of the binomial expansion $T_k = \sum_j \binom{k}{2j} T_1^{k-2j} (mU_1^2)^j$ are nonnegative, so $T_k \geq T_1^k$. Hence the term $-\log_{10} 2$ in Theorem I.2.1 can be dropped:

$$T_k \geq (m^{3/2}/m')^{m'}.$$

The proof uses only that m is odd and squarefree and that $m' \mid k$. So the same bound holds for kernels $m \equiv 3 \pmod{8}$, which carry the pairs with middle $\equiv 2 \pmod{4}$ (Lemma I.1.4).

In words. A slightly better bound holds, without the factor $1/2$, and it holds for both kinds of even middles.

Corollary I.2.3 (kernel bound). [proved] *Let m and k be as in Theorem I.2.1. If $m \nmid U_1$, then $T_k > m^{4.5}/27$. If $m \mid U_1$, then $T_k \geq T_1 > m^{3/2}$.*

Proof. If $m \nmid U_1$, then $m' > 1$, and m' is odd, so $m' \geq 3$. By Remark I.2.2, $\ln T_k \geq f(m')$ with $f(x) = x(1.5 \ln m - \ln x)$. Since $f'(x) = 1.5 \ln m - \ln(ex)$, the function f increases on $[3, m]$ as soon as $m^{3/2}/e > m$, that is, for $m \geq 8$. As $m' \leq m$, we get $\ln T_k \geq f(3) = 4.5 \ln m - 3 \ln 3$, which is the claim. The only kernel below 8 is $m = 7$, where $m' = 7$; there the smallest term is $T_7(7) \geq T_1(7)^7 = 8^7$, far above $7^{4.5}/27$. If $m \mid U_1$, then $U_1 \geq m$, so $T_1 > \sqrt{m}U_1 \geq m^{3/2}$, and $T_k \geq T_1$. $\square$

In words. If m does not divide U_1 , the middle is larger than $m^{4.5}/27$. If m divides U_1 , we only know that the middle is larger than $m^{1.5}$.

Corollary I.2.4 (two wings). [proved] *Let $H > 0$. Let $n < 10^H$ be the middle of a triple of consecutive powerful numbers, or the middle $n \equiv 0 \pmod{4}$ of a pair of powerful numbers at distance two, and let $m = \text{sqfree}(n^2 - 1)$. Then*

(i) either $m \nmid U_1$ and $m < (27 \cdot 10^H)^{2/9}$,

(ii) or $m \mid U_1$ and $m < 10^{2H/3}$.

Proof. By Lemmas I.1.1 and I.1.3, $n = T_k(m)$ for a lattice point (m, k) . In case $m \nmid U_1$, Corollary I.2.3 gives $m^{4.5}/27 < n < 10^H$; in case $m \mid U_1$ it gives $m^{3/2} < n < 10^H$. $\square$

In words. Every triple below 10^H has its kernel in one of two ranges. We call them wing (i), for kernels that do not divide U_1 , and wing (ii), for kernels that do. Wing (ii) allows much larger kernels.

The condition that separates the wings, $m \mid U_1$ or not, has an exact description by Wieferich primes in number fields, due to Fellini and Murty [30, Theorem 1.1]. We use only the different height behavior of the two wings.

Wing (ii) is the weak side. For a kernel with $m \mid U_1$ only the bound $n > m^{3/2}$ of Corollary I.2.3 is available. All such kernels up to $1.5 \cdot 10^{12}$ are known, and up to $5.325 \cdot 10^{13}$ if Reinhart's extended search missed nothing (Remark I.2.5); above that we know nothing about their size beyond Corollary I.2.3. A lower bound for $T_1(m)$ that grows faster with m in this class would close this gap, and we do not have one. Park [59] proves lower bounds for fundamental units that hold for almost all members of certain families. A statement about almost all members does not exclude a single exception, so it does not close the gap.

Remark I.2.5 (Reinhart's list). [literature] The squarefree m with $m \mid U_1(m)$ are the members of OEIS A135735 [56]. Reinhart proved by computer search that the list is complete up to $1.5 \cdot 10^{12}$ [61, Remark 5.5]. His later search covers the squarefree m up to $5.325 \cdot 10^{13}$ and found four further members, but he writes that he does not claim that these are all the members in that range, for lack of an independent double check, and that it is likely that all were found [62, Section 3]. We therefore keep two bounds apart: below $1.5 \cdot 10^{12}$ the list is complete; below $5.325 \cdot 10^{13}$ it is complete if the extended search missed nothing. Every statement that uses the larger bound says so. Of the members $m \equiv 7 \pmod{8}$, 3 lie below the smaller bound and 4 below the larger one.

In words. A published list of the dangerous kernels exists. Up to $1.5 \cdot 10^{12}$ its author proved it complete. Up to $5.325 \cdot 10^{13}$ he searched everything, but could not afford a second, independent run, and says so. We keep the two levels apart, and every statement below names the one it uses.

3 The reduction theorem

Theorem I.3.1 (reduction). [computed] *Among the 10 132 126 squarefree $m \equiv 7 \pmod{8}$ with $m \leq 10^8$, exactly 83 pass the discard criterion described below, and 79 of these have an odd multiple k of m' with $T_k(m) < 10^{2000}$. Together they have exactly 445 such indices k . For 74 of them T_k is odd. For each of the remaining 371 there is a prime $p < 2 \cdot 10^5$ that divides T_k exactly once. Consequently, there is no triple of consecutive powerful numbers whose middle n satisfies $n < 10^{2000}$ and $\text{sqfree}(n^2 - 1) \leq 10^8$.*

Proof. By Lemma I.1.1, such a triple is a lattice point (m, k) with $m \leq 10^8$ and $T_k < 10^{2000}$ at which T_k is even and powerful. An odd T_k fails the first condition. A prime that divides

T_k exactly once shows that T_k is not powerful. It remains to check that the list of points is complete, which is the computation below. $\square$

In words. We checked every possible address with a kernel up to 10^8 and a middle below 10^{2000} . At each of the 445 addresses, either the middle is odd, or a small prime divides it only once. In both cases the middle is not powerful, so no triple lives there. The theorem says nothing about kernels above 10^8 or middles above 10^{2000} .

The computation. Put $H = 2000$. For each $m \leq 10^8$ the continued fraction of $\sqrt{m}$ is computed twice: modulo m , which gives $U_1 \bmod m$ and hence m' by Lemma I.1.2, and in scaled floating point, which gives $\log_{10} T_1$; on a sample of 3531 kernels, all up to $2 \cdot 10^4$ and the others drawn at random up to 10^8 , its relative error against the exact value is below $3 \cdot 10^{-16}$. A kernel is discarded when $m' > (H + 0.31)/\log_{10} T_1 + 1$. This is rigorous: for every admissible k , $\log_{10} T_k \geq k \log_{10} T_1 - 0.31 \geq m' \log_{10} T_1 - 0.31$, so a discarded kernel has no index with $T_k < 10^H$; the extra 1 is a margin against rounding. It absorbs any relative error in $\log_{10} T_1$ up to $\log_{10} T_1 / (H + 0.31)$, and since $T_1 \geq 4$ for every $m \equiv 7 \pmod{8}$, an error below $3 \cdot 10^{-4}$ is enough, and the measured error is far below that. For the 83 surviving kernels the unit is computed exactly and compared with the modular route, and the run stops on any mismatch. Each $T_k < 10^H$ is computed exactly and tested for parity and for a prime $p < 2 \cdot 10^5$ with $v_p(T_k) = 1$. The discard criterion was also rechecked without floating point, from the bound $\log_{10} T_1 \geq (\ell - 1) \log_{10} \varphi - \log_{10} \sqrt{5}$, where ℓ is the period length of the continued fraction and φ the golden ratio; the numerators of the convergents grow at least like Fibonacci numbers. This was done for a random sample of 530 discarded kernels up to 10^8 and for every discarded kernel up to $2 \cdot 10^5$: none is contradicted, and the smallest margin by which a discarded kernel exceeds 10^H is 21.5 orders of magnitude. Three versions of the scan, which differ in speed and not in the criterion, give identical lists of points and witnesses. The list of points with their witnesses is part of the data deposit. Figure 3 shows the numbers.

An example. Of the 445 points, the one with the smallest middle is $(m, k) = (7, 7)$, the example of Mollin and Walsh [51, p. 111]. There $\varepsilon_7 = 8 + 3\sqrt{7}$, so $7 \nmid U_1$ and $m' = m$, and

$$n = T_7(7) = 130\,576\,328 = 2^3 \cdot 29 \cdot 197 \cdot 2857.$$

Since $n \equiv 0 \pmod{4}$ and m' divides k , Lemma I.1.3 shows that $n - 1$ and $n + 1$ are powerful. But the prime 29 divides n exactly once, so n is not powerful, and $(n - 1, n, n + 1)$ is not a triple. The prime 29 is the witness recorded for this point.

Remark I.3.2 (weaker bounds with less work). [proved] The same method gives weaker bounds with less work. The list OEIS A135735 contains all squarefree m with $m \mid U_1(m)$ up to $1.5 \cdot 10^{12}$, and up to $5.325 \cdot 10^{13}$ if Reinhart's extended search missed nothing (Remark I.2.5). A kernel with $m \mid U_1$ that is not on this list is therefore larger than $1.5 \cdot 10^{12}$, and Corollary I.2.3 gives $n > m^{3/2} > 1.83 \cdot 10^{18}$ for every triple with such a kernel; under the assumption on the extended search the bound is $3.88 \cdot 10^{20}$. The enumeration of consecutive powerful pairs behind the b-file of OEIS A060355 is described by Mian and Siddique as exhaustive below 10^{22} but not certified [55, 49]; since every triple contains two such pairs, it excludes triples

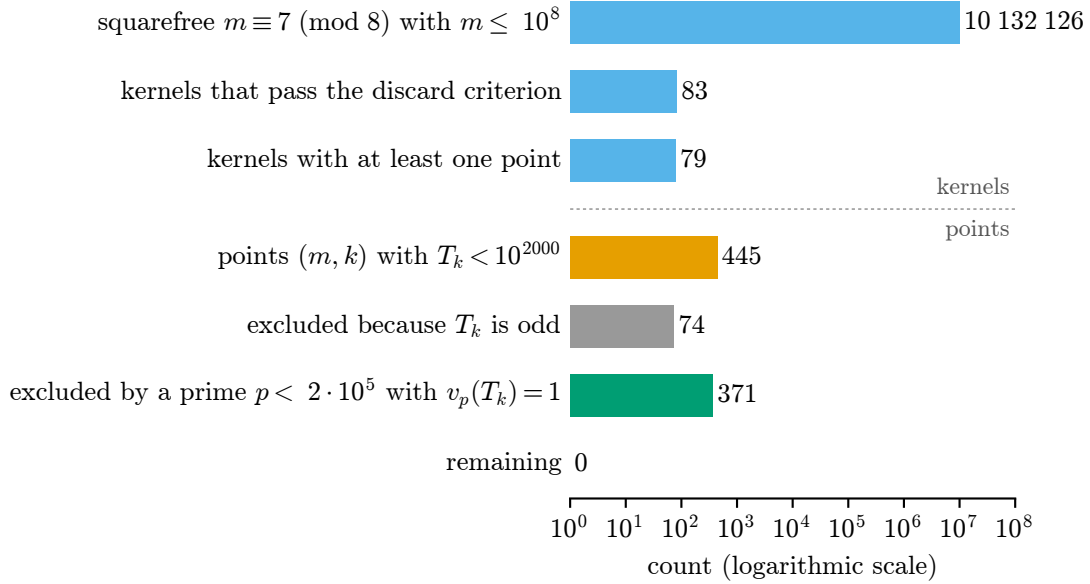


Figure 3: **No point of the box remains.** From all kernels to no candidate (Theorem I.3.1). The horizontal axis is logarithmic. The upper three bars count kernels, the lower ones count points (m, k) ; the two exclusions add up to all points, and none remains.

below that bound on the same terms. The bound above is stated as Proposition I.6.2(d); here it only illustrates the method.

In words. A short list plus the height bound already pushes some triples beyond $1.83 \cdot 10^{18}$. This is only an illustration: other searches reach further, and the reduction theorem is the actual result of this section.

4 The exponent condition and the witness reduction

For a prime p that does not divide m , this section describes at which indices k it divides T_k , and with which exponent. This turns the question whether T_k is powerful into a question about finitely many primes.

Lemma I.4.1 (Bennett and Walsh [8, proof of Lemma 3.3, p. 3484]; Lehmer [41]). [literature] *Let p be a prime with $p \nmid m$. Either p divides no T_k , or there is a smallest index $\alpha_m(p) \geq 1$ with $p \mid T_{\alpha_m(p)}$. In the second case, $p \mid T_k$ if and only if k is an odd multiple of $\alpha_m(p)$.*

We call $\alpha_m(p)$ the *rank* of p . Bennett and Walsh state the lemma in the proof of their Lemma 3.3 and characterize the rank in their Lemma 5.1 [8, p. 3486], where they call it quite standard and refer to Lehmer.

In words. A prime appears in the sequence $T_1, T_2, T_3, \dots$ for the first time at its rank α . After that it divides exactly the terms $T_\alpha, T_{3\alpha}, T_{5\alpha}, \dots$, and no others.

Lemma I.4.2 (Bennett and Walsh [8, Theorem 1.2]). [literature] *Let m and $b > 1$ be squarefree. There is at most one index k with $T_k = bx^2$ for an integer x , and if it exists, then $k = \alpha_m(b)$, the smallest k with $b \mid T_k$.*

By Lemma I.4.1, $\alpha_m(b)$ exists if and only if every prime of b has a rank and all these ranks have the same 2-adic valuation; then $\alpha_m(b)$ is their least common multiple.

In words. A term T_k can be b times a square for at most one index k , and this index is the first place where b divides the sequence.

Lemma I.4.3 (valuation identity). [proved] *Let p be an odd prime with $p \nmid m$ and $p \mid T_r$. Then $v_p(T_{rt}) = v_p(T_r) + v_p(t)$ for every odd t .*

Proof. Here $v_p(z)$, for $z \in \mathbb{Z}[\sqrt{m}]$, is the largest e with $p^e \mid z$ in $\mathbb{Z}[\sqrt{m}]$, the least p -adic valuation of the two coefficients of z . It is not multiplicative when p splits, but it does not change under multiplication by a unit or by an element congruent to 1 modulo p , and $v_p(tz) = v_p(t) + v_p(z)$ for $t \in \mathbb{Z}$; this is all that the standard proof of the lifting-the-exponent lemma uses. Put $w = \varepsilon_m^{2r}$ and $u = w + 1 = \varepsilon_m^r \cdot 2T_r$, so $v_p(u) = v_p(T_r) \geq 1$. For odd t ,

$$2T_{rt} = \varepsilon_m^{-rt} (w^t + 1) = \varepsilon_m^{-rt} (1 - (1 - u)^t),$$

and the lifting-the-exponent lemma in $\mathbb{Z}[\sqrt{m}]$ gives $v_p(1 - (1 - u)^t) = v_p(u) + v_p(t)$. It applies because p is odd, $p \nmid m$, and p divides u but not $1 - u$; the proof is the same as for integers. $\square$

In words. If a prime p divides T_r , then in T_{rt} it appears exactly as often as in T_r , plus as often as it divides t . So a prime that appears once in T_r appears once in T_{rt} , unless it also divides t .

Proposition I.4.4 (exponent-3 condition). [proved] *Let T_k be an even powerful number with k odd and $m' \mid k$, and let $b = \text{sqfree}(T_k)$, so that $T_k = b^3 z^2$ for an integer z . If $b > 1$, then $k = \alpha_m(b)$, and for every odd prime $p \mid b$*

$$v_p(T_{\alpha_m(p)}) + v_p(\alpha_m(b)/\alpha_m(p)) \geq 3.$$

Proof. We have $T_k = b(bz)^2$, so $k = \alpha_m(b)$ by Lemma I.4.2. Now let $p \mid b$ be odd. Then $p \mid T_k$, so $k = \alpha_m(p)t$ with t odd by Lemma I.4.1, and $v_p(T_k) = v_p(T_{\alpha_m(p)}) + v_p(t)$ by Lemma I.4.3. As p divides the powerful number T_k to an odd power, this power is at least 3. $\square$

For general Lucas sequences the valuation identity is standard; see Sanna [68]. Ballot [5, p. 265, (2)] printed it with the term $(\nu - 1) + v_p(n)$, and the errata [4] corrects this to $\nu + v_p(n/\rho)$. The corrected form is the one in Lemma I.4.3. On 56 cases of our sequence, the corrected form holds in all of them and the printed form in 0; on 50 cases of the Fibonacci numbers, the corrected form holds in all of them.

In words. If the middle T_k is powerful, then every prime that divides it an odd number of times divides it at least three times. This gives a condition on the first term $T_{\alpha(p)}$ in which the prime appears. Proposition I.4.4 does not prove the conjecture of Mollin and Walsh [51, p. 111] that T_k is never an even powerful number for odd $k > 1$; it says what a counterexample would have to satisfy.

Theorem I.4.5 (the b -box). [computed] *There is no triple of consecutive powerful numbers whose middle n has kernel $m \leq 10^6$ and $\text{sqfree}(n) \leq 10^4$, whatever the size of n .*

Proof. If $\text{sqfree}(n) = 1$, then n is a square, and Theorem I.5.4 gives $m \mid U_1$; no such kernel with T_1 even lies below 10^6 : the kernels $m \mid U_1$ below this bound are listed completely [61, Remark 5.5], and the only one with $m \equiv 7 \pmod{8}$, 209 991, has T_1 odd (Theorem I.5.4; OEIS [56]). Otherwise let $b = \text{sqfree}(n) > 1$ and $n = T_k(m)$. By Proposition I.4.4, $k = \alpha_m(b)$, so $\alpha_m(b)$ must exist, be odd and be a multiple of m' , and b^3 must divide $T_{\alpha_m(b)}$. We checked all pairs. There are 101 329 squarefree $m \equiv 7 \pmod{8}$ with $m \leq 10^6$; for 32 118 of them T_1 is odd, so they carry no even middle. The others give 421 010 513 pairs (m, b) with squarefree $b \leq 10^4$, $b = 1$ included. In 343 504 538 pairs some prime of b has no rank, in 16 603 302 the ranks have different 2-adic valuations, in 11 242 224 the index $\alpha_m(b)$ is even, and in 49 659 524 it is not a multiple of m' ; this class contains the pairs with $b = 1$, where $k = 1$ while $m' > 1$ for every such kernel with T_1 even [62]. For each of the remaining 925 pairs, b^3 does not divide $T_{\alpha_m(b)}$; this is tested modulo b^3 . $\square$

In words. If the kernel is at most 10^6 and the part of the middle with odd exponents is at most 10^4 , there is no triple, however large the middle is. This result has no height limit, unlike Theorem I.3.1.

Theorem I.4.6 (reduction to a witness statement). [proved] *Let $\mathcal{L}$ be the set of lattice points (m, k) with $T_1(m)$ even. For $(m, k) \in \mathcal{L}$ one has $4 \mid T_k(m)$ and $\gcd(T_k(m), m) = 1$. The following statements are equivalent.*

- (i) *There are no three consecutive powerful numbers.*
- (ii) *For every $(m, k) \in \mathcal{L}$ there is a prime p with $v_p(T_k(m)) = 1$, a witness.*
- (iii) *For every $(m, k) \in \mathcal{L}$ there is an odd prime $p \nmid m$ such that k is an odd multiple of $\alpha_m(p)$, $v_p(T_{\alpha_m(p)}(m)) = 1$, and $p \nmid k/\alpha_m(p)$.*

In particular, if $T_k(m)$ is powerful, then $p^2 \mid T_k(m)$ for every prime p with $\alpha_m(p) = k$.

Proof. By Lemmas I.1.1 and I.1.3, a triple exists if and only if $T_k(m)$ is powerful for some $(m, k) \in \mathcal{L}$. For every $(m, k) \in \mathcal{L}$ the number T_k is even (proof of Lemma I.1.4), $U_k = mw$ by Lemma I.1.2 with w odd, and $T_k^2 = 1 + m^3 w^2 \equiv 1 + 7 \equiv 0 \pmod{8}$, because $m \equiv 7 \pmod{8}$; hence $4 \mid T_k$. From $T_k^2 \equiv 1 \pmod{m}$ we get $\gcd(T_k, m) = 1$. So T_k is powerful if and only if no odd prime $p \nmid m$ has $v_p(T_k) = 1$, which proves (i) $\Leftrightarrow$ (ii). For (ii) $\Leftrightarrow$ (iii): by Lemma I.4.1 an odd prime $p \nmid m$ divides T_k exactly when k is an odd multiple of $\alpha_m(p)$, and then $v_p(T_k) = v_p(T_{\alpha_m(p)}) + v_p(k/\alpha_m(p))$ by Lemma I.4.3. So $v_p(T_k) = 1$ exactly when $v_p(T_{\alpha_m(p)}) = 1$ and $p \nmid k/\alpha_m(p)$. For $\alpha_m(p) = k$ the identity reads $v_p(T_k) = v_p(T_{\alpha_m(p)})$, which gives the last claim. $\square$

In words. Three consecutive powerful numbers do not exist exactly when every address has a witness: a prime that divides the middle only once. The last sentence says what a counterexample would need. Every prime that appears in T_k for the first time would have to divide it at least twice, a coincidence of the kind known as a Wieferich condition.

The reduction is due to Mollin and Walsh, who state it and conjecture its conclusion [51, p. 111]. The mechanism has a published general form: Fellini and Murty [30, Lemma 5.6] split $\alpha^n - 1$ into a squarefree and a powerful part in the ring of integers of a number field and show that every prime of the squarefree part is a base- α non-Wieferich prime. Kym [38, Corollary 6.6 and Remark 6.7] gives the closest published form for Lucas sequences: primitive prime divisors in a coprime product must appear with exponent at least 2. Kym studies a different equation and not consecutive powerful numbers. What is specific here is the translation into the addresses (m, k) and the finite bookkeeping of Section 6.

Lemma I.4.7 (the T_1 -part of a lattice point). [proved] *Let m be odd and squarefree with $T_1 = T_1(m)$ even, and let k be odd. Then $T_1 \mid T_k$ and T_k/T_1 is odd, so $v_2(T_k) = v_2(T_1)$. For every odd prime $p \mid T_1$ one has $\alpha_m(p) = 1$ and $v_p(T_k) = v_p(T_1) + v_p(k)$. Consequently:*

- (a) *the 2-adic valuation of the middle of any triple with kernel m is $v_2(T_1(m))$, which is at least 2 for $m \equiv 7 \pmod{8}$;*
- (b) *if T_k is powerful, then k is divisible by every odd prime p with $v_p(T_1) = 1$.*

Proof. We have $T_k = \sum_{j=0}^{(k-1)/2} \binom{k}{2j} T_1^{k-2j} (mU_1^2)^j$, and every term contains T_1^{k-2j} with $k-2j \geq 1$, so $T_1 \mid T_k$. In T_k/T_1 the term with $j = (k-1)/2$ equals $k(mU_1^2)^{(k-1)/2}$, which is odd because k, m and U_1 are odd; U_1 is odd since T_1 is even and $T_1^2 - mU_1^2 = 1$. Every other term contains T_1^{k-2j-1} with an even exponent at least 2 and is even. So T_k/T_1 is odd. The p -adic formula is the valuation identity of Lemma I.4.3 with $\alpha_m(p) = 1$. Part (a) follows, where $v_2(T_1) \geq 2$ for $m \equiv 7 \pmod{8}$ because $T_1^2 = 1 + mU_1^2 \equiv 1 + 7 \equiv 0 \pmod{8}$, and part (b) follows because $v_p(T_k) = 1 + v_p(k) \geq 2$ forces $p \mid k$. $\square$

In words. All terms T_k with odd k contain T_1 as a factor, with the same power of 2. Every odd prime that divides T_1 only once must also divide the index k of a powerful middle. For $m = 7$ it gives nothing, because $T_1 = 8$; the ladder of Theorem I.5.1 applies the same idea at the index $k_0 = m'$ instead of at 1, and for $m = 7$ its first step is the one of Mollin and Walsh.

Corollary I.4.8 (the divisor count of the Wieferich conditions). [proved] *Let $m > 1$ be squarefree and let k be odd. For every divisor $d \geq 5$ of k there is a prime p_d with $\alpha_m(p_d) = d$, and:*

- (i) *every odd prime $p \nmid m$ with $\alpha_m(p) = d$ satisfies $p \equiv \pm 1 \pmod{4d}$, so $p \geq 4d - 1$; in particular no prime dividing d has rank d ;*
- (ii) *distinct divisors d give distinct primes p_d , and each p_d divides $T_k(m)$; hence the number $\omega(T_k(m))$ of distinct prime factors satisfies $\omega(T_k(m)) \geq \#\{d \mid k : d \geq 5\}$, which is $\tau(k) - 1$ if $3 \nmid k$ and $\tau(k) - 2$ if $3 \mid k$;*
- (iii) *if $T_k(m)$ is powerful, then for every divisor $d \geq 5$ of k and every prime p with $\alpha_m(p) = d$, in particular for p_d , either $p^2 \mid T_d(m)$ or $p \mid k/d$; the second case forces $k \geq d(4d - 1)$.*

Proof. Put $\alpha = T_1 + U_1\sqrt{m}$ and $\beta = T_1 - U_1\sqrt{m}$. They are the distinct real roots of $z^2 - 2T_1z + 1$, so $D_n = (\alpha^n - \beta^n)/(\alpha - \beta) = U_n/U_1$ is a Lucas sequence in the sense of Carmichael, and $D_{2n} = D_n \cdot 2T_n$. *Existence.* Let $d \geq 5$ divide k ; then d is odd and $2d \geq 10$.

By Carmichael's theorem [17], in the form of Yabuta [87, Theorem 1], the term D_{2d} has a primitive divisor p : $p \mid D_{2d}$ and $p \nmid D_j$ for $0 < j < 2d$. Since $D_{2d} = D_d \cdot 2T_d$ and $p \nmid D_d$, we get $p \mid 2T_d$; since $p \nmid D_2 = 2T_1$, the prime p is odd, so $p \mid T_d$. If p divided some T_j with $j < d$, then $p \mid 2T_j \mid D_{2j}$ with $2j < 2d$, which is impossible; so $\alpha_m(p) = d$. Also $p \nmid m$, because $p \mid m$ would give $T_j^2 = 1 + mU_j^2 \equiv 1 \pmod{p}$ for all j . (i) $p \mid T_d$ means $\varepsilon_m^{2d} \equiv -1 \pmod{p}$, and d is the smallest index with this property, so ε_m has order $4d$ modulo p . The elements of norm one in $(\mathbb{F}_p[\sqrt{m}])^\times$ form a cyclic group of order $p - (\frac{m}{p})$, and ε_m has norm one, so $4d \mid p - (\frac{m}{p})$ and $p \equiv \pm 1 \pmod{4d}$. This uses only $\alpha_m(p) = d$, so it holds for every prime of rank d . As $p \geq 4d - 1 > d$, no prime dividing d has rank d . (ii) The rank determines d , so distinct d give distinct primes. Since k/d is odd, Lemma I.4.1 gives $p_d \mid T_k$. The odd divisors of k below 5 are 1 and 3. (iii) Let p be any prime with $\alpha_m(p) = d$. As above $p \nmid m$, and $p \mid T_k$ as in (ii). If T_k is powerful, then $v_p(T_k) \geq 2$, and $v_p(T_k) = v_p(T_d) + v_p(k/d)$ by Lemma I.4.3. So $p^2 \mid T_d$ or $p \mid k/d$; in the second case $k/d \geq p \geq 4d - 1$ by (i). The prime p_d is one such prime. $\square$

In words. Every divisor $d \geq 5$ of k brings its own new prime into T_k . For T_k to be powerful, each of these primes must pass a Wieferich-type test, unless it also divides k/d , which forces $k \geq d(4d - 1)$ and is therefore impossible when d is large compared with $\sqrt{k}$. So a counterexample needs many coincidences at once, each at a different prime; and every further prime of the same rank d must pass the same test. When k is a prime, the count gives only a single condition.

The argument is standard, and it is in print for the Fibonacci and Lucas numbers. The earliest occurrence we found is Luca [44, p. 267], who derives $\omega(F_n) \geq \tau(n) - 2$ from Carmichael's theorem in the same way. Bugeaud, Luca, Mignotte and Siksek [16, proof of Theorem 5] use the same step. Pongsriiam [60, Lemmas 2.5(i) and 3.4(i)] states it, including the version for the companion sequence used above, and Somer and Křížek [74, Lemma 2.7] collect the general facts for Lucas sequences, with the congruence modulo $2n$. What we did not find in the literature is the statement for the norm-one Pell sequences and the sharpening from modulus $2d$ to modulus $4d$, which follows from $N(\varepsilon_m) = 1$; neither is a new mechanism. Mollin and Walsh [51, p. 111] already point to the theory of Lucas and Lehmer [45, 41] for this step.

Theorem I.4.9 (Granville [33, Theorem, p. 216]). [literature] *If there are no three consecutive powerful numbers, then there are infinitely many primes p with $p^2 \nmid 2^p - 2$.*

In words. If triples never exist, then infinitely many primes are not Wieferich primes to base 2. Nobody knows how to prove this last statement without extra assumptions. So a proof that triples never exist would also settle an open problem about Wieferich primes.

That infinitely many primes are non-Wieferich is known under the *abc* conjecture. Silverman [72, Theorem 1] proved it for a fixed rational base; Kotyada and Muthukrishnan [37, Theorem 1.1] for the fundamental unit of a real quadratic field of class number one; and Fellini and Murty [30, Theorem 1.2] for number fields, under the *abc* conjecture of Masser. Fellini and Murty also derive it from the finiteness of base- α super-Wieferich primes [30, Theorem 1.3]; in the rational case this is Theorem 1.4 of Murty and Séguin [52]. All of these split $\alpha^n - 1$ into a squarefree and a powerful part, as this section does. The same splitting, applied to the terms of a Lucas sequence, shows under the *abc* conjecture that there are only finitely many powerful terms: Ribenboim and Walsh [64, Corollary 2], and Yabuta [88, Theorem 2.3], who uses the primitive divisor theorem of Bilu, Hanrot and Voutier [9]. For the

sequences $T_k(m)$ this gives, under *abc*, finitely many k for each fixed m . The *abc* conjecture also implies that there are only finitely many triples [1, Section 6], [79, Section 1]. None of these conditional results is effective.

Granville's proof is a construction [33, pp. 216–217]. If every prime $p > p_0$ satisfied $p^2 \mid 2^p - 2$, it would produce a triple whose middle is a power of 2, and in fact one with middle A^n for every $n \geq 1$, where A is the number of his construction, so that the theorem already holds if there are only finitely many triples; Granville states it under the conjecture of Mollin and Walsh. In our notation such a middle is $T_k(m) = 2^a$. By Lemma I.4.7 the number T_1 divides 2^a and T_k/T_1 is odd, so $T_k = T_1$ and $k = 1$. Proposition I.6.7 treats this case numerically.

Remark I.4.10 (what the witness condition does not follow from). [proved] The witness condition does not follow from coprimality. If n and $n+1$ are powerful, then $\gcd(n, n+2) \mid 2$ and $\gcd(n+1, n+2) = 1$, so every odd prime factor of $n+2$ divides neither n nor $n+1$. This says which primes occur in $n+2$, but nothing about their exponents. An example is $n = 9800$: both n and $n+1$ are powerful, and $n+2 = 2 \cdot 13^2 \cdot 29$, so the new prime 13 appears twice. If n is even and powerful, then $4 \mid n$, so $n+2 \equiv 2 \pmod{4}$ and $v_2(n+2) = 1$; this is the reason for the parity filter of Theorem I.3.1, which keeps only kernels with T_1 even. The difficulty lies entirely in the case of odd n , where the witness must be odd. The step from “a new prime divides $n+2$ ” to “some prime divides $n+2$ exactly once” is equivalent to the non-existence of a triple, and so, by Theorem I.4.6, to the witness statement (ii) there, which asks for the witness in the middle of a triple instead of in its outer number. An argument of this shape has been posted [13]; this remark concerns that step and is not an assessment of the preprint.

In words. Knowing that a prime is new in $n+2$ does not tell us how often it appears there. The pair 9800, 9800 + 1 shows that a new prime can appear twice in the next number.

Remark I.4.11 (where a witness in an outer number can sit). [proved] Let m be odd and squarefree with $T_1 = T_1(m)$ even, let k be odd, and let γ be a square root of $\varepsilon = T_1 + U_1\sqrt{m}$ in an extension field, so that γ^{-1} is a square root of ε^{-1} . Then $(\gamma \pm \gamma^{-1})^2 = \varepsilon + \varepsilon^{-1} \pm 2 = 2(T_1 \pm 1)$, and in the same way $(\gamma^k \pm \gamma^{-k})^2 = 2(T_k \pm 1)$. The quotients

$$V_k = \frac{\gamma^k + \gamma^{-k}}{\gamma + \gamma^{-1}} = \sum_{j=0}^{k-1} (-1)^j \gamma^{k-1-2j}, \quad W_k = \frac{\gamma^k - \gamma^{-k}}{\gamma - \gamma^{-1}} = \sum_{j=0}^{k-1} \gamma^{k-1-2j}$$

are polynomials with integer coefficients in $(\gamma + \gamma^{-1})^2 = 2(T_1 + 1)$: the terms j and $k-1-j$ carry the same sign and combine to $\gamma^r + \gamma^{-r}$ with r even, which is an integer polynomial in $\gamma^2 + \gamma^{-2} = (\gamma + \gamma^{-1})^2 - 2$. Hence V_k and W_k are integers, and squaring gives

$$T_k + 1 = (T_1 + 1) V_k^2, \quad T_k - 1 = (T_1 - 1) W_k^2.$$

Reducing modulo a prime $p \mid T_1 + 1$ amounts to setting $(\gamma + \gamma^{-1})^2 = 0$, that is $\gamma^2 = -1$, where $V_k = k\gamma^{k-1} = \pm k$; reducing modulo a prime $p \mid T_1 - 1$ amounts to $(\gamma - \gamma^{-1})^2 = 0$, that is $\gamma^2 = 1$, where $W_k = k$. So $V_k \equiv \pm k \pmod{p}$ for every prime $p \mid T_1 + 1$, and $W_k \equiv k \pmod{p}$ for every prime $p \mid T_1 - 1$; all these primes are odd because T_1 is even.

Now let $n+1 = T_k(m)$, so that n and $n+2$ are the outer numbers. Every prime that divides n or $n+2$ but not $T_1 \mp 1$ appears there with an even exponent, so the squarefree parts

of the outer numbers are those of $T_1 - 1$ and $T_1 + 1$, whatever k is. An odd prime q divides $n + 2$ exactly once if and only if q divides $T_1 + 1$ exactly once and $q \nmid k$, and likewise for n and $T_1 - 1$; if $q \mid k$, then $q \mid V_k$ and the exponent is at least 3. Such a prime divides m and not U_1 , because $(T_1 - 1)(T_1 + 1) = mU_1^2$ and q divides exactly one of the two coprime factors, to an odd power; so it divides m' . This settles whether the structure forces an odd witness in $n + 2$ as it forces the prime 2 when n is even: the candidates are exactly the odd primes that divide $T_1(m) + 1$ exactly once, each of them is a witness unless it divides k , and on a lattice point, where $m' \mid k$ (Lemma I.1.3), every candidate divides k . If $T_1 + 1$ is a square there is no candidate at all, and $n + 2$ is powerful for every odd k . This is the case for every prime kernel $m \equiv 7 \pmod{8}$, by an elementary argument: if T_1 were odd, then U_1 would be even and the coprime numbers $(T_1 - 1)/2$ and $(T_1 + 1)/2$ would have product $m(U_1/2)^2$, so one of them would be a square a^2 and the other mb^2 with $b \geq 1$; the case $a^2 - mb^2 = -1$ is impossible for $m \equiv 3 \pmod{4}$, and $a^2 - mb^2 = 1$ would be a solution smaller than the fundamental one. So T_1 is even, $T_1 - 1$ and $T_1 + 1$ are coprime with product mU_1^2 , one of them is a square, and it is not $T_1 - 1 \equiv 3 \pmod{4}$. Hence no odd prime is forced in the way the prime 2 is, and for the outer numbers the question returns to Theorem I.4.6, which places the witness in the middle.

In words. The two neighbors of T_k inherit the primes of odd exponent from the two neighbors of T_1 ; everything else they gain comes in squares. So an odd witness in an outer number can only be a prime of $T_1 \pm 1$, and on a lattice point none of these primes is available.

Remark I.4.12 (two elementary constraints, and the size of the gap). [proved] First, if a prime q divides N and $q > N^{1/2}$, then $v_q(N) = 1$. So a powerful number has no prime factor above its square root, and a large prime factor is always a witness. Second, $n^2 - (n - 1)(n + 1) = 1$ is an *abc* equation. For a triple, all three numbers are powerful, so the radical of $(n - 1)n(n + 1)$ is at most $((n - 1)n(n + 1))^{1/2} < n^{3/2}$, while the largest term is n^2 . Hence an *abc* inequality with exponent below $4/3$ would exclude all large triples. (The *abc* conjecture states that for every $\epsilon > 0$ there is a constant C_ϵ with $c < C_\epsilon \text{rad}(abc)^{1+\epsilon}$ for all coprime positive integers with $a + b = c$, where rad is the product of the distinct prime factors.) The exponents derived from Baker's explicit form of the conjecture are larger than $4/3$: $7/4$ [39, Theorem 1], 1.72 [22, Theorem 1.4], and 1.7 [21, Theorem 4.1]; all three assume that explicit conjecture. The explicit form does give a finite bound: under it, Chim, Shorey and Sinha [22, Theorem 1.8] show that the middle of every triple is below 10^{51075} , and they cite a bound 10^{20000} attributed to Trudgian. Without any hypothesis, the best known bounds in the *abc* problem are exponential in a power of the radical [77].

In words. A very large prime factor is always a witness. The *abc* conjecture in a strong enough form would rule out all large triples. The explicit forms that have been worked out are themselves conjectures, and they give exponents too large for this. What is proved without any assumption is much weaker still.

Remark I.4.13 (the cyclotomic axis). [proved] If one member of a triple is a proper power x^r with $r \geq 2$, its neighbors are $x^r \pm 1$, and these split into values $\Phi_e(\pm x)$ of cyclotomic polynomials. Each factor gives further conditions on the possible witnesses. These conditions apply only when a member is a proper power, and they are treated in Part III. For $r = 2$, the number $x^2 + 1$ is powerful exactly when $x^2 - b^3y^2 = -1$ has a solution with b squarefree; this equivalence is due to De Koninck, Doyon and Luca [26, (1)–(2)].

In words. If one number of a triple is a square, a cube or a higher power, its neighbors split into smaller factors, and each factor brings its own conditions on a witness. Part III follows this route.

For a middle that is a cube, four shapes of the neighbors have been excluded, with primes p and q : Chan [19] excludes $x^3 - 1 = p^3 y^2$ together with $x^3 + 1 = q^3 z^2$, She [71] excludes $x^3 - 1 = p^2 a^3$ together with $x^3 + 1 = q^2 b^3$, and [69, Theorem 1] excludes the two mixed shapes, $x^3 - 1 = p^2 a^3$ with $x^3 + 1 = q^3 z^2$ and $x^3 - 1 = p^3 y^2$ with $x^3 + 1 = q^2 b^3$. That paper reduces both mixed shapes to the single equation $t^6 + t^3 + 1 = 3w^2$, whose rational solutions it determines through an elliptic quotient of rank 0. Each of these results excludes infinitely many candidate middles at once, by an equation and not by a computation up to a bound.

5 Exclusion regions and square-type middles

Theorem I.3.1 covers kernels up to 10^8 and middles below 10^{2000} . This section adds three results of a different shape: a strip of small kernels where every possible middle is huge, a box of larger kernels built from small primes, and a restriction for middles of a special form.

Theorem I.5.1 (the strip). [computed] *There are 206 squarefree $m \equiv 7 \pmod{8}$ with $m \leq 2000$. For 52 of them T_1 is odd, and no triple has such a kernel. For each of the other 154, the middle $n = T_k(m)$ of a triple with kernel m would have more than $8.4 \cdot 10^7$ decimal digits. For $m = 7$ it would have more than $10^{17.5}$ decimal digits.*

Proof. If T_1 is odd, then T_k is odd for every odd k (proof of Lemma I.1.4), so T_k is not the middle of a triple. For the other kernels we climb a ladder of two steps. It rests on Lemma I.4.1 and Lemma I.4.3: if a prime p divides T_{k_0} exactly once and k is an odd multiple of k_0 , then $v_p(T_k) = 1 + v_p(k/k_0)$, so T_k can be powerful only if p divides k/k_0 . Start with the smallest admissible index $k_0 = m'$. The primes that divide T_{k_0} exactly once force k to be a multiple of a larger index k_1 , and the same argument at k_1 gives an index k_2 . So $T_k \geq T_{k_2}$ whenever T_k can be powerful, and T_{k_2} has the stated number of digits. In the first run such primes were found at both steps for all but 14 kernels; a deeper search found them for these 14 as well, but for 2 of them only at the first step; there the bound from the first step is already larger than the smallest one. All other bounds are larger than the smallest one too, which occurs at $m = 223$. $\square$

For $m = 7$ the first step is due to Mollin and Walsh [51, p. 111]: the primes 29, 197 and 2857 divide $T_7(7)$ exactly once, so k must be a multiple of $7 \cdot 29 \cdot 197 \cdot 2857$.

In words. Of the kernels up to 2000, the 52 with T_1 odd carry no triple at all. For the other 154 a triple is not ruled out, but its middle would be gigantic: it would have more than $8.4 \cdot 10^7$ digits. For $m = 7$ even the number of its digits would be larger than $10^{17.5}$.

Remark I.5.2 (the bound of Chim, Shorey and Sinha). [literature] Chim, Shorey and Sinha [22, p. 440] state the lower bound $10^{3 \cdot 10^{13}}$ for the middle when $m \in \{15, 23, 31, 39, 47, 55, 87\}$, following the arguments of Mollin and Walsh, and do not give the computation. For $m = 15$, 31 and 87 the two-step ladder gives a larger bound, for $m = 23$ and 47 a smaller one. For $m = 39$ and 55 the number T_1 is odd, so these kernels carry no triple at all.

In words. For the kernels named by Chim, Shorey and Sinha, our ladder gives a larger bound for three and a smaller one for two; two of them carry no triple at all, because T_1 is odd.

Theorem I.5.3 (smooth kernels). [computed] *No triple of consecutive powerful numbers with middle $n < 10^{2000}$ has a kernel m with $10^8 < m \leq 10^{15}$ whose prime factors are all at most 100 and which has at least 4 prime factors. There are 889 437 such squarefree $m \equiv 7 \pmod{8}$. For 356 888 of them the period ℓ of the continued fraction of $\sqrt{m}$ exceeds 9573, which forces $T_1 > 10^{2000}$. None of the remaining 532 549 has an admissible index k with $T_k(m) < 10^{2000}$.*

Proof. The numerators A_j of the convergents of $\sqrt{m}$ satisfy $A_j \geq F_{j+2}$, where F_j are the Fibonacci numbers, and T_1 is the numerator at the end of the first period; the period is even because $x^2 - my^2 = -1$ has no integer solution for $m \equiv 3 \pmod{4}$. So $T_1 \geq F_{\ell+1} \geq \varphi^{\ell-1}$, and $\ell > 9573$ gives $T_1 > 10^{2000}$. For the other kernels the computation of Theorem I.3.1 finds no admissible index below 10^{2000} . $\square$

In words. Theorem I.3.1 stops at kernels up to 10^8 . For one kind of larger kernel, up to 10^{15} and with all prime factors at most 100, we can still rule out every triple with a middle below 10^{2000} . (Such a kernel above 10^8 automatically has at least 4 prime factors, since any 4 primes up to 100 have a product below 10^8 .)

Theorem I.5.4 (square-type middles). [computed] *Let $(n-1, n, n+1)$ be a triple of consecutive powerful numbers with $n = T_k(m)$. If n is a square or twice a square, in particular if n is a fourth power, then $k = 1$ and $m \mid U_1$. Hence either $m > 1.5 \cdot 10^{12}$, or m is one of 209 991, 4 099 215 and 117 477 414 815; if Reinhart's extended search missed nothing (Remark I.2.5), either $m > 5.325 \cdot 10^{13}$, or m is one of these three or 39 028 039 587 479. None of the four carries a triple with a square-type middle, and T_1 is odd for 209 991 and 117 477 414 815, so that these two kernels carry no triple at all (Lemma I.1.4).*

Proof. If $n = X^2$, then $X^4 - mU_k^2 = 1$. By Cohn's theorem [23], quoted as Theorem 1.1 of Bennett and Walsh [8], the only possible solutions are $X^2 = T_1$ and $X^2 = 2T_1^2 - 1 = T_2$. So $k \in \{1, 2\}$, and k is odd (Lemma I.1.1), so $k = 1$. If $n = 2X^2$, then $T_k = 2X^2$, and by Theorem 1.2 of Bennett and Walsh [8] at most one index k has this property, namely the smallest k with $2 \mid T_k$. As T_1 is even for a triple, this is $k = 1$. A fourth power is a square. In all cases $k = 1$ is a multiple of m' , so $m' = 1$ and $m \mid U_1$ by Lemma I.1.2. The squarefree m with $m \mid U_1(m)$ are listed in OEIS A135735, completely up to $1.5 \cdot 10^{12}$ [61, Remark 5.5] and, under the stated assumption, up to $5.325 \cdot 10^{13}$ [62] (Remark I.2.5); the four values above are the members with $m \equiv 7 \pmod{8}$, the first three below the smaller bound. For $m = 209 991$ and for $m = 117 477 414 815$ the number T_1 is odd (computed with exact integers), so these two kernels carry no triple at all by Lemma I.1.4. For the other two, a prime divides T_1 exactly once: 701 for $m = 4 099 215$ and 5 for $m = 39 028 039 587 479$. So $n = T_1$ is not powerful. $\square$

Cohn's theorem sharpens Ljunggren's bound of at most two solutions for $x^4 - Dy^2 = 1$ [42]. Cohn obtains the single case with two solutions, $D = 1785$, from Ljunggren's result that $y^2 = 2x^4 - 1$ has only the solutions $x = 1$ and $x = 13$ [43].

In words. If the middle of a triple is a square, twice a square, or a fourth power, then its kernel must divide U_1 . Such kernels are known completely up to $1.5 \cdot 10^{12}$, and up to $5.325 \cdot 10^{13}$ if a search that its author could not repeat missed nothing; none of the known ones carries a

triple with such a middle. So a triple of this special shape would need an enormous kernel. The theorem does not exclude these middles; it restricts where they could be.

Remark I.5.5 (the share of the candidates covered). [computed] Theorem I.5.4 applies to a fixed share of the candidates for a middle. The middle of a triple is powerful and divisible by 4 (Lemma I.1.4), so we compare with all powerful n with $4 \mid n$. The number of powerful $n \leq x$ is asymptotic to $(\zeta(3/2)/\zeta(3))\sqrt{x}$ [28, 6]. The condition $4 \mid n$ multiplies this by $(\sqrt{2} + 1)/(2\sqrt{2} + 1)$: in the Dirichlet series of the powerful numbers, the Euler factor at 2 changes from $1 + 2^{-2s}/(1 - 2^{-s})$ to $2^{-2s}/(1 - 2^{-s})$, and at $s = 1/2$ their quotient is this number. Among these n , the squares number about $\sqrt{x}/2$ and the twice squares about $\sqrt{x}/(2\sqrt{2})$. Hence, as $x \rightarrow \infty$, the share of squares tends to

$$\frac{(2\sqrt{2} + 1)\zeta(3)}{2(\sqrt{2} + 1)\zeta(3/2)} = 0.3648,$$

the share of twice squares to 0.2580, and together to 0.6228. A count up to 10^{14} gives 0.6252. Cubes and higher odd powers have share tending to 0.

In words. Among all powerful numbers divisible by 4, which are the candidates for a middle, a little more than three fifths have the special shape of Theorem I.5.4. For these, a triple would need a kernel beyond $1.5 \cdot 10^{12}$, and beyond $5.325 \cdot 10^{13}$ under the assumption of Remark I.2.5.

6 Data

Proposition I.6.1 (exponents at the rank). [measured] *Consider the 750 kernels $m \equiv 7 \pmod{8}$ with $m \leq 10^4$ and $T_1(m)$ even, and the primes $3 \leq p \leq 3000$ that have a rank $\alpha_m(p)$; this gives 120 373 pairs (m, p) . Among them $v_p(T_{\alpha_m(p)}) \geq 2$ holds 470 times and $v_p(T_{\alpha_m(p)}) \geq 3$ holds 52 times. The model in which every pair has $v_p \geq 2$ with probability $1/p$ and $v_p \geq 3$ with probability $1/p^2$, independently, predicts 486 and 49.1.*

The model is the usual heuristic for Wieferich primes: Crandall, Dilcher and Pomerance record that the probability that p is a Wieferich prime to base 2 is sometimes estimated as about $1/p$ [25], and Ryan [67, Section 1] states it in the same form for the Wieferich locus of an arbitrary algebraic unit, conditional on an equidistribution of Fermat quotients that is open even for base 2. In our setting the unit is ε_m and the event is $p^2 \mid T_{\alpha_m(p)}(m)$, the Wieferich condition of Proposition I.6.10.

Both counts lie within one standard deviation of the prediction. The independence in the model was tested rather than assumed: all events of one kernel come from the same unit ε_m , so they might cluster. Let k_m be the number of events $v_p \geq 2$ at the kernel m and λ_m the sum of $1/p$ over its pairs. The statistic $S = \sum_m (k_m - \lambda_m)^2$ is 411.7, against 436.2 ± 28.2 in 4000 simulations of the independent model: $z = -0.87$, one-sided empirical $p = 0.80$. No clustering within kernels is detected. The kernels with 0, 1, 2, 3, 4 events number 396, 255, 83, 15, 1. Split by prime, the largest deviation is $p = 181$ at 2.4 standard deviations, about what testing each prime with an expected count of at least 1 separately produces by chance; primes with a smaller expected count were not ranked.

The power of the test was measured as well, with data that keep the frequencies of the events but give a common shock to a fraction ρ of the kernels (40 runs for each ρ). Full

coupling, $\rho = 1$, lifts the mean z to about 652. At $\rho = 0.2$ the coupling is detected in 90 % of the runs, at $\rho = 0.05$ in 48 %. So a coupling of strength 0.2 or more would very likely have been seen, and a coupling of strength 0.05 or less cannot be decided from these data.

In words. For a prime p the chance that p^2 divides the first term of the Pell sequence divisible by p should be about $1/p$. Over the 120 373 pairs (m, p) this simple rule predicts the observed number of such squares and cubes well. We also checked that the squares do not come in bunches at single kernels, and we measured how strong a bunching the check would have seen.

Proposition I.6.2 (no triple below $1.83 \cdot 10^{18}$, and the pairs at distance two with middle divisible by four). [computed]

- (a) *The middles $n \equiv 0 \pmod{4}$ of pairs $n - 1, n + 1$ of powerful numbers with $n < 10^{2000}$ and $\text{sqfree}(n^2 - 1) \leq 10^8$ are exactly the numbers $T_k(m)$ of the 371 lattice points of Theorem I.3.1 with T_k even.*
- (b) *Alekseyev's table for OEIS A076445 [54] has 33 terms up to its largest, which has 67 digits. Up to that term they are exactly the pairs at distance two of our computation with kernel at most 10^8 , for middles $\equiv 0$ and $\equiv 2 \pmod{4}$ alike. Exactly 9 of them have a middle $\equiv 0 \pmod{4}$.*
- (c) *The list in (a) contains every middle $n \equiv 0 \pmod{4}$ of a pair at distance two with $n < 1.83 \cdot 10^{18}$, and every such middle with $n < 3.88 \cdot 10^{20}$ if Reinhart's extended search missed nothing (Remark I.2.5). It contains every such middle with $n < 10^{36}/27$ whose kernel does not divide U_1 , and every such middle with $n < 10^{36}/27$ if no squarefree $m \equiv 7 \pmod{8}$ with $m \mid U_1(m)$ and $1.5 \cdot 10^{12} < m < 10^{24}/9$ has $T_1(m) < 10^{36}/27$.*
- (d) *Consequently, with Theorem I.3.1, no triple of consecutive powerful numbers has a middle below $1.83 \cdot 10^{18}$, and none below $3.88 \cdot 10^{20}$ if Reinhart's extended search missed nothing.*

Proof. (a) is Lemma I.1.3 together with the enumeration of Theorem I.3.1: a lattice point with T_k odd gives no middle divisible by four. (b) is a comparison of the two lists; we compared our own values with the table through their hashes, so no value of the table enters our data. (c) Let n be such a middle and m its kernel. If $m \nmid U_1$, Corollary I.2.4(i) gives $m < (27n)^{2/9}$, which is at most 10^8 for $n < 10^{36}/27$. If $m \mid U_1$, part (ii) gives $m < n^{2/3}$ and, since $n = T_k(m) \geq T_1(m)$, also $T_1(m) \leq n$. For $n < 1.83 \cdot 10^{18}$ this forces $m < 1.5 \cdot 10^{12}$, the range in which Reinhart's list of such kernels is complete [61, Remark 5.5]; its 3 kernels $m \equiv 7 \pmod{8}$ all have $T_1(m) \geq 10^{94}$, so none of them occurs. For $n < 3.88 \cdot 10^{20}$ the same holds with the 4 kernels of the extended list [62], if that list is complete (Remark I.2.5). For $n < 10^{36}/27$ the same argument leaves exactly the kernels named in the condition. (d) The middle of a triple is such a middle (Lemma I.1.1), so by (c) it is in the list of (a), and every point of that list is excluded in Theorem I.3.1. $\square$

In words. Our list of the middles divisible by four is complete in the box of Theorem I.3.1, and it agrees with the longest published table in both directions. How far it is complete beyond the box depends on the second kind of kernel, the ones dividing U_1 : they are known with

proof only up to $1.5 \cdot 10^{12}$, which gives completeness up to $1.83 \cdot 10^{18}$ without any assumption, and up to $3.88 \cdot 10^{20}$ if the extended search missed nothing.

Bloom [11] states that no triple has a middle below $7.38 \cdot 10^{28}$, by OEIS A076445. This rests on the terms listed there, and the entry does not state that its list is complete; Mian and Siddique make the same point [49].

Proposition I.6.3 (ladders for the known kernels dividing U_1). [computed] *Table 5 gives, for the two known squarefree kernels $m \equiv 7 \pmod{8}$ with $m \mid U_1$ and $T_1(m)$ even, the ladder of Theorem I.5.1: at each stage the primes $p \leq P$ with $v_p(T_k) = 1$ at the current index k , and then the smallest index a triple with this kernel could still have, together with the number of digits its middle would have at least. The other two known kernels, $m = 209\,991$ and $m = 117\,477\,414\,815$, have T_1 odd (with 95 and 11 621 digits) and give no triple by parity.*

kernel m	digits of T_1	P	witnesses, stage by stage	$k \geq$	digits of $n \geq$
4 099 215	228	$2 \cdot 10^5$	701, 100469, 157049	$1.1 \cdot 10^{13}$	$2.5 \cdot 10^{15}$
39 028 039 587 479	1 880 030	$3 \cdot 10^3$	5, 7, 71, 1657; 19, 29, 421, 569, 1987; 379, 2089, 2203, 2437	$4.5 \cdot 10^{30}$	$8.6 \cdot 10^{36}$

Table 5: The ladder for the two known kernels $m \equiv 7 \pmod{8}$ with $m \mid U_1$ and $T_1(m)$ even. The witnesses of one stage are separated by commas, the stages by semicolons. The last two columns are rounded down.

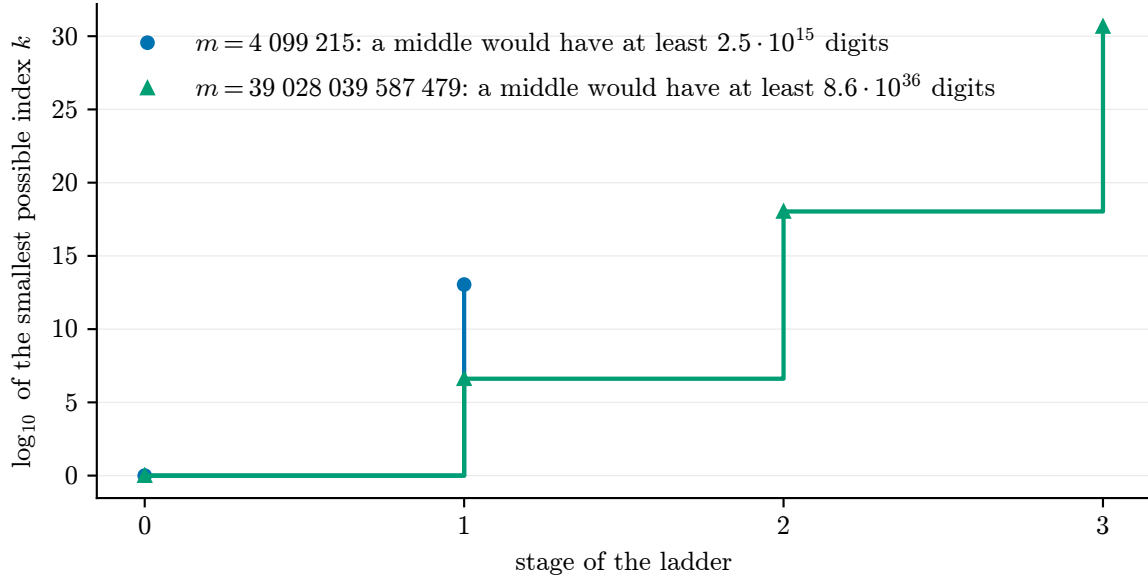


Figure 4: **Each stage of a ladder pushes the smallest possible index up.** The ladder of Table 5, stage by stage. Each step multiplies the smallest possible index k by the product of the witnesses of that stage; the last point is the smallest index that remains possible.

How the table is computed. Let k_0 be an index with $v_p(T_{k_0}) = 1$ for an odd prime p , and let T_k be powerful with $k_0 \mid k$ and k/k_0 odd. By Lemma I.4.3, $v_p(T_k) = 1 + v_p(k/k_0) \geq 2$, so

$p \mid k/k_0$. Starting at $k_0 = 1$, the product of the witnesses of a stage therefore divides k/k_0 , and the next stage starts at k_0 times that product. The witnesses are found from T_1 and U_1 modulo p^2 , without large integers. The ladder stops at the first stage without a witness below P . $\square$

In words. These two kernels are dangerous in the sense of Section 2, because they divide U_1 ; the other two known kernels divide U_1 as well, but their T_1 is odd. For each of them the table shows how far out a triple would have to be: the middle would have at least the number of digits in the last column. For $m = 4\,099\,215$ the indices $k = 1, 3, 5, 7$ also lie in the box of Theorem I.3.1, where they are excluded by the witnesses 701, 11, 19, 29.

Reinhart [62, Def. 1.4, p. 6] asks whether $4\,099\,215$ or $39\,028\,039\,587\,479$, or even 7, admits an index k with T_k powerful and $m \mid U_k$ (his condition (C), Question I.9.4). Our first witnesses 701 and 5 for these two kernels are his own. The table does not decide the question; it only says how large a positive answer would have to be.

Proposition I.6.4 (the smallest witnesses). [computed] *For each of the 371 lattice points of Theorem I.3.1 that are excluded by a witness, let p be the smallest witness and $\alpha_m(p)$ its rank. In 8 points the smallest witness is primitive, $\alpha_m(p) = k$; in all others it is a prime whose rank divides k with odd cofactor. The smallest witness is 29 in 79 points, 19 in 78, 11 in 45, 3 in 43, 5 in 33 and 7 in 25; its median is 19 and its maximum 114 997. In every point $v_p(T_{\alpha_m(p)}) = 1$ and $p \nmid k/\alpha_m(p)$.*

The valuation identity of Lemma I.4.3 was checked in every point. As controls, the point (7, 7) has the witness 29 of rank 7, and the identity predicts $v_{197}(T_{1379}(7)) = 2$, which the exact computation confirms.

In words. A triple would have to survive the test of Theorem I.4.6 not only at the primes that first appear at its index k , but at every small prime whose rank divides k . In practice the lattice points fail at small primes: the typical witness is 19 or 29, and only at 8 of the points is the smallest witness a prime that first appears at k itself.

Remark I.6.5 (the condition modulo 36 and the witnesses). [computed] By Beckon's condition (Remark I.8.1) the middle n of a triple satisfies $n \equiv 0, 8$ or $28 \pmod{36}$; the earliest derivation we located is Beckon's [7, Theorem 0.1]. Proposition 3(i) of [69] refines it to 39 classes modulo 900 and 1209 classes modulo 44 100. Rizzi [65, Proposition 5.1] derives the condition modulo 36 again and notes that it agrees with Beckon's; the same paper reports a search up to 10^{10} that finds fourteen pairs and no triple [65, Chapters 4 and 6]. Of the 371 middles of Proposition I.6.4, 328 satisfy the condition, and the 43 that do not are exactly the points whose smallest witness is 3.

This is no coincidence. For the middle n of a pair, that is, with $n - 1$ and $n + 1$ powerful, and $4 \mid n$, the condition fails exactly when n is divisible by 3 but not by 9, that is, when $v_3(T_k) = 1$; then 3 is a witness, and the smallest one. So the condition modulo 36 is the case $p = 3$ of the witness reduction. Since the two computations work by different methods, their agreement on the same points checks both.

In words. A known rule says what the middle of a triple must look like modulo 36. Our witness search already contains this rule: the middles that break it are exactly those where the prime 3 appears only once.

Proposition I.6.6 (how many conditions there are, and how often they are met). [computed]

- (a) For the 15 kernels of class 7 with the most lattice points and all primes $p \leq 10^6$ that have a rank d , the congruence $p \equiv \pm 1 \pmod{4d}$ of Corollary I.4.8(i) holds at all 425 362 pairs (p, d) .
- (b) Over the 445 lattice points of Theorem I.3.1, the indices k have 2149 divisors $d \geq 5$ in total (median 4 per point, maximum 16). For 1682 of them (78.3 %) one has $k < d(4d - 1)$, so by Corollary I.4.8(iii) a powerful T_k would need $p_d^2 \mid T_d$ at that d . Per point the number of such conditions has median 3, mean 3.8 and maximum 12.
- (c) For the 25 kernels of class 7 with the most lattice points (378 points), every divisor of every k was taken as a rank, and for each rank the first 5 primes of that rank below $3 \cdot 10^6$ were tested. Of the 2536 rank occurrences (a point together with a divisor of its index, so that $d = 1$ and $d = 3$ are included and one pair (m, d) counts once for every point whose index d divides), 2025 have a prime below this bound, and 1988 of these (98.17 %) carry a witness; 37 carry only primes with exponent at least 2. Testing only the smallest prime of each rank gives 1936 of the same 2025 ranks (95.60 %) and 89 with exponent at least 2 only.

The 511 rank occurrences without a prime below the bound are undecided; for $d \geq 5$ they are not known to be free of witnesses, since their smallest prime lies higher, and among them are the occurrences of $d = 1$ at kernels with $T_1(m)$ a power of 2, which have no odd prime at all. Parts (b) and (c) report rates. No proof in this paper depends on them, and Corollary I.4.8 holds independently of them; the single ranks behind (c) are decided, since one prime of exponent 1 settles its rank by Theorem I.4.6.

In words. Part (a) checks a proved rule on 425 362 cases. Part (b) counts how many separate tests a triple would have to pass at a lattice point: typically a few, sometimes a dozen. Part (c) shows how often such a test is actually decided by a small prime: almost always, and more often when more primes per rank are tried.

Proposition I.6.7 (middles that are powers of two). [computed] *For every integer a with $2 \leq a \leq 100\,000$, at least one of $2^a - 1$ and $2^a + 1$ is not powerful. In particular no triple has a middle 2^a in this range, that is, a middle of this form with up to 30 103 digits.*

The computation. For each prime $p \leq 10^6$ let e be the order of 2 modulo p . The prime p divides $2^a - 1$ exactly when $e \mid a$, and then $v_p(2^a - 1) = v_p(2^e - 1) + v_p(a/e)$; so p is a witness for $2^a - 1$ exactly when $e \mid a$, $p^2 \nmid 2^e - 1$ and $p \nmid a/e$. The prime p divides $2^a + 1$ exactly when e is even and a is an odd multiple of $e/2$, and then $v_p(2^a + 1) = v_p(2^{e/2} + 1) + v_p(2a/e)$; so p is a witness for $2^a + 1$ exactly when e is even, a is an odd multiple of $e/2$, $p^2 \nmid 2^{e/2} + 1$ and $p \nmid 2a/e$. The loop runs over the primes rather than over the exponents. Among the 99 999 exponents, 8957 (8.96 %) leave $2^a - 1$ without a witness below the bound and 2511 (2.51 %) leave $2^a + 1$ without one; no exponent leaves both without one. $\square$

The exponents without a witness are undecided; they are not known to be powerful: their prime factors lie above the search bound. Equivalently, $2^a - 1$ is proved not powerful for 91 042 of the exponents and $2^a + 1$ for 97 488. As controls, the computation of the orders

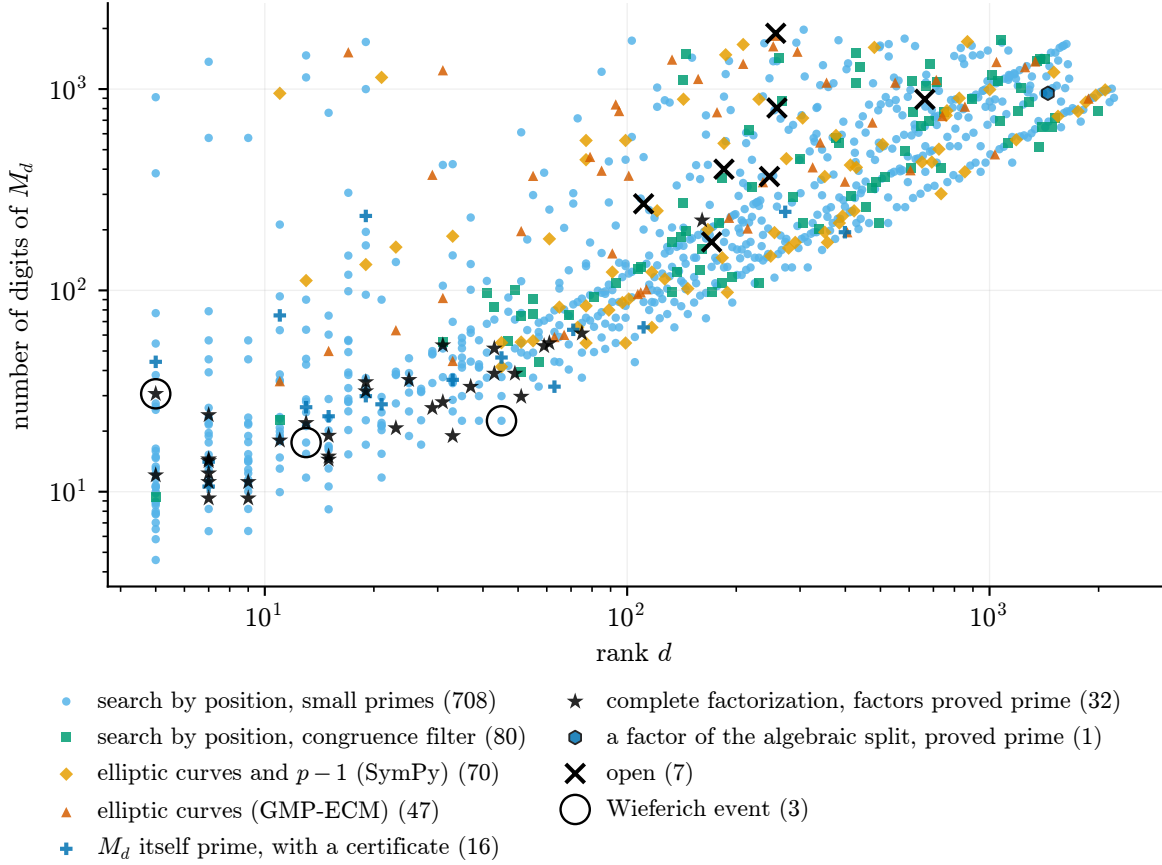


Figure 5: **Every building block, settled or open.** The 961 building blocks M_d of Proposition I.6.8, one point each: the rank d against the number of digits, both on logarithmic scales. Color and shape show how each block was shown not to be powerful; the classes are the rows of Table 6, and every class except the open blocks is a certified witness. The 7 open blocks (crosses) have at least 173 digits each, and no search found a factor of them. Circles mark the 3 blocks with a Wieferich event below $3 \cdot 10^6$ (Proposition I.6.10). The terms are explained in Table 4.

returns exactly the two Wieferich primes 1093 and 3511 below the bound, and $2^3 + 1 = 9$, which is powerful, receives no witness. Question I.9.3 asks for all a .

In words. If the middle of a triple were a power of two, then $2^a - 1$ and $2^a + 1$ would both be powerful. For every a up to 100 000 a small prime appears exactly once in one of the two numbers. The search runs over primes, which makes it cheap.

Proposition I.6.8 (the primitive part over an explicit finite set). [computed] *Let $\mathcal{P}$ be the set of the 445 lattice points of Theorem I.3.1; all of them are excluded there. For a kernel m of a point of $\mathcal{P}$ and an odd $d \geq 5$ dividing an index k with $(m, k) \in \mathcal{P}$, let M_d be the part of $T_d(m)$ supported on the primes of rank exactly d . There are 961 such pairs (m, d) , on the 79*

kernels of $\mathcal{P}$.

(a) Settled: 954 of the 961, each by a certified witness.

(b) Open: 7 of the 961.

A witness here is a prime p of rank d with $v_p(M_d) = 1$; it shows that M_d is not powerful. For some blocks the witness comes from a complete factorization, and for one from the algebraic split of Remark I.6.13. For a prime p of rank exactly d one has $v_p(M_d) = v_p(T_d)$ by definition; for the Möbius dual $\prod_{e|d} T_e^{\mu(d/e)}$, the product with which the computations work, the same holds at every prime of rank exactly d [66, Theorem 2.2(c)]; the restriction to odd d is the range in which the dual of the companion sequence is integral [66, Theorem 2.3]. Since every point of $\mathcal{P}$ is already excluded, the proposition excludes no further triple. It asks, on these pairs, whether a primitive part can be powerful (Question I.9.2). For 21 of the kernels $T_1(m)$ is odd; their points are excluded by parity alone and enter only as instances of that question.

how the pair was settled	first 25 kernels	other 54	all
certified witness: search by position, small primes	605	103	708
certified witness: search by position, congruence filter	64	16	80
certified witness: elliptic curves and $p - 1$ (SymPy)	70	0	70
certified witness: elliptic curves (GMP-ECM)	27	20	47
certified witness: M_d itself prime, with a certificate	8	8	16
certified witness: complete factorization, factors proved prime	32	0	32
certified witness: a factor of the algebraic split, proved prime	1	0	1
open	4	3	7

Table 6: How the 961 pairs of Proposition I.6.8 were settled. The first 25 kernels are those with the most lattice points; for the other 54, and for the pairs of the first 25 with $t_{35} < 1$, the elliptic curve method ran to a smaller effort, see Table 7.

How the pairs were settled. A search by position. Every prime $p \leq 3 \cdot 10^6$ of rank d was tested, and later, restricted by Corollary I.4.8(i) to $p \equiv \pm 1 \pmod{4d}$, every such prime up to 10^9 . For the pairs that remain open, this search also shows that no prime of rank d with $v_p(M_d) = 1$ lies below its bound. *A search by size.* The elliptic curve method, first with SymPy [48] and then with GMP-ECM [90], looks for a factor by its size rather than by its position; it shows only what it finds. The effort spent on the pairs that remain open is in Table 7. *Complete factorization.* Small primitive parts were factored completely, and some are themselves prime, which no search for a proper divisor can see. One pair, $(m, d) = (319, 5)$, has no witness below the first bound and has its own row in Table 6; its witnesses come from the complete factorization of $M_5(319)$ given after Proposition I.6.10, whose large factors are proved prime. *Checks.* Every witness was recomputed on a second, independent route (the powers of ε_m with the rank taken from the order of the unit, or the Lucas sequence $V_n(2T_1, 1)$, whichever differs from the route that found it). Every witness of the 954 pairs in the first part of (a) is certified: below ψ_{12} by the Miller–Rabin test with the first twelve prime bases, which is deterministic there [75], and above it by Pocklington’s criterion from a factorization of $p - 1$ [14, Theorem 4 and Corollary 1], each certificate checked by a separate program. The prime factors of the complete factorizations and the factor from the algebraic split are

certified in the same way: below ψ_{12} by the deterministic test, and above it 18 of them, with up to 75 digits, by Pocklington’s criterion, and 6, with up to 477 digits, by elliptic curve certificates [31, 2], which PARI/GP [80] produced and a separate program of ours checked step by step. $\square$

m	d	digits	curves at $B_1 =$			t_{35}	t_{40}
			$5 \cdot 10^4$	10^6	$3 \cdot 10^6$		
399	171	173	300	1071	2753	8.36	1.10
407	111	269	300	1071	2753	8.36	1.10
39	247	367	300	1071	2753	8.36	1.10
111	185	400	300	1071	2753	8.36	1.10
407	259	805	300	1071	2753	8.05	1.05
663	663	889	300	1071	38	1.05	0.11
20 303	257	1893	300	209	–	0.22	0.02

Table 7: The 7 open pairs of Proposition I.6.8 and the effort of GMP-ECM on each. The columns t_{35} and t_{40} give the fraction of the curves the program expects to need for a factor of 35 and of 40 digits, for the second-stage bound it chose. A fraction of 1 still misses a factor of that size with probability about $1/e$, so “none found” describes the effort, not the factors. The kernels 399, 663 and 20 303 are among the other 54; the rest are among the first 25. The blocks that the searches left open on the kernels $m = 7$ and $m = 23$, where $T_1(m) + 1$ is a square, are settled by the algebraic split of Remark I.6.13 and are not listed.

In words. Theorem I.3.1 already rules out all 445 lattice points in its search range. This proposition asks a different question: can a building block M_d be powerful at all? We take all 79 kernels of the lattice points and look at every building block that occurs there, 961 in total. A number is powerful when each of its prime factors appears at least twice. So one prime that appears exactly once is enough to show that a block is not powerful. We found such a prime, and proved that it is prime, for 954 blocks; for some of them we split the whole block into its prime factors, and for one the algebraic split of Remark I.6.13 gives the factor. Every prime used is proved prime, the large ones by certificates that our own program checks. For the last 7 blocks we found no such prime. Finding nothing only means that our search did not succeed, not that no such prime exists. The 7 open blocks are in the deposit with their decimal expansions; a prime factor of any one of them with exponent one settles that block, and a complete factorization settles it either way.

A different question, measured. Among the 1341 triples (m, d, p) of the search by position with $p \leq 3 \cdot 10^6$ (for each rank, the first five primes found by the search), $v_p = 1$ in 1338 cases, $v_p = 2$ in 3 and $v_p \geq 3$ in none. The heuristic $P(v_p \geq j) = p^{1-j}$ predicts 2.38 triples with $v_p \geq 2$. The observations are consistent with it and do not confirm it.

Remark I.6.9 (where this stands in the literature). [literature] Ribenboim and Walsh [64, Corollary 2] show that the abc conjecture implies that a Lucas sequence of positive discriminant has only finitely many powerful terms, and Yabuta [88] extends this to negative discriminant. These statements concern the terms rather than their primitive parts, and they are conditional and not effective, so they neither contain nor are contained in the computation above. Closer to the question is [66, Corollary 5.1]: the primitive parts of the Fibonacci

numbers are squarefree for all $n \neq 6$ if and only if there is no Wall–Sun–Sun prime. Square-free is a far stronger property than not powerful; the question asked here is the weaker one, and it is the one a finite computation can address.

In words. If the *abc* conjecture is true, only finitely many terms of such sequences are powerful, but nobody knows which, and the statement is about whole terms, not their building blocks. For the Fibonacci numbers, even the question whether all building blocks are squarefree is tied to an open problem about special primes.

Proposition I.6.10 (Wieferich primes of the unit and the primitive parts). [proved] *Fix a squarefree $m \equiv 7 \pmod{8}$ with $T_1(m)$ even. For odd $d \geq 5$ let M_d be the part of $T_d(m)$ supported on the primes of rank exactly d . Call an odd prime $p \nmid m$ that divides some $T_k(m)$ a Wieferich prime for ε_m if $p^2 \mid T_{\alpha_m(p)}(m)$. The following are equivalent:*

- (i) ε_m has only finitely many Wieferich primes of odd rank;
- (ii) M_d is squarefree for all but finitely many odd $d \geq 5$.

In that case M_d is not powerful for all but finitely many odd $d \geq 5$. Moreover, let $p \nmid m$ be an odd prime that has a rank α , let $u_n = U_n/U_1$ be the Lucas sequence of the first kind for $(P, Q) = (2T_1, 1)$, and let $e = (m/p)$. Then $p \nmid U_1$ and $v_p(u_{p-e}) = v_p(T_\alpha)$. So the Wieferich primes for ε_m are exactly the odd Lucas–Wieferich primes of this sequence that divide some T_k .

Proof. A prime p dividing M_d has rank exactly d , so $v_p(M_d) = v_p(T_d) = v_p(T_{\alpha_m(p)})$; in the language of Möbius duals this is [66, Theorem 2.2(c)]. Hence p divides M_d exactly once when p is not a Wieferich prime for ε_m . (i) $\Rightarrow$ (ii): a Wieferich prime of odd rank d affects only M_d , so under (i) all M_d outside finitely many odd d are squarefree. (ii) $\Rightarrow$ (i): a Wieferich prime of odd rank $d \geq 5$ makes M_d non-squarefree, so under (ii) such primes have one of finitely many ranks, and each rank carries finitely many primes, since M_d is a positive integer; the ranks 1 and 3 carry only the prime divisors of T_1 and T_3 . For the last assertion, $M_d > 1$ for odd $d \geq 5$, because T_d has a primitive prime divisor [17], in the form of [87] used in the proof of Corollary I.4.8; and a squarefree integer above 1 is not powerful. For the last statement, $p \mid T_\alpha$ gives $\varepsilon_m^{2\alpha} + 1 = 2\varepsilon_m^\alpha T_\alpha \equiv 0 \pmod{p}$ in $\mathbb{Z}[\sqrt{m}]$, and the proof of Corollary I.4.8(i) shows that ε_m has order 4α modulo the primes above p and that $4\alpha \mid p - e$; in particular $p \nmid U_1$, because $p \mid U_1$ would give $\varepsilon_m^2 \equiv 1$. Write $y = \varepsilon_m^{4\alpha}$ and $M = (p - e)/(2\alpha)$, so that $\varepsilon_m^{2(p-e)} = y^M$ and $y \equiv 1 \pmod{p}$. Here $p \nmid M$, because $p \nmid p - e$, and $y^M - 1 = (y - 1)(1 + y + \cdots + y^{M-1})$ with the second factor congruent to M modulo p . Multiplication by an element of $\mathbb{Z}[\sqrt{m}]$ whose norm is prime to p does not change v_p (see the proof of Lemma I.4.3), so $v_p(y^M - 1) = v_p(y - 1)$. Next, $y - 1 = (\varepsilon_m^{2\alpha} + 1)(\varepsilon_m^{2\alpha} - 1)$ with $\varepsilon_m^{2\alpha} - 1 \equiv -2 \pmod{p}$ of norm prime to p , so $v_p(y - 1) = v_p(2\varepsilon_m^{2\alpha} T_\alpha) = v_p(T_\alpha)$. Finally $\varepsilon_m^{p-e} \cdot u_{p-e} \cdot 2U_1\sqrt{m} = \varepsilon_m^{2(p-e)} - 1$, and the norms of ε_m^{p-e} and $2U_1\sqrt{m}$ are prime to p , so $v_p(u_{p-e}) = v_p(\varepsilon_m^{2(p-e)} - 1) = v_p(T_\alpha)$. $\square$

In words. A prime p is a Wieferich prime (to base 2) if p^2 divides $2^{p-1} - 1$. Below 10^6 there are exactly two, 1093 and 3511 (the control in Proposition I.6.7). Fix one m with $T_1(m)$ even. The rank of a prime p is the first k with p dividing $T_k(m)$. We call p a Wieferich prime

for m if p^2 already divides the term at its rank. If this rank d is odd and at least 5, then p^2 divides the block M_d , so M_d is not squarefree. Proposition I.6.10 says: the blocks M_d with odd d are squarefree from some point on exactly when only finitely many such primes have odd rank. In that case only finitely many blocks with odd d can be powerful. Such primes do exist. Among the primes below $3 \cdot 10^6$ we found 15 of odd rank $d \geq 5$ on those of the 79 kernels of Proposition I.6.8 that have T_1 even. Since finitely many are allowed, these do not settle the condition. For 12 of them the block is not powerful: another prime divides it exactly once. For the other 3 we do not know. Whether a block can be powerful remains open in general. For the Fibonacci numbers, the same kind of condition is the open question whether a Wall–Sun–Sun prime exists.

For the Fibonacci numbers, Ross, Shen and Cai show that the Möbius duals $\prod_{d|n} F_d^{\mu(n/d)}$ are squarefree for all $n \neq 6$ exactly when there is no Wall–Sun–Sun prime [66, Corollary 5.1]. The proposition is the analog for ε_m , restricted to odd rank and allowing finitely many exceptions; it is elementary, and its content is the identification of the hypothesis.

Remark I.6.11 (the Wieferich events of the units). [computed] That hypothesis is not empty for our units. Among the primes below $3 \cdot 10^6$ the 79 units of Proposition I.6.8 carry 54 Wieferich events, that is, pairs of a unit and a prime (40 distinct primes). Of these, 18 have odd rank $d \geq 5$, and 15 of those lie on units with T_1 even, where the proposition applies; Table 8 lists them. So the squarefree statement for all odd d fails for these units. In 11 of the 15 cases M_d is nevertheless not powerful, since a prime of rank d below $3 \cdot 10^6$ divides it exactly once, and $M_5(319) = 20161^2 \cdot 1\,375\,502\,561 \cdot 158\,585\,313\,121$ with both large factors proved prime. For the remaining 3 the question is not settled here; they lie outside the finite set of Proposition I.6.8. Conversely, the 3 events whose block lies in that set, the circles in Figure 5, are all settled; the last column of Table 8 marks them. On the 25 units of the first search there are 16 events. Part III finds more on the same units (Remark III.4.1) because it also counts primes that divide no $T_k(m)$. Whether a primitive part can be powerful at all is Question I.9.2.

Remark I.6.12 (Lucas–Wieferich primes). [computed] The condition in (i) has a name in the computational literature. For the Lucas sequence (u_n) of the first kind attached to $(P, Q) = (2T_1, 1)$, a prime p with $u_{p-e} \equiv 0 \pmod{p^2}$, where e is the Legendre symbol of the discriminant, is a Lucas–Wieferich prime [47, p. 2088]. That condition is placed on the sequence of the first kind and ours on the companion sequence, but for primes that have a rank the two agree, by the last statement of Proposition I.6.10. The computation agrees as well: on the 16 events of the first search of Proposition I.6.8 (the 25 units with the most lattice points) the two conditions select the same primes; as controls, the 3 Wieferich primes 13, 31, 1546463 of the Pell sequence are recognized, and 25 primes that divide their term exactly once are not. This checks the computation on one sequence; the identity itself is proved above.

Remark I.6.13 (a second primitive prime factor). [computed] If every block M_d with $d \geq 5$ had two distinct prime factors, a powerful block would need two primes of rank d that each divide $T_d(m)$ at least twice, two simultaneous events of the kind counted in Proposition I.6.1. For Lehmer numbers there is a theorem of this kind: Juričević [35, Theorems 1.1 and 2.1], making a result of Schinzel effective, proves that the n -th term of a real Lehmer pair has at least two primitive prime divisors when $n > 4$, $n \neq 6$ and $n/(\eta\kappa)$ is odd, where η and κ

unit m	rank d	prime p	M_d not powerful by	in Prop. I.6.8
15	45	181	witness	yes
31	39	157	witness	no
79	253	4049	witness	no
143	13	311	witness	yes
223	45	179	witness	no
319	5	20 161	complete factorization	yes
383	21545	258 539	witness	no
447	413	24 781	witness	no
455	9	37	witness	no
615	807	3229	not settled	no
1343	5	19	witness	no
1535	63	251	witness	no
2135	13	53	witness	no
2135	37	149	not settled	no
123 879	9	37	not settled	no

Table 8: The Wieferich events of odd rank $d \geq 5$ below $3 \cdot 10^6$ on the units ε_m of Proposition I.6.8 with $T_1(m)$ even. “Witness”: a prime of rank d below $3 \cdot 10^6$ divides M_d exactly once. “Not settled”: not decided by our computations. The last column says whether (m, d) is one of the 961 pairs of Proposition I.6.8.

are invariants of the pair, apart from an explicit list of exceptions with $n \leq 30$. Here M_d is the primitive part of the term of index $2d$ of the Lucas sequence $u_n(2T_1, 1) = U_n/U_1$, whose terms of even index are $2T_1$ times the Lehmer numbers of the pair with $L = 4T_1^2$ and $M = 1$. For this pair $\eta = \kappa = 1$, so the condition asks for $2d$ to be odd, and the theorem does not apply to any block.

A second pair. If $T_1(m) + 1 = x^2$ is a square, then $\varepsilon = \gamma^2$ with $\gamma + \gamma^{-1} = x\sqrt{2}$, and (γ, γ^{-1}) is a real Lehmer pair with $L = 2x^2$ and $M = 1$, for which $\kappa = 2$ and $\eta = 2$. Its term of index $4d$ is the term of index $2d$ above, so its primitive divisors are the primes of rank d , and for odd $d \geq 5$ the quotient $4d/(\eta\kappa) = d$ is odd; none of the exceptional triples has these values. Hence on such a kernel every block M_d with $d \geq 5$ has at least two distinct prime factors. The split behind it is elementary: for odd d , $T_d + 1 = (xw_d)^2$ with $w_d = (\gamma^d + \gamma^{-d})/(\gamma + \gamma^{-1})$ an integer, so $T_d = (s - 1)(s + 1)$ with $s = xw_d$, and the odd number M_d is the product of the coprime factors $\gcd(M_d, s - 1)$ and $\gcd(M_d, s + 1)$; that M_d has two distinct prime factors is the content of the theorem. This is a splitting of the same kind as the Aurifeuillian factorization of the Lucas numbers, where the primitive part of L_{5n} , for odd n , is the product of two coprime factors [15, (2.9)–(2.11)]. The case occurs for 26 of the 79 kernels of Proposition I.6.8, which carry 361 of the 961 blocks, and for every prime kernel $m \equiv 7 \pmod{8}$: the class number is odd, so $x^2 - my^2 = 2$ is solvable (Remark I.4.11 gives an elementary proof), and we checked it for the 171 prime kernels below 5000. We also checked the conclusion on each of these 361 blocks: M_d is composite and not a prime power.

On the other kernels the theorem does not apply, and the data show that a block can have a single prime factor: of 163 blocks M_d with $d \geq 5$ and small $T_d(m)$ that we factored completely, 18 are prime, none of them on a kernel of the kind above, the largest at $d = 59$,

and 5 of these have $2d \geq 31$, so that it is the parity condition and not the size of n that fails. So on the 26 kernels a powerful block needs two rare events, and on the others no argument of this kind forces two events: a block can have a single prime factor, and a powerful block could then be the power of a single prime.

The split also settles blocks that the searches of Proposition I.6.8 left open. For $(m, d) = (23, 161)$ both factors $\gcd(M_d, s - 1)$ and $\gcd(M_d, s + 1)$, with 111 and 112 digits, are prime, so M_d is their product and is not powerful. For $(m, d) = (7, 1449)$ one factor, with 477 digits, is prime; since the two factors are coprime, it divides M_d exactly once. All three primes are proved by elliptic curve certificates (proof of Proposition I.6.8).

In words. One could hope that a building block always has at least two different primes, so that a powerful block would need two rare events at once. A theorem of this type exists for a related sequence. On the kernels for which $T_1 + 1$ is a square, it applies after a change of the sequence, and there every block has at least two primes. On the other kernels its condition is never met by our blocks, and in our data some blocks are themselves prime. So there one rare event may be all that a powerful block needs.

7 Profile of a counterexample

This section collects in one place what is known about a triple, if one exists. Every clause is proved above or computed in Section 5 or 6, and the clauses that depend on Reinhart's extended search say so; nothing new is claimed here except clause (viii).

Theorem I.7.1 (profile). [computed] *Suppose that $n - 1$, n and $n + 1$ are all powerful, let $m = \text{sqfree}(n^2 - 1)$, and let (m, k) be the address of n .*

- (i) $m \equiv 7 \pmod{8}$ is squarefree, $T_1(m)$ is even, k is an odd multiple of m' , $n = T_k(m)$, and $4 \mid n$.
- (ii) $n \geq (m^{3/2}/m')^{m'}$. If $m \nmid U_1$, then $n > m^{9/2}/27$ and $k \geq 3$; if $m \mid U_1$, then $n \geq T_1 > m^{3/2}$.
- (iii) $v_2(n) = v_2(T_1(m))$, and k is divisible by every odd prime p with $v_p(T_1(m)) = 1$.
- (iv) Every odd prime $p \mid n$ is coprime to m , k is an odd multiple of $\alpha_m(p)$, and $v_p(T_{\alpha_m(p)}) + v_p(k/\alpha_m(p)) \geq 2$. Every prime p with $\alpha_m(p) = k$ satisfies $p^2 \mid n$.
- (v) Let $b = \text{sqfree}(n)$. If $b > 1$, then $k = \alpha_m(b)$, and $v_p(T_{\alpha_m(p)}) + v_p(\alpha_m(b)/\alpha_m(p)) \geq 3$ for every odd prime $p \mid b$.
- (vi) If n is a square, twice a square, or a fourth power, then $k = 1$, $m \mid U_1$, and $m > 1.5 \cdot 10^{12}$ or m is one of 209 991, 4 099 215 and 117 477 414 815, which carry no triple with such a middle; if Reinhart's extended search missed nothing (Remark I.2.5), then $m > 5.325 \cdot 10^{13}$ or m is one of the four known values.
- (vii) *Explicit exclusions.* Either $m > 10^8$ or $n \geq 10^{2000}$. If $m \leq 10^6$, then $\text{sqfree}(n) > 10^4$. If $m \leq 2000$, then n has more than $8.4 \cdot 10^7$ digits. If $10^8 < m \leq 10^{15}$ and m is a product of at least 4 primes up to 100, then $n \geq 10^{2000}$. If $m \mid U_1$ and $m \leq 1.5 \cdot 10^{12}$, then

$m = 4099215$, which forces $k \geq 1.1 \cdot 10^{13}$ (the other members of Reinhart's list below this bound, 209991 and 117477414815, have T_1 odd and carry no triple); if Reinhart's extended search missed nothing, the same holds for $m \leq 5.325 \cdot 10^{13}$ with the further kernel 39028039587479 and $k \geq 4.5 \cdot 10^{30}$. In either case n has at least $2.5 \cdot 10^{15}$ digits. In all cases $n > 1.83 \cdot 10^{18}$ (Proposition I.6.2(d)).

(viii) If $T_1(m) + 1$ is a square, then $n + 1$ is a square. This happens for 26 of the 79 kernels of Proposition I.6.8 and for every prime $m \equiv 7 \pmod{8}$ (Remark I.6.13).

Proof. (i) Lemmas I.1.1, I.1.3 and I.1.4, and Theorem I.4.6 for $4 \mid n$. (ii) Remark I.2.2 and Corollary I.2.3; if $m \nmid U_1$ then $m' \geq 3$, so $k \geq 3$. (iii) Lemma I.4.7. (iv) Theorem I.4.6 and Lemma I.4.3. (v) Lemma I.4.2 and Proposition I.4.4. (vi) Theorem I.5.4. (vii) Theorems I.3.1, I.4.5, I.5.1 and I.5.3; the kernels 209991 and 117477414815 have T_1 odd (Theorem I.5.4 and Proposition I.6.3), and the bounds for the other two come from the ladder of Theorem I.5.1 applied to these kernels, as reported in Proposition I.6.3. (viii) Remark I.6.13: $T_k + 1 = (xw_k)^2$ for odd k if $T_1 + 1 = x^2$. $\square$

In words. The theorem lists the conditions on a counterexample that this part proves or computes: clauses (i) to (v) are proved, (vi) and (vii) rest on computations. Its kernel m and its index k must pass every test proved in this paper at the same time: the parity, the height bounds, the exponent conditions, and the explicit searches.

The kernels with $m' = 1$, for which every odd k is admissible, are exactly those with $m \mid U_1$ (Lemma I.1.2); they are listed in OEIS A135735, completely up to $1.5 \cdot 10^{12}$ [61, Remark 5.5] and, if Reinhart's extended search missed nothing, up to $5.325 \cdot 10^{13}$ (Remark I.2.5). They are the weakest point for an unconditional lower bound on n : an unknown kernel of this kind above $1.5 \cdot 10^{12}$ gives only $n > m^{3/2} > 1.83 \cdot 10^{18}$ (the discussion after Corollary I.2.4 records that no stronger bound is available without hypotheses, and Remark I.4.12 discusses the *abc* route). The enumeration behind the b-file of OEIS A060355 is described by Mian and Siddique as exhaustive below 10^{22} but not certified [55, 49]. The terms of OEIS A076445 recorded by Bloom reach $7.38 \cdot 10^{28}$ [11, 54], and Alekseyev's table, which we compare in Proposition I.6.2(b), goes up to a term with 67 digits; neither list is known to be complete.

8 Routes that do not close the problem

Several natural approaches do not decide the problem. This section records why, so that the reasons are available together.

Remark I.8.1 (congruences). [literature] Beckon [7, Theorem 0.1] showed that the smallest member n of a triple $(n, n + 1, n + 2)$ of powerful numbers satisfies $n \bmod 36 \in \{7, 27, 35\}$. Proposition 3 of [69] refines this: modulo ℓ^2 the proportion of admissible residues is $1/4$ for $\ell = 2$, $1/3$ for $\ell = 3$ and $(\ell^2 - 3\ell + 3)/\ell^2$ for $\ell \geq 5$, and the density of admissible residues tends to 0 like $(\log L)^{-3}$. Every factor is positive, so for every finite modulus some residue class survives, and no finite sieve of this type decides the problem. Over the primes up to 10^5 the product of these proportions is about $2.67 \cdot 10^{-4}$. Congruences remain useful as search filters: by Corollary I.4.8(i), a prime of rank d satisfies $p \equiv \pm 1 \pmod{4d}$, so a search for such primes needs only 2 of the $2d$ odd residue classes modulo $4d$.

In words. Conditions modulo a fixed number can never rule out all triples: some remainders always survive. They thin out the candidates more and more, but never to zero. They are still good filters for computer searches.

Remark I.8.2 (Yokoi's representation). [computed] Mollin and Walsh [51, p. 112] suggest that Yokoi's representation of the fundamental unit [89, Proposition 1] may help. In our notation it gives, for a prime $p \equiv 3 \pmod{4}$ dividing both m and U_1 , that p^2 divides $T_1 - 1$ or $T_1 + 1$. The identity $T_1^2 - 1 = mU_1^2$ gives more: p^3 divides $(T_1 - 1)(T_1 + 1)$, and since p is odd, p^3 divides one of the two factors. So the representation adds no new constraint. We checked this on all 123 such pairs (m, p) with T_1 even and $m \leq 20000$.

In words. An idea that Mollin and Walsh suggested gives nothing new here: the condition it yields already follows from the equation $T_1^2 - 1 = mU_1^2$.

Remark I.8.3 (small units). [computed] A search beyond Theorem I.3.1 might target kernels with a small fundamental unit, where many indices fit below a height bound. The kernels $m = t^2 - 2$ with t odd and $m = t^2 - 1$ with $4 \mid t$ have the units $(t^2 - 1) + t\sqrt{m}$ and $t + \sqrt{m}$. There $U_1 = t$ with t odd, resp. $U_1 = 1$, is prime to m , so $m' = m$, and by Theorem I.2.1 the first admissible index $k = m$ is already far out. Among the 251 such squarefree kernels $m \equiv 7 \pmod{8}$ with $t \leq 399$, only 17 have a first candidate $T_m(m)$ with fewer than 2000 digits, and all of them have $m \leq 1023$, inside the range of Theorem I.3.1. For these two kinds of kernel, small units therefore bring no new kernels into reach.

In words. A small unit sounds helpful, but it makes m' as large as m , and then the first possible middle is enormous. For the two kinds of kernel we checked, nothing new appears.

The route through the *abc* conjecture is discussed in Remark I.4.12: the uniform explicit exponents are too weak, and the full explicit form gives only a conditional bound.

Remark I.8.4 (the polynomial analog). [literature] Over the polynomials in one variable over a field of characteristic 0, not even two consecutive nonconstant powerful polynomials exist. If f and $f + 1$ are both powerful of degree $n \geq 1$, the theorem of Mason and Stothers [78, 46], applied to $(f + 1) - f = 1$, gives $n \leq \deg \text{rad}(f(f + 1)) - 1 \leq n/2 + n/2 - 1$, a contradiction; see Snyder [73] for a short proof of the theorem. In the integers there are infinitely many consecutive powerful pairs, such as $(8, 9)$ and $(288, 289)$ [32]. The proof uses the derivative, which has no analog for integers. In characteristic p the statement fails as well: x^p and $x^p + 1 = (x + 1)^p$ are consecutive p -th powers, because the derivative of a p -th power vanishes.

In words. For polynomials the problem is easy, even for pairs, because one can take derivatives. For whole numbers there is no derivative, and pairs of consecutive powerful numbers do exist. So the polynomial proof cannot be copied.

9 Open questions

Question I.9.1 (finiteness of the kernels of wing (ii)). [open] Are there only finitely many squarefree $m \equiv 7 \pmod{8}$ with $m \mid U_1(m)$ and $T_1(m)$ even?

These are the kernels of wing (ii) in Corollary I.2.4, the weakest point for an unconditional bound (Section 7). Walsh [86, p. 97] asks the corresponding question for all squarefree m with $m \mid U_1(m)$.

Question I.9.2 (the primitive part of $T_d(m)$). [open] Let $m \equiv 7 \pmod{8}$ be squarefree with $T_1(m)$ even, let $d \geq 5$ be odd, and let M_d be the part of $T_d(m)$ supported on the primes of rank exactly d . Can M_d be powerful?

The case of a prime d is, in our view, the hardest, since Corollary I.4.8 then gives a single condition. Section 6 examines the question on a finite set of pairs (m, d) . For the Fibonacci numbers, Fathi [29, Theorems 1.2–1.6] shows for several infinite families of indices n , among them every prime power $n > 2$, that F_n has a prime factor that first appears at n and occurs to an odd power. For these n the primitive part is not a square; this is weaker than not being powerful. Granville [34, Theorem 3] proves, for recurrences $x_{n+2} = bx_{n+1} + cx_n$ with $c \equiv 2 \pmod{4}$ and further conditions, that the terms have a primitive prime factor to an odd power. Our recurrence $T_{n+2} = 2T_1T_{n+1} - T_n$ has $c = -1$, and the question here asks for exponent one, not for an odd exponent.

Question I.9.3 (a middle that is a power of 2). [open] Can $2^a - 1$ and $2^a + 1$ both be powerful for some $a \geq 2$?

This is the case of a middle that is a power of 2, the case that Granville’s construction produces (Section 4). Proposition I.6.7 answers it for all exponents up to a bound. She [71, p. 7] conjectures that $x^n - 1$ is never powerful for $x > 1$ and $n > 2$, which would answer the question.

Question I.9.4 (Reinhart’s condition (C)). [open] Does Reinhart’s condition (C) [62, Definition 1.4] hold for some squarefree d , that is, is there an index k with $T_k(d)$ powerful and $d \mid U_k(d)$? For $d \mid U_1(d)$ the second condition holds for every k .

Reinhart asks this in particular for $d = 4\,099\,215$ and $d = 39\,028\,039\,587\,479$, and even for $d = 7$ [62, p. 6]. For odd k , Theorems I.5.1 and I.7.1(vii) show how large such an index would have to be.

An expectation of the author, not a claim. If a triple exists, we expect it to belong to a special case that is singled out by an additional equation, such as a middle that is a proper power, rather than to a lattice point without such structure. The reason is the model of Proposition I.6.1: at a point without additional structure every condition of Corollary I.4.8 is one more rare event, so a triple there would need many independent coincidences, while an algebraic identity can satisfy several conditions at once, as the Pell equations do for pairs. The special shapes treated so far, the cube middles of [19, 71, 69], the square-type middles of Theorem I.5.4 and the powers of 2 of Proposition I.6.7, are excluded or restricted. The expectation says where a triple would most likely be found, not whether one exists.

In words. These are the questions this part leaves open. The first asks whether the dangerous kernels of wing (ii) are finite in number. The second asks whether a building block of the middle can be powerful. The third asks whether the middle can be a power of 2, which Proposition I.6.7 rules out for small exponents. The fourth is Reinhart’s question for single kernels, including the three of Table 5. Whether the structure forces an odd witness in an outer number is settled by Remark I.4.11: it does not, beyond the primes of $T_1 \pm 1$.

The type of every statement

Every statement of this part carries one of six labels, printed after its number. The table lists them; it is generated from the text by the build and checked against a second, independent

reading of the source.

statement	title	type
Lemma I.1.1	Mollin and Walsh [51, Theorem, p. 110]	literature
Lemma I.1.2	Walker [85, Theorems 2.2(3) and 2.5]; Mollin and Walsh [51, p. 111]	literature
Lemma I.1.3	Sentance [70, Theorem and Converse, p. 272]	literature
Lemma I.1.4	Sentance [70, p. 272]; Mollin and Walsh [50, Lemma 1.1]	literature
Theorem I.2.1	height inequality	proved
Remark I.2.2	the constant in the height bound	proved
Corollary I.2.3	kernel bound	proved
Corollary I.2.4	two wings	proved
Remark I.2.5	Reinhart's list	literature
Theorem I.3.1	reduction	computed
Remark I.3.2	weaker bounds with less work	proved
Lemma I.4.1	Bennett and Walsh [8, proof of Lemma 3.3, p. 3484]; Lehmer [41]	literature
Lemma I.4.2	Bennett and Walsh [8, Theorem 1.2]	literature
Lemma I.4.3	valuation identity	proved
Proposition I.4.4	exponent-3 condition	proved
Theorem I.4.5	the b -box	computed
Theorem I.4.6	reduction to a witness statement	proved
Lemma I.4.7	the T_1 -part of a lattice point	proved
Corollary I.4.8	the divisor count of the Wieferich conditions	proved
Theorem I.4.9	Granville [33, Theorem, p. 216]	literature
Remark I.4.10	what the witness condition does not follow from	proved
Remark I.4.11	where a witness in an outer number can sit	proved
Remark I.4.12	two elementary constraints, and the size of the gap	proved
Remark I.4.13	the cyclotomic axis	proved
Theorem I.5.1	the strip	computed
Remark I.5.2	the bound of Chim, Shorey and Sinha	literature
Theorem I.5.3	smooth kernels	computed
Theorem I.5.4	square-type middles	computed
Remark I.5.5	the share of the candidates covered	computed
Proposition I.6.1	exponents at the rank	measured
Proposition I.6.2	no triple below $1.83 \cdot 10^{18}$, and the pairs at distance two with middle divisible by four	computed
Proposition I.6.3	ladders for the known kernels dividing U_1	computed
Proposition I.6.4	the smallest witnesses	computed
Remark I.6.5	the condition modulo 36 and the witnesses	computed
Proposition I.6.6	how many conditions there are, and how often they are met	computed
Proposition I.6.7	middles that are powers of two	computed
Proposition I.6.8	the primitive part over an explicit finite set	computed
Remark I.6.9	where this stands in the literature	literature
Proposition I.6.10	Wieferich primes of the unit and the primitive parts	proved
Remark I.6.11	the Wieferich events of the units	computed
Remark I.6.12	Lucas–Wieferich primes	computed

statement	title	type
Remark I.6.13	a second primitive prime factor	computed
Theorem I.7.1	profile	computed
Remark I.8.1	congruences	literature
Remark I.8.2	Yokoi's representation	computed
Remark I.8.3	small units	computed
Remark I.8.4	the polynomial analog	literature
Question I.9.1	finiteness of the kernels of wing (ii)	open
Question I.9.2	the primitive part of $T_d(m)$	open
Question I.9.3	a middle that is a power of 2	open
Question I.9.4	Reinhart's condition (C)	open

The types: *proved*, proved in this part; the proof may use published results (15); *literature*, due to the cited source, sometimes with a proof given here for completeness (12); *computed*, the result of an exact computation, or a proof that uses one of our computations (19); *measured*, a measurement, with a model that is not proved (1); *open*, an open question (4).

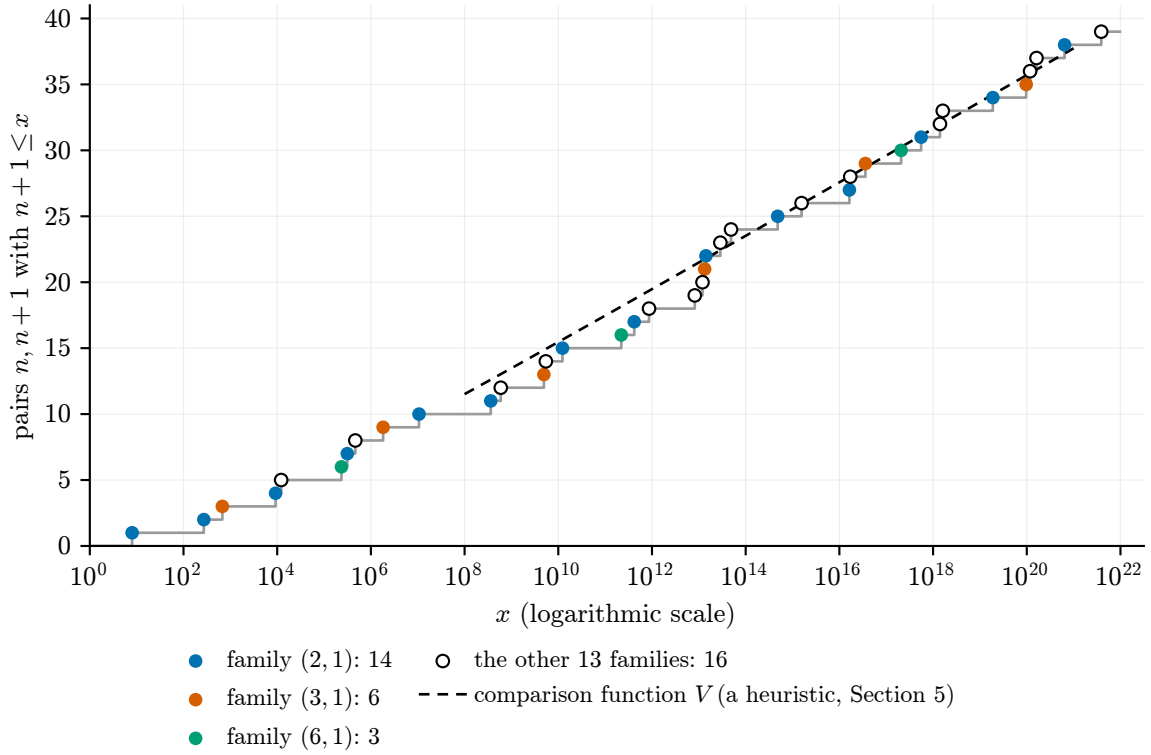


Figure 6: **The listed pairs up to n_0 , and the family each comes from.** The 39 pairs $n, n+1$ of powerful numbers with $n \leq n_0$ in the list of Proposition II.3.1(b), which is complete below 2^{52} , as a staircase: each step is one pair, and each point is colored by the family of Section 2 that contains it; the families with at least three pairs have their own color. The positions come from the family formulas of Proposition II.2.3, and at every power of ten from 10^8 to 10^{21} the staircase has exactly the number of pairs of our count, of which Table 12 shows every second one. The dashed line is the comparison function of Section 5, a heuristic yardstick, drawn from 10^8 on, where that table begins.

Part II: pairs

1 Introduction

A positive integer is *powerful* if every prime that divides it divides it at least twice. There are infinitely many n such that n and $n+1$ are both powerful; the solutions of Pell equations give them, as Walker recounts [85, p. 111]. Erdős asked two further questions [12]: whether all such pairs come from Pell equations, that is, whether n or $n+1$ must be a square, and whether the number $N_1(x)$ of such n with $n+1 \leq x$ is bounded by a power of $\log x$. The first question has a negative answer: Golomb's pair 12167, 12168 has no square, and Walker showed that its family is infinite [85, p. 111]; Walker also described all pairs through Pell

equations [85, Sections 2 and 3]. The second question is open. The best upper bound we know is $N_1(x) \ll x^{29/100+\epsilon}$, from a preprint of Reuss [63, Theorem 4], after $x^{2/5}(\log x)^2$ and $x^{7/19} \log x$ by Chan [18, Theorems 3 and 4] and x^γ for every $\gamma > 61/180$ by Blomer and Schöbel [10, Theorem 3], all three for every fixed distance; Tao [79, Corollary 2.11] proves $x^{2/5+o(1)}$ for pairs at every fixed distance and discusses the question. The pairs at distance two, $n-1$ and $n+1$ powerful, are the objects of Part I, where their middles n carry the question of three consecutive powerful numbers; a question of Golomb about two consecutive odd powerful numbers was answered by Sentance [70, p. 272].

This part collects what we can prove, compute and measure about the count of pairs, at distance one and at distance two. It establishes an explicit lower bound of the order $\log x$, with a computed constant, checks the known list of pairs, and places the counting problem for distance two in the kernels that divide their own U_1 , of which only 3 with T_1 even have been found below $5.325 \cdot 10^{13}$, and which contain the case in which Tao's argument is weakest, under a squarefreeness assumption.

Main results.

- *An explicit lower bound* (Proposition II.4.1): $\liminf N_1(x)/\log x \geq 1.6822$, from the families of Walker with a square and squarefree part at most 10^6 , and from those without a square and $bd \leq 10^6$. That there are infinitely many pairs is classical; we did not find an explicit constant in the literature.
- *The count at distance two is a sum over towers* (Proposition II.6.2), and so $P_2(10^h) \geq 0.595h - 122$ for every h (Corollary II.6.3). In the box of kernels up to 10^8 and middles below 10^{2000} the list of pairs at distance two is complete: 371 with middle divisible by four (Part I) and 832 with middle $\equiv 2 \pmod{4}$ (Proposition II.6.1).
- *Localization* (Proposition II.6.4): up to $O(N^{2/9})$, the pairs at distance two with middle at most N are exactly the first pairs of the kernels m with $m \mid U_1(m)$, so every bound $E_3(N) \ll N^\theta$ on the number of these kernels gives $P_2(N) \ll N^{\max(2/9, \theta)}$. The case in which the argument of Tao is weakest lies among these kernels when the numbers n_2 and m_2 of Remark II.6.6 are squarefree.
- *The families with a square overtake the comparison function* (Proposition II.5.2): the count of pairs with a square in the families with squarefree part up to 10^4 exceeds the comparison function of Section 5 for every $x \geq e^{1000}$, and at every row of the table from 10^{130} on. So the agreement of such a model with the counts up to n_0 , which we also record, is a fact about that range and not a law.

How to read this part. Every statement is followed by a paragraph *In words* that says what it means with as few formulas as possible; a reader who wants the results can read the statements and these paragraphs. A reader who wants to check can read the proofs, the paragraphs *The computation*, and the tables; every number we computed is generated from the data files of the deposit or from a cited constant, and every numbered statement carries a label that says whether it is proved, taken from the literature, computed, measured, a heuristic, or open (see the table at the end). A statement that combines several kinds carries the label of the part that needs the most trust: a proof that uses our own computation is

labeled computed, and a proof that uses a published result is labeled proved. Table 11 collects the notation.

Plan. Section 2 recalls Walker’s families and their periods. Section 3 lists the pairs up to 2^{52} from our own enumeration and checks the published list beyond it inside the families. Section 4 establishes the lower bound. Section 5 sets up the comparison function, marks it as a heuristic, shows that the square families overtake it, and states the convergence question. Section 6 treats the pairs at distance two: the second class of middles, the tower sum, the lower bound, the localization and the growth question. Section 7 puts the pairs at distance one on one Pell equation per kernel and tests the third neighbor. Section 8 collects the open questions.

2 Pell families

2.1 Pairs as solutions of one equation

As in Part I, $\text{sqfree}(N)$ is the product of the primes that divide N to an odd power, so that $N = \text{sqfree}(N) r^2$ for a positive integer r .

Lemma II.2.1 (powerful numbers). [proved] *A positive integer n is powerful if and only if $n = bY^2$ with $b = \text{sqfree}(n)$ and $b \mid Y$. This is the usual description of powerful numbers, in the form used below.*

Proof. Write $n = bY^2$ with $b = \text{sqfree}(n)$. For a prime $p \mid b$ we have $v_p(n) = 1 + 2v_p(Y)$, which is at least 2 exactly when $p \mid Y$. For a prime $p \mid n$ with $p \nmid b$ we have $p \mid Y$ and $v_p(n) = 2v_p(Y) \geq 2$ in any case. $\square$

In words. Every number is its squarefree part times a square. The number is powerful exactly when the squarefree part divides the root of the square part; for example $72 = 2 \cdot 6^2$ is powerful because $2 \mid 6$, and $12 = 3 \cdot 2^2$ is not because $3 \nmid 2$.

Proposition II.2.2 (Walker [85, Sections 2 and 3]). [literature] *Let n and $n + 1$ be powerful, and put $b = \text{sqfree}(n)$, $d = \text{sqfree}(n + 1)$, $Y^2 = n/b$, $X^2 = (n + 1)/d$. Then*

$$dX^2 - bY^2 = 1, \quad b \mid Y, \quad d \mid X.$$

Conversely, every solution in positive integers of this equation with b, d squarefree, $b \mid Y$ and $d \mid X$ comes from exactly one such pair, namely $n = bY^2$.

Proof. The first part is Lemma II.2.1 applied to n and to $n + 1$. Conversely, bY^2 and dX^2 are consecutive, and they are powerful by Lemma II.2.1, because $\text{sqfree}(bY^2) = b$ and $\text{sqfree}(dX^2) = d$ for squarefree b and d . The number n determines b, d, X and Y . $\square$

In words. A pair of consecutive powerful numbers is the same thing as a solution of $dX^2 - bY^2 = 1$ with two divisibility conditions. Golomb [32] sorted the pairs into two types, as Walker recounts in his introduction [85, p. 111], and Walker described both types through equations of this shape.

We call the set of pairs with a given (b, d) the *family* (b, d) . The order matters: in the family (b, d) the smaller number has squarefree part b . As $X^2 - Y^2 = 1$ has no solution in positive integers, b and d are not both 1. If $b = 1$ or $d = 1$, one of the two numbers is a square (Walker's type I); if $b, d \geq 2$, neither is (type II). By Proposition II.2.2 every pair lies in exactly one family.

2.2 The pairs inside one family

For type I with $d = 1$ the equation is Pell's equation $X^2 - bY^2 = 1$. We use the notation of Part I with $m = b$: $\varepsilon_b = T_1 + U_1\sqrt{b}$ is its fundamental solution, $\varepsilon_b^k = T_k + U_k\sqrt{b}$, and $b' = \prod_{p|b, p \nmid U_1} p$.

Proposition II.2.3 (Walker [85, Theorems 2.2, 2.5, 3.2 and 3.5]). [literature]

- (i) Let $b \geq 2$. The pairs of the family $(b, 1)$ are $n = bU_k^2$, $n + 1 = T_k^2$ for exactly the $k \geq 1$ with $b' \mid k$.
- (ii) Let $d \geq 2$. The family $(1, d)$ needs a solution of $Y^2 - dX^2 = -1$. If there is none, or if d is even, the family is empty. Otherwise let $\eta = y_0 + x_0\sqrt{d}$ be the fundamental solution, $\eta^k = Y_k + X_k\sqrt{d}$ for odd k , and $g = d/\gcd(d, x_0)$. The pairs are $n = Y_k^2$, $n + 1 = dX_k^2$ for exactly the odd multiples k of g .
- (iii) Let $b, d \geq 2$, and let $\alpha = x_0\sqrt{d} + y_0\sqrt{b}$ be the smallest solution of $dX^2 - bY^2 = 1$ if there is one. All solutions are $\alpha^j = X\sqrt{d} + Y\sqrt{b}$ with j odd. If d is even and x_0 is odd, or b is even and y_0 is odd, the family is empty. Otherwise its pairs come from exactly the odd multiples j of ℓ , the product of the odd primes that divide bd but not x_0y_0 .

Proof. Part (i) is Lemma I.1.2 for $m = b$: $b \mid U_k$ if and only if $b' \mid k$. Part (ii) follows in the same way. For odd k the binomial theorem gives $X_k = \sum_{i \text{ odd}} \binom{k}{i} y_0^{k-i} x_0^i d^{(i-1)/2} \equiv k y_0^{k-1} x_0 \pmod{p}$ for a prime $p \mid d$, and $y_0^2 \equiv -1 \pmod{p}$, so y_0 is invertible modulo p . Hence $p \mid X_k$ if and only if $p \mid kx_0$. If d is even, then y_0 is odd, $dx_0^2 = y_0^2 + 1 \equiv 2 \pmod{8}$, so x_0 is odd and no X_k with k odd is even. If d is odd, $d \mid X_k$ if and only if $g \mid k$. Part (iii) is Walker's Theorems 3.2 and 3.5; the description of all solutions as odd powers of α is his Theorem 3.1, which he takes from [84, Theorem 9]. $\square$

In words. Inside one family the pairs come at regular steps. For the family $(b, 1)$ one takes every b' -th power of the Pell solution, for $(1, d)$ every $2g$ -th power starting at the g -th, and for type II the same with ℓ . We call b' , g and ℓ the *period* of the family. Walker proved all three parts; the proof of (i) and (ii) above is short enough to give here.

For the computations we used our own forms of these statements and checked them: part (i) against direct iteration modulo b for all 6082 squarefree $b \leq 10\,000$, and our residue class form of part (iii) against direct iteration modulo bd for all 955 solvable equations with $b, d \geq 2$ and $bd \leq 3000$; all agreed. For the families in Table 10 we also computed b' , g and ℓ from Walker's definitions and found the periods of our computation.

In words. The formulas were not only cited: a computer followed the powers one by one and found the pairs exactly where the formulas for parts (i) and (iii) say.

2.3 The families of the known pairs

Each family grows geometrically. Here and below, $\log$ is the natural logarithm; Table 10 gives $\log_{10}$ for readability. If n_1 is its first pair, the j -th is about $n_1 F^{j-1}$, where $\log F = 2b' \log \varepsilon_b$ for the family $(b, 1)$, $\log F = 4g \log \eta$ for $(1, d)$, and $\log F = 4\ell \log \alpha$ for type II. Throughout this part $n_0 = 3\,887\,785\,221\,910\,670\,811\,499$ is term number 39 of the list in Section 3, so $n_0 \approx 3.89 \cdot 10^{21}$. Table 10 lists the families of the 39 pairs of that list with $n \leq n_0$: 16 families, 32 pairs with a square and 7 without. The first pairs of the families $(46, 1)$, $(6, 1)$, $(1, 5)$ and $(3, 7)$ are the four examples in Walker's paper [85, pp. 113 and 116].

family (b, d)	period	$\log_{10} n_1$	$\log_{10} F$	pairs
(2, 1)	1	0.90	1.53	14
(3, 1)	3	2.83	3.43	6
(23, 2)	1	4.09	9.37	2
(6, 1)	3	5.37	5.97	3
(1, 5)	5	5.67	12.54	2
(46, 1)	1	8.77	9.37	2
(7, 2)	7	9.73	20.67	1
(5, 1)	5	11.94	12.54	1
(430, 1)	1	12.91	13.52	1
(79, 23)	1	13.08	27.36	1
(3, 7)	7	13.69	28.58	1
(10, 1)	5	15.19	15.79	1
(7, 1)	7	16.23	16.83	1
(14, 1)	7	20.07	20.67	1
(1022, 57)	1	20.21	41.62	1
(19, 35)	5	21.59	44.38	1

Table 10: The families of the pairs $n, n+1$ of powerful numbers with $n \leq n_0 \approx 3.89 \cdot 10^{21}$, ordered by their first pair n_1 . The period is b' , g or ℓ of Proposition II.2.3; F is the growth factor from one pair of the family to the next. Each row is checked against three computations: the count in the list of pairs, the count predicted from the family, and, for type II, the pairs generated from the family and factored.

In words. Most of the known pairs sit in a few families. The family $(2, 1)$, which starts with 8 and 9, alone has 14 of them, because its growth factor is small. Most other families contribute a single pair so far.

2.4 Notation and terms at a glance

Table 11 collects the symbols and words of this part, with the place where each is defined.

3 Enumeration

Proposition II.3.1 (the known pairs). [computed]

(a) *There are exactly 26 pairs $n, n+1$ of powerful numbers with $n+1 \leq 2^{52}$.*

symbol or term	meaning	defined in
powerful	every prime factor occurs at least twice	Lemma II.2.1
$\text{sqfree}(N)$	the product of the primes that divide N to an odd power	§2
family (b, d)	the pairs $n, n+1$ with $\text{sqfree}(n) = b$ and $\text{sqfree}(n+1) = d$	§2
type I, type II	one of $n, n+1$ is a square ($b = 1$ or $d = 1$); neither is ($b, d \geq 2$)	§2
$\varepsilon_b, T_k, U_k, b'$	as in Part I, with the kernel $m = b$	Proposition II.2.3
η, g	the smallest solution $y_0 + x_0\sqrt{d}$ of $y^2 - dx^2 = -1$, and $g = d/\gcd(d, x_0)$	Proposition II.2.3
α, ℓ	the smallest solution $x_0\sqrt{d} + y_0\sqrt{b}$ of $dX^2 - bY^2 = 1$, and the product of the odd primes of bd that do not divide x_0y_0	Proposition II.2.3
period	b', g or ℓ : the step between the powers that give pairs	Proposition II.2.3
growth factor F	the factor from one pair of a family to the next; $\log F = 2b' \log \varepsilon_b, 4g \log \eta$ or $4\ell \log \alpha$	§2
n_0	term number 39 of the list of Section 3; the range of Table 10 is $n \leq n_0$	§2
$N_1(x)$	the number of pairs with $n+1 \leq x$	Proposition II.4.1
pair at distance two, middle	powerful numbers $n-1, n+1$; the number n between them	§6
kernel m, m' , lattice point	as in Part I: $m = \text{sqfree}(n^2 - 1)$, and $n = T_k(m)$ with k an odd multiple of m'	Lemma I.1.4
$P_2(N), P_2(N; M)$	the number of middles $n \leq N$, all of them or those with kernel at most M	§6
discard criterion	a proved lower bound for $T_k(m)$ that removes a kernel before any T_k is computed	Theorem I.3.1
tower	the middles $T_k(m)$ of one kernel m with $T_1(m)$ even, one for each odd multiple k of m'	§6
$w(m), c_M$	the rate $1/(2m' \log_{10} \varepsilon_m)$ of one tower in pairs per decimal digit, and the sum of these rates up to M	Proposition II.6.2
$E_J(N)$	the number of kernels with $m' < J$ and $T_1(m)$ even whose first middle is at most N	Proposition II.6.4
index m' , wing (ii), rung	the product of the primes of m that do not divide U_1 ; the kernels with $m \mid U_1$; one pair of a tower	Part I
$v_p(N), c_1, c_2, a_p$	the exponent of p in N ; the constants of $Q(x)$; the proportion of n with $v_p(n) = 1$	§5
$N_Q^*(L)$	the count of the pairs with a square from the family periods, by the floor formula	Proposition II.5.2
$Q(x), \rho(x)$	the number of powerful numbers up to x , and the derivative of its two main terms	§5
$C, V(x)$	the local correction factor and the comparison function, a heuristic yardstick	Definition II.5.1
$N_Q(x)$, born	the pairs with a square in the families with $b, d \leq 10^4$; a family is born at the logarithm of its first pair	Proposition II.5.2
D, R , third number	$D = \text{sqfree}(a(a+1))$, $a(a+1) = DR^2$; the neighbor $a-1$ or $a+2$ that is not excluded by 2	Proposition II.7.1

Table 11: Symbols and terms of Part II.

- (b) Let n_0 be term number 39 of the b-file of OEIS A060355 [55], computed by D. Johnson, which lists such pairs by their smaller member n . The families of Section 2, with a square and squarefree part at most 10^6 or without a square and $bd \leq 10^6$, generate exactly 39 pairs with $n \leq n_0$ from their fundamental solutions, each checked by complete factorization and by the identity $dX^2 - bY^2 = 1$ with $b \mid Y$ and $d \mid X$. The first 26 of them are the pairs in (a), and the 39 pairs are exactly the first 39 terms of the b-file (Proposition II.3.2). That no pair with $n \leq n_0$ lies outside these families is a statement of that enumeration, not of ours.

The computation. (a) We listed all powerful numbers below 2^{52} block by block, the block k being $[2^k, 2^{k+1})$, each number written as a^2b^3 with b squarefree; this form is unique. The size of every block agrees with the count $\sum_b \lfloor \sqrt{x/b^3} \rfloor$ over squarefree b , taken at the two ends of the block. Inside a block the pairs are neighbors at distance one. A pair across two blocks would be $(2^k - 1, 2^k)$, and $2^k - 1$ is not powerful for $2 \leq k \leq 63$, by complete factorization. (b) The pairs are generated by the recursion of the Pell equation from the fundamental solution of each family, as in the proof of Proposition II.3.2; both members of every pair are factored completely, and the identity is checked with the squarefree parts b and d . The comparison with the b-file is made at run time against a copy that the reader supplies; the terms of the b-file are not stored in the data deposit. $\square$

In words. Up to 2^{52} we found all pairs ourselves. Beyond that we used a published list and checked that each of its entries really is a pair; we did not check that the list misses nothing.

Mian and Siddique [49] describe the larger enumerations as exhaustive below 10^{22} but not certified. We rely on it only where we say so. The next statement checks it inside the families of Section 2, independently of the list.

Proposition II.3.2 (completeness inside the families). [computed]

- (a) In the families with a square, $(b, 1)$ and $(1, d)$ with $b, d \leq 10^4$, the pairs with $n \leq n_0$ are exactly the 32 pairs with a square among the terms of Proposition II.3.1(b). They lie in 10 families.
- (b) In the families without a square with $bd \leq 10^6$, the pairs with $n \leq n_0$ are exactly the 7 pairs without a square among those terms.

The computation. For every family in the range we generated its pairs from Proposition II.2.3 up to n_0 and compared them with the terms, in both directions: no generated pair is missing from the terms, and no term with a square (in (a)) or without a square (in (b)) is missing from the generated pairs. For (b) each generated pair was also factored completely, and both numbers are powerful. $\square$

In words. Every one of the 39 known pairs comes out of our own family computation, and within the families we searched our computation finds no pair that the list lacks. So the list is complete for these families. Outside them, for example a family $(b, 1)$ with $b > 10^4$ whose first pair lies below n_0 , a missing pair would not be noticed here.

4 An explicit lower bound

Proposition II.4.1 (explicit lower bound). [computed] *Let $N_1(x)$ count the pairs $n, n+1$ of powerful numbers with $n+1 \leq x$. Then*

$$\liminf_{x \rightarrow \infty} \frac{N_1(x)}{\log x} \geq 1.6822.$$

The 676 094 nonempty families with a square and squarefree part at most 10^6 contribute 1.3685, and the 275 371 nonempty families without a square and with $bd \leq 10^6$ contribute 0.3137.

Proof. Let F be the growth factor of a nonempty family, as in Section 2. We first show that its j -th pair satisfies $n_j + 1 \leq F^j$. In the family $(b, 1)$ the j -th pair has $n_j + 1 = T_{jb'}^2$ by Proposition II.2.3(i), and $T_k = (\varepsilon_b^k + \varepsilon_b^{-k})/2 \leq \varepsilon_b^k$, so $n_j + 1 \leq \varepsilon_b^{2jb'} = F^j$. In the family $(1, d)$ the j -th pair comes from $k = (2j-1)g$, and for odd k the conjugate of η^k is $-\eta^{-k}$, so $X_k\sqrt{d} = (\eta^k + \eta^{-k})/2 \leq \eta^k$ and $n_j + 1 = dX_k^2 \leq \eta^{2(2j-1)g} \leq \eta^{4jg} = F^j$. For type II the j -th pair comes from the exponent $(2j-1)\ell$, the conjugate of $\alpha^i = X\sqrt{d} + Y\sqrt{b}$ is $X\sqrt{d} - Y\sqrt{b} = \alpha^{-i}$, and the same computation gives $n_j + 1 = dX^2 \leq \alpha^{4j\ell} = F^j$.

Hence a family has at least $\lfloor \log x / \log F \rfloor \geq \log x / \log F - 1$ pairs with $n+1 \leq x$. Different families have different pairs by Proposition II.2.2. So for every finite set S of nonempty families

$$N_1(x) \geq \sum_{f \in S} \left(\frac{\log x}{\log F_f} - 1 \right), \quad \liminf_{x \rightarrow \infty} \frac{N_1(x)}{\log x} \geq \sum_{f \in S} \frac{1}{\log F_f}.$$

For the two sets named in the statement we computed the sums; the total is rounded down. $\square$

In words. Each family on its own produces pairs at a steady rate on a logarithmic scale: the number of its pairs below x grows like $\log x$ divided by the logarithm of its growth factor. Since no pair belongs to two families, the rates add up. Adding them for all families up to the stated sizes gives the constant. Families outside these ranges can only increase it.

5 A heuristic comparison model, and the convergence question

This section compares the count $N_1(x)$ of Section 4 with a function that a simple probabilistic picture suggests. The picture is a heuristic. We use the function only as a yardstick, and we do not claim that it predicts anything; the only statement of this section that is not a heuristic is Proposition II.5.2, and the lower bound of Section 4 does not depend on the picture.

5.1 The comparison function

Let $Q(x)$ be the number of powerful numbers up to x . Erdős and Szekeres found the main term, and Bateman and Grosswald proved $Q(x) = c_1\sqrt{x} + c_2x^{1/3} + O(x^{1/6})$ [28], [6, Theorem 1], with $c_1 = \zeta(3/2)/\zeta(3)$ and $c_2 = \zeta(2/3)/\zeta(2)$, so that $c_2 < 0$. Let $\rho(x) = c_1/(2\sqrt{x}) + c_2/(3x^{2/3})$ be the derivative of the two main terms.

Definition II.5.1 (comparison function; a heuristic). [heuristic] For a prime p let $a_p = (p-1)/p^2$, the proportion of integers n with $v_p(n) = 1$, and let

$$C = \prod_p \frac{1 - 2a_p}{(1 - a_p)^2}, \quad V(x) = C \int_8^x \rho(t)^2 dt.$$

The picture behind V is this. If “ n is powerful” and “ $n+1$ is powerful” were independent events of probability $\rho(n)$ each, the expected number of pairs with $n+1 \leq x$ would be about $\int_8^x \rho(t)^2 dt$, which grows like $(c_1^2/4) \log x = 1.1808 \log x$. The events are not independent: a prime p cannot divide both n and $n+1$, so the events $v_p(n) = 1$ and $v_p(n+1) = 1$ exclude each other. Under the picture, the proportion of n with $v_p(n) \neq 1$ and $v_p(n+1) \neq 1$ is $1 - 2a_p$ instead of $(1 - a_p)^2$, and the factor C collects this correction over all primes, in the manner of the Hardy–Littlewood constants. Two assumptions enter here and are not derived: that the primes act independently, and that these proportions, which are densities over all integers, may be applied to powerful numbers; among powerful numbers the share of n with $4 \mid n$ has a different limit (Remark I.5.5). Numerically $C = 0.7482$, so $V(x)$ grows like $0.8835 \log x$. Heuristics with a local factor at each prime are used for related counting problems, for example in [83].

In words. Powerful numbers get rarer as they grow, at a known rate. If two neighbors were powerful independently of each other, one could compute how many pairs to expect. They are not independent, because a prime never divides two neighbors at once, and C is one simple correction for that, not a derived one. The result is a curve to compare with, not a prediction.

Table 12 sets V against the count. Up to n_0 the list has 39 pairs; it is complete below 2^{52} and inside the families of Proposition II.3.2, and beyond that it is described as exhaustive but not certified [49]. The comparison function gives 38.94, and the independent picture gives 52.04. The constant C and the function ρ contain nothing fitted to the data: we computed V before we compared it with the terms beyond 2^{52} , where the count is 13 against 12.06 for V and 16.12 for the independent picture. The agreement is a fact about this range, within a scatter of about $\sqrt{N}$ for a count of N pairs; beyond 2^{52} the comparison does not separate V from the independent picture. The next statement shows that the agreement does not last.

$\log_{10} x$	pairs counted	$V(x)$	independent picture
8	10	11.5	15.4
10	14	15.5	20.7
12	18	19.5	26.0
14	24	23.5	31.4
16	26	27.6	36.9
18	31	31.6	42.3
20	35	35.7	47.7

Table 12: The number of pairs $n, n+1$ with $n+1 \leq x$, the comparison function $V(x)$ of Definition II.5.1, and the uncorrected integral $\int_8^x \rho^2$. The counts up to 2^{52} are our own enumeration; beyond it they come from the terms of Proposition II.3.1(b).

5.2 The families with a square overtake the comparison function

Let $N_Q(x)$ be the number of pairs $n, n+1$ with $n+1 \leq x$ that lie in a family $(b, 1)$ or $(1, d)$ with $b, d \leq 10^4$. By Proposition II.2.3 these families are known exactly. For a family f with first pair n_1 and growth factor F put $t_f = \log n_1$; we say the family is *born* at $\log x = t_f$. By the formulas of Section 2, n_j+1 is $(\alpha^k + \alpha^{-k})^2/4$ times a constant, with α the unit of the family and k growing by the period from one pair to the next; so $\log(n_j+1) = \log(n_1+1) + (j-1) \log F - \delta_j$ with $0 \leq \delta_j < 2 \log 2$, the correction coming from the decreasing terms α^{-k} . The count

$$N_Q^*(L) = \sum_f \max\left(0, \left\lfloor \frac{L - t_f}{\log F_f} \right\rfloor + 1\right),$$

over the families with $b, d \leq 10^4$, therefore agrees with $N_Q(e^L)$ except that a pair within $1/n_1$ of the boundary on the logarithmic scale may fall on the wrong side.

Proposition II.5.2 (the families with a square overtake the comparison function). [computed]

(a) Table 13 gives $N_Q^*(L)$ and $V(e^L)$ for 8 values of $L = \log x$ up to 1000. On a finer grid than the table, N_Q^* first exceeds V at $L = 192.9$, that is, at $x \approx 10^{83.8}$; at the next row of the table it is below V again, and from $L = 300$ on it is above V at every row of the table.

(b) For every $L \geq 1000$, $N_Q(e^L) > V(e^L)$.

$L = \log x$	$N_Q^*(L)$	$V(e^L)$	independent picture	rate of the families born
49.7	32	38.9	52.0	0.7049
75.0	54	61.3	81.9	0.7744
100.0	76	83.4	111.4	0.8136
150.0	124	127.5	170.4	0.8755
200.0	171	171.7	229.5	0.9011
300.0	265	260.1	347.6	0.9300
500.0	468	436.8	583.7	0.9765
1000.0	997	878.5	1174.1	1.0345

Table 13: The pairs with a square in the families with $b, d \leq 10^4$, counted exactly from the family periods, against the comparison function. The last column is $\sum 1/\log F$ over the families born by L , the rate at which N_Q grows there.

Proof. (a) is the computation of the table. (b) Let $\mathcal{F}$ be the set of the 103 families with $b, d \leq 10^4$ that are born by $L = 1000$, and let $s = \sum_{f \in \mathcal{F}} 1/\log F_f = 1.0345$. Write $y_f(L) = (L - t_f)/\log F_f$. The j -th pair of f has $\log(n_j+1) \leq L$ as soon as $\log(n_1+1) + (j-1) \log F_f \leq L$, that is, $j \leq (L - \log(n_1+1))/\log F_f + 1$; so f has more than $(L - \log(n_1+1))/\log F_f \geq y_f(L) - 1/(n_1 \log F_f)$ pairs with $n+1 \leq e^L$, and the last term is below 0.05, because $n_1 \geq 8$ and $\log F_f \geq 2 \log(2 + \sqrt{3}) > 2.6$ for every family. On the other hand, each term of $N_Q^*(1000)$ is at most $y_f(1000) + 1$. So for $L \geq 1000$,

$$N_Q(e^L) > \sum_{f \in \mathcal{F}} y_f(L) - 0.05 |\mathcal{F}| \geq (N_Q^*(1000) - |\mathcal{F}|) - 0.05 |\mathcal{F}| + s(L - 1000)$$

and $0.05|\mathcal{F}| < 6$, so

$$N_Q(e^L) > 894 - 6 + s(L - 1000).$$

On the other side, $\frac{d}{dL}V(e^L) = C\rho(e^L)^2e^L = C(c_1/2 + (c_2/3)e^{-L/6})^2 < Cc_1^2/4 = 0.8835$, because $c_2 < 0$. Hence $V(e^L) \leq V(e^{1000}) + 0.8835(L - 1000)$ with $V(e^{1000}) = 878.5$. Since $894 - 6 > 878.5$ and $s > 0.8835$, the claim follows. The value of $V(e^{1000})$ is rounded to one decimal and the product C is truncated; truncating C increases V , and the margin is larger than the rounding, so both errors are on the safe side. $\square$

In words. The comparison curve grows at a fixed rate on a logarithmic scale. The families with a square alone grow faster once enough of them have been born, and from 10^{130} on, at every row of the table, they produce more pairs than the curve gives for all pairs together; for all $x \geq e^{1000}$ this is proved. So the curve matches the counts only in the range we can see, and it must fail beyond it. It is a yardstick for the present, not a law.

5.3 The convergence question

By Proposition II.4.1, $\liminf N_1(x)/\log x$ is at least the sum of $1/\log F$ over any finite set of families. The sum over the families with a square and squarefree part at most B is 0.6055 for $B = 10$ and 0.8949 for $B = 100$; Table 14 continues it and adds the families without a square with $bd \leq B$.

$\log_{10} B$	with a square, squarefree part $\leq B$	without a square, $bd \leq B$	sum
3	1.1044	0.1783	1.2827
4	1.2235	0.2385	1.4619
5	1.3110	0.2831	1.5941
6	1.3685	0.3137	1.6822

Table 14: The partial sums $\sum 1/\log F$ over the families of Section 2. The last entry of the last column is the constant of Proposition II.4.1.

Question II.5.3 (convergence of the rates). [open] Does $\sum_f 1/\log F_f$, taken over all families, converge?

The increase per decade of B shrinks in both columns of the table. We counted where the last increase comes from: in the decade $10^5 < B \leq 10^6$, families with $\log \varepsilon \leq 100$ supply 79 % of it among the families with a square and 77 % among those without, and the largest one percent of the families supply 71 % and 72 %. So in the last decade we computed, the increase is carried by a small share of the families, those with small units, not by many small contributions. Whether this continues is not known.

A positive answer would not by itself give an asymptotic formula for $N_1(x)$. Families keep being born, and each newborn family adds a pair at once; turning the sum of the rates into the rate of $N_1(x)$ needs control of how many families are born below every height, which we do not have. The towers of Section 6 pose the same question for distance two, as Question II.6.7.

This is also where an answer to the second question of Erdős could come from. Every pair belongs to exactly one family, and a family has about $\log x/\log F$ pairs below x . So $N_1(x)$ is governed by two quantities: how many families are born below x , and the sum of their rates, which is Question II.5.3. If the number of families born below x were bounded by a power of

$\log x$, so would be $N_1(x)$, and the answer to the question of Erdős would be yes. We know no such bound, and no bound for the number of families better than the upper bounds for $N_1(x)$ itself recalled in the introduction. We do not know whether it holds.

In words. Every family contributes a rate, and the rates add up. Whether the total of all rates is finite is open. Our partial sums grow more and more slowly, but they are driven by a small share of the families, those with small units, so a few more decades would not settle it. Erdős's question would be answered by showing that new families are born rarely enough; we cannot show this.

6 Pairs at distance two

A *pair at distance two* is a pair $n - 1, n + 1$ of powerful numbers, and n is its *middle*. By Lemma I.1.4 the middle is even, and $n = T_k(m)$ for exactly one squarefree odd $m \geq 3$ with $T_1(m)$ even and one odd multiple k of m' . The kernel satisfies $m \equiv 7 \pmod{8}$ when $4 \mid n$, and $m \equiv 3 \pmod{8}$ when $n \equiv 2 \pmod{4}$. We write

$$P_2(N) = \#\{n \leq N : n - 1 \text{ and } n + 1 \text{ are powerful}\},$$

and $P_2(N; M)$ for the same count restricted to kernels $m \leq M$. Proposition I.6.2 lists the middles divisible by four in a box. The next statement does the same for the other middles, and the rest of the section counts both. Figure 7 shows all these pairs at once.

6.1 The middles that are not divisible by four

Proposition II.6.1 (the middles that are not divisible by four). [computed]

- (a) Among the 10132121 squarefree $m \equiv 3 \pmod{8}$ with $m \leq 10^8$, exactly 86 pass the discard criterion of Theorem I.3.1 for the height 10^{2000} ; the largest of them is 3926299. Their lattice points (m, k) with $T_k(m) < 10^{2000}$ number 883. For 51 of them $T_1(m)$ is odd, and the other 832 are exactly the middles $n \equiv 2 \pmod{4}$ of pairs at distance two with $n < 10^{2000}$ and $\text{sqfree}(n^2 - 1) \leq 10^8$.
- (b) These 832 middles have 57 different kernels, and the kernel $m = 3$ alone gives 583 of them.
- (c) The list in (a) contains every middle $n \equiv 2 \pmod{4}$ with $n < 1.83 \cdot 10^{18}$, every such middle with $n < 3.88 \cdot 10^{20}$ if Reinhart's extended search missed nothing (Remark I.2.5), and every such middle with $n < 10^{36}/27$ whose kernel does not divide U_1 .

Proof. (a) is the computation of Theorem I.3.1, run with the residue class 3 in place of 7, together with Lemma I.1.4: a lattice point with T_1 odd has T_k odd and gives no pair. (b) is a count in the list of (a). (c) Let n be such a middle and m its kernel. Remark I.2.2 gives the bound $T_k \geq (m^{3/2}/m')^{m'}$ in this class too, so the proof of Corollary I.2.3 applies: if $m \nmid U_1$, then $m^{4.5}/27 < n$, and $m < 10^8$ for $n < 10^{36}/27$. If $m \mid U_1$, then $m^{3/2} < T_1(m) \leq n$, so $n < 1.83 \cdot 10^{18}$ forces $m < 1.5 \cdot 10^{12}$, the range in which Reinhart's list of such kernels is complete [61, Remark 5.5], and $n < 3.88 \cdot 10^{20}$ forces $m < 5.325 \cdot 10^{13}$, the range of his extended search [62]. The list has two kernels $m \equiv 3 \pmod{8}$, 1752299 below the smaller

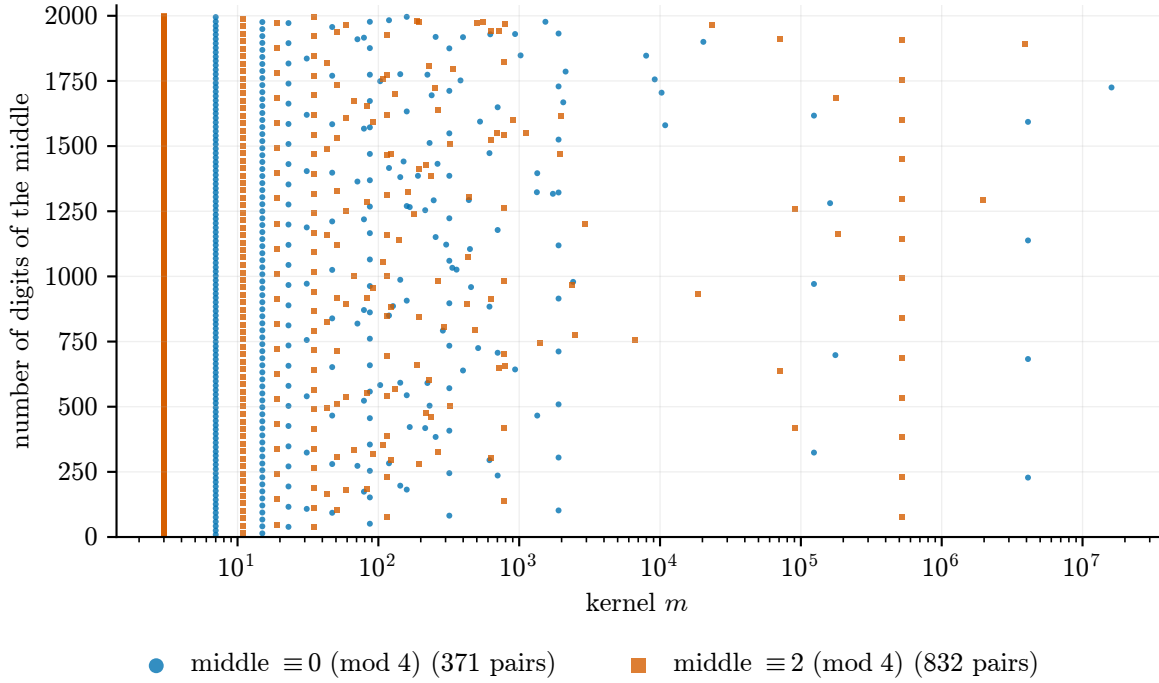


Figure 7: **The pairs at distance two stand in towers.** The pairs at distance two with middle below 10^{2000} and kernel at most 10^8 , each placed at its kernel m and the number of digits of its middle. The pairs of one kernel stand in a column at equal distances, the towers of Proposition II.6.2. The kernel $m = 3$ alone gives 583 of them. The column at $m = 525\,003$, with $m' = 3$, is the kernel from the discussion of Question II.6.7.

bound and $20\,256\,129\,307\,923$ below the larger one. The first lies in the box of (a); its T_1 has 117 digits and is odd. For the second, T_1 is even and has 99 134 digits, so its first middle is far above $3.88 \cdot 10^{20}$. $\square$

In words. The middles of the form $4j + 2$ come from the kernels m of the form $8j + 3$. In the box of kernels up to 10^8 and middles below 10^{2000} there are 832 of them, and most come from a single kernel, $m = 3$. None of them is the middle of a triple, since such a middle is divisible by 2 exactly once.

6.2 The count is a sum over towers

For a kernel m with $T_1(m)$ even, the middles $T_k(m)$ with k an odd multiple of m' form the *tower* over m .

Proposition II.6.2 (the count is a sum over towers). [proved] *For all integers $N, M \geq 1$,*

$$P_2(N; M) = \sum_{\substack{3 \leq m \leq M \\ \text{odd, squarefree} \\ T_1(m) \text{ even}}} \#\{k \text{ an odd multiple of } m' : k \log \varepsilon_m \leq \log(2N)\}.$$

Proof. By Lemma I.1.4 the middles $n \leq N$ with kernel $m \leq M$ correspond one to one to the pairs (m, k) in the sum with $T_k(m) \leq N$. It remains to show that $T_k \leq N$ exactly when $\varepsilon_m^k \leq 2N$. We have $2T_k = \varepsilon_m^k + \varepsilon_m^{-k}$ with $0 < \varepsilon_m^{-k} < 1$. If $T_k \leq N$, then $\varepsilon_m^k < 2T_k \leq 2N$. If $\varepsilon_m^k \leq 2N$, then $2T_k < 2N + 1$, and since $2T_k$ is an integer, $2T_k \leq 2N$. $\square$

In words. Each kernel carries a tower of pairs, one on every rung k , and the rungs are evenly spaced on a logarithmic scale: from one rung to the next the middle grows by the factor $\varepsilon_m^{2m'}$. Counting the pairs below a height is counting rungs below that height, tower by tower.

We evaluated the right side for the kernels that pass the discard criterion in both classes and compared it with the lattice points that the searches of Theorem I.3.1 and Proposition II.6.1 found. Table 15 shows six heights; the two agree at every height in both classes. The last two columns are a negative control: the same count with the step $\log T_1$ in place of $\log \varepsilon_m$, an error of an early version of our program, gives larger counts, which are wrong.

h	pairs, counted and by the tower sum		negative control (step $\log T_1$)	
	middle $\equiv 0 \pmod{4}$	middle $\equiv 2 \pmod{4}$	middle $\equiv 0 \pmod{4}$	middle $\equiv 2 \pmod{4}$
100	14	37	21	67
250	39	97	53	168
500	83	199	111	344
1000	176	410	224	695
1500	270	616	345	1049
2000	371	832	467	1404

Table 15: Pairs at distance two with middle at most 10^h and kernel at most 10^8 . The counted pairs and the tower sum of Proposition II.6.2 agree in every row.

Each tower adds one pair every time $\log_{10} N$ grows by $2m' \log_{10} \varepsilon_m$. So $P_2(10^h; M)$ grows with h at the average rate

$$c_M = \sum_{\substack{3 \leq m \leq M \\ \text{odd, squarefree} \\ T_1(m) \text{ even}}} w(m), \quad w(m) = \frac{1}{2m' \log_{10} \varepsilon_m},$$

pairs per decimal digit. For the towers of Table 15 the rate is 0.1817 for the middles divisible by four and 0.4134 for the others. This rate belongs to the kernels up to M ; it is not a rate for P_2 itself, because kernels above M add towers of their own. Whether c_M stays bounded as M grows is Question II.6.7.

Corollary II.6.3 (lower bound at every height). [computed] *For every integer $h \geq 1$,*

$$P_2(10^h) \geq 0.595h - 122.$$

For the middles divisible by four alone, the count is at least $0.1817h - 60$.

Proof. Take the 122 kernels $m \leq 10^8$ with $T_1(m)$ even that pass the discard criterion, 60 of them with $m \equiv 7 \pmod{8}$ and 62 with $m \equiv 3 \pmod{8}$. By Proposition II.6.2 the tower over m has a pair with middle at most 10^h for every odd j with $j \leq y_m$, where $y_m = (h + \log_{10} 2)/(m' \log_{10} \varepsilon_m)$. There are $\lfloor (y_m + 1)/2 \rfloor > y_m/2 - 1/2$ such j , and $y_m/2 \geq h w(m)$. Different kernels give different middles. Summing over the kernels gives more than

$h \sum_m w(m) - 122/2$ pairs. The sum of the weights is at least 0.595, and at least 0.1817 over the first class, which gives both bounds with half the subtracted constants; the statement keeps the larger ones. $\square$

In words. Every tower keeps producing pairs forever, one per rung. Adding up the rungs of the towers we know gives a lower bound for the number of pairs below any height, however large. Any single tower already gives a bound of this shape; the constant here is the sum over the 122 towers that pass the discard criterion. For heights below 10^{2000} the count over these towers is exactly the number of pairs with kernel at most 10^8 , by Proposition II.6.2.

That there are infinitely many pairs at distance two is classical [70, p. 272]; Corollary II.6.3 only puts a number on how fast they appear.

6.3 Where the pairs live

For an odd $J \geq 3$ let

$$E_J(N) = \#\{m \text{ odd, squarefree} : m' < J, T_1(m) \text{ even}, T_{m'}(m) \leq N\},$$

the kernels whose first rung is a middle at most N and whose index m' is below J . For $J = 3$ these are the kernels with $m \mid U_1$ and $T_1(m)$ even; those with $m \equiv 7 \pmod{8}$ form wing (ii) of Part I (Corollary I.2.4).

Proposition II.6.4 (localization of the pair count). [proved] *For every integer $N \geq 2$ and every odd $J \geq 3$,*

$$P_2(N) \leq 6J (J^J N)^{2/(3J)} + \frac{2}{\log 3} (J^J N)^{1/(3J)} \log N + \frac{2 \log N}{\log 3} E_J(N), \quad E_J(N) \leq P_2(N).$$

In particular $P_2(N) \leq 18 (27N)^{2/9} + \frac{2}{\log 3} (27N)^{1/9} \log N + \frac{2 \log N}{\log 3} E_3(N)$. Moreover,

$$E_3(N) \leq P_2(N) \leq E_3(N) + 41 N^{2/9} + 3 N^{1/9} \log N \leq E_3(N) + 51 N^{2/9}.$$

Proof. By Lemma I.1.4 every middle $n \leq N$ is $T_k(m)$ for one kernel m and one odd multiple k of m' , with $T_1(m)$ even. For a fixed kernel, $T_k \geq T_1^k$ and $T_1 > \sqrt{m}$, so there are fewer than $2 \log N / \log m$ indices k with $T_k \leq N$.

Kernels with $m' \geq J$. By Remark I.2.2, $N \geq T_k \geq (m^{3/2}/m')^{m'}$. The function $f(x) = x(1.5 \log m - \log x)$ increases on $[3, m]$ for $m \geq 8$, as in the proof of Corollary I.2.3, and for $m = 7$ one checks $f(3) < f(5) < f(7)$ directly; for $m = 3$ we have $m' = 3 = J$, and $m = 5$ does not occur, because $m \equiv 3$ or $7 \pmod{8}$. So $N \geq (m^{3/2}/J)^J$, that is, $m \leq M_J := (J^J N)^{2/(3J)}$. Using $\sum_{3 \leq m \leq M} 1/\log m \leq \sqrt{M}/\log 3 + 2M/\log M$ and $\log M_J \geq (2/(3J)) \log N$, these kernels contribute at most

$$2 \log N (2M_J/\log M_J + \sqrt{M_J}/\log 3) \leq 6JM_J + (2/\log 3) \sqrt{M_J} \log N$$

middles.

Kernels with $m' < J$. Then $T_{m'}(m) \leq T_k(m) \leq N$, and $T_{m'}$ is even because T_1 is even and m' is odd. So m is counted by $E_J(N)$, and it contributes fewer than $2 \log N / \log 3$ middles.

Finally, $m \mapsto T_{m'}(m)$ maps the kernels counted by $E_J(N)$ to different middles at most N ; each $T_{m'}(m)$ is a middle by the converse direction of Lemma I.1.4. So $E_J(N) \leq P_2(N)$.

For the last statement take $J = 3$. The kernels with $m' \geq 3$ contribute at most $18(27N)^{2/9} + (2/\log 3)(27N)^{1/9} \log N$ middles, as above. A kernel with $m' < 3$ has $m' = 1$, because m' is odd, so $m \mid U_1$; its middles are the T_k with k odd, and the first of them, T_1 , is counted by $E_3(N)$. A second middle $T_k \leq N$ with $k \geq 3$ requires $T_1^3 \leq T_3 \leq N$. Since $m \mid U_1$, we have $U_1 \geq m$ and $T_1^2 = 1 + mU_1^2 > m^3$, so $m < N^{2/9}$, and the number of odd $k \geq 3$ with $T_k \leq N$ is at most $\log N/(2 \log T_1) < \log N/(3 \log m)$. With the bound for $\sum 1/\log m$ above and $M = N^{2/9}$, these kernels contribute at most $3N^{2/9} + N^{1/9} \log N/(3 \log 3)$ further middles. Adding up, with $18 \cdot 27^{2/9} + 3 < 41$ and $(2/\log 3) 27^{1/9} + 1/(3 \log 3) < 3$, gives the first bound, and $N^{1/9} \log N \leq (9/e) N^{2/9}$ gives the second. $\square$

In words. There are two kinds of kernels. Those with a large index m' must be small, because their towers start high; together they give only at most of the order $N^{2/(3J)}$ pairs. All other pairs come from the kernels whose index is small, whose number is the open point, and each of these gives only a logarithmic number of pairs. So the question of how many pairs there are is a question about how many kernels with small index there are. A finer count shows that a kernel with $m \mid U_1$ almost always gives only its first pair: a second one needs $T_1^3 \leq N$, which only kernels below $N^{2/9}$ can afford. So the pairs and these kernels agree up to $N^{2/9}$.

Corollary II.6.5 (what the localization says, and what it does not say). [proved]

- (a) For every fixed odd $J \geq 3$, all but $O_J(N^{2/(3J)})$ of the middles counted by $P_2(N)$ have a kernel with $m' < J$.
- (b) Since $E_J(N) \leq P_2(N)$, Proposition II.6.4 gives no bound for $P_2(N)$ by itself. Conversely, every bound $E_J(N) \ll N^\theta$ gives $P_2(N) \ll N^{\max(2/(3J), \theta)} \log N$. For $J = 3$ the factor $\log N$ is not needed: every bound $E_3(N) \ll N^\theta$ gives $P_2(N) \ll N^{\max(2/9, \theta)}$, and for $\theta \geq 2/9$ the two bounds are equivalent.

Proof. Both parts are read off from Proposition II.6.4. $\square$

In words. The inequality moves the problem, it does not solve it: counting pairs is as hard as counting the kernels with small index, and for those we have no better bound than the one for all pairs.

Remark II.6.6 (where Tao's argument is weakest). [proved] For distance two, Chan [18, Theorem 3] proves $P_2(N) \ll N^{2/5}(\log N)^2$, Tao [79, Corollary 2.11] proves $P_2(N) \ll N^{2/5+o(1)}$ within a more general statement, and Blomer and Schöbel [10, Theorem 3] prove $P_2(N) \ll N^\gamma$ for every $\gamma > 61/180$, for every fixed nonzero difference. Tao [79, Remark 2.12] notes that his argument is weakest when $n - 1 = n_1^2 n_2^3$ and $n + 1 = m_1^2 m_2^3$ with n_1, n_2, m_1, m_2 all of size about $N^{1/5}$. If moreover n_2 and m_2 are squarefree, the kernel is $m = n_2 m_2$, of size about $N^{2/5}$. A kernel with $m \nmid U_1$ would give $n > m^{4.5}/27$ by Corollary I.2.3 (with Remark I.2.2 when $n \equiv 2 \pmod{4}$), which exceeds N for large N ; so $m \mid U_1$, and $k = 1$, because $k \geq 3$ would give $n \geq T_1^3 > m^{9/2}$. Precisely, if $n \leq N$ and $m \geq (27N)^{2/9}$, then $m \mid U_1(m)$ and $k = 1$. So, under the assumption that n_2 and m_2 are squarefree, the case in which Tao's argument is weakest lies in the set counted by $E_3(N)$, the first rungs of kernels that divide U_1 . We do not claim the converse: E_3 also counts kernels of other sizes.

In words. The bounds cited above come from counting solutions of one equation. Tao says where his counting is weakest, and that place lies inside the kernels of the localization when the numbers n_2 and m_2 are squarefree.

6.4 How fast does the count grow?

Question II.6.7 (growth of the tower rate). [open] Does $c_M = \sum_{m \leq M} w(m)$, the sum over odd squarefree m with $T_1(m)$ even of $w(m) = 1/(2m' \log_{10} \varepsilon_m)$, stay bounded as $M \rightarrow \infty$?

The partial sums increase with M , so either they converge or they tend to infinity. We separate what is proved, what is measured and what is heuristic.

Proved and computed. Proposition II.6.2 writes $P_2(N; M)$ as an exact tower sum, and Corollary II.6.3 gives $P_2(10^h) \geq 0.595h - 122$. Neither answer to the question is excluded. An answer would not give an asymptotic formula for $P_2(N)$ by itself: that also needs to know how many kernels have a tower that reaches below N , and this is governed by the same unknown kernels.

Measured. Table 16 gives c_M for $M = 10, 10^2, \dots, 10^7$ in both classes. In both, the increase from one decade to the next becomes smaller. Single kernels matter: in the second class the kernel $m = 525\,003$, with $m' = 3$, alone gives about 45 % of the increase from $M = 10^5$ to $M = 10^6$.

$\log_{10} M$	$c_M, m \equiv 7 \pmod{8}$	$c_M, m \equiv 3 \pmod{8}$
1	0.0594	0.2914
2	0.1341	0.3666
3	0.1711	0.4027
4	0.1936	0.4212
5	0.2047	0.4366
6	0.2151	0.4513
7	0.2217	0.4566

Table 16: The partial sums c_M of Question II.6.7, over all kernels up to M with $T_1(m)$ even, without a height limit.

Heuristic, not a claim. The quotient m/m' is the product of the primes of m that divide U_1 . A simple model says that a prime $p \mid m$ divides U_1 with probability $1/p$. For the kernels up to 10^7 with T_1 even and the primes $p \leq 31$, the measured frequency divided by $1/p$ lies between 0.94 and 1.01 in the first class and between 0.91 and 1.04 in the second; $m' = m$ holds for 87.1 % and 86.3 % of the kernels. Under the model, with independence over the primes of m , and if $\log \varepsilon_m$ is of the order $\sqrt{m}$, the expected contribution of a kernel to c_M is about $m^{-3/2+o(1)}$, and the expected sum converges. The model is checked only for $p \leq 31$, and $\log \varepsilon_m$ is far below $\sqrt{m}$ in families such as $m = t^2 - 2$ (Remark I.8.3). The sum could still diverge through kernels with a small index m' ; whether they occur often enough is exactly what the question asks. The example above shows why more decades of the same computation would say little: a single such kernel can supply a large part of a decade.

In words. Every kernel adds a small amount to the rate at which pairs appear. The question is whether these amounts add up to a finite total. Our numbers grow more and more slowly, which is compatible with a finite total and equally with a slowly growing one; one kernel with a small index can change the picture, so the data do not decide it.

7 Distance one along the Pell route

Section 2 describes a pair $a, a+1$ by the two squarefree parts b and d . This section uses their product instead. That puts every pair on the equation $x^2 - Dy^2 = 1$ for one kernel D , as Part I does for the middles at distance two, and it makes the question of Part I easy to ask: a triple is a pair whose third neighbor is powerful as well.

Proposition II.7.1 (pairs on one Pell equation). [proved] *Let $a \geq 1$, $T = 2a + 1$ and $D = \text{sqfree}(a(a+1))$, and write $a(a+1) = DR^2$. Then $T^2 - D(2R)^2 = 1$, and a and $a+1$ are both powerful if and only if $D \mid R$.*

Proof. We have $T^2 - 1 = 4a(a+1) = 4DR^2$. Since a and $a+1$ are coprime, every prime factor of $a(a+1)$ divides exactly one of them, with the same exponent; so both are powerful if and only if $a(a+1)$ is. By Lemma II.2.1, $a(a+1) = DR^2$ is powerful if and only if $D \mid R$. $\square$

In words. Two neighbors are both powerful exactly when their product is. The product is a squarefree number D times a square R^2 , and it is powerful exactly when D divides R . So the pairs with a given D are found among the solutions of one Pell equation.

For a family (b, d) of Section 2 one has $D = bd$ and $R = XY$, so Proposition II.7.1 is Proposition II.2.2 written with one kernel instead of two. The solutions of $x^2 - Dy^2 = 1$ with x odd carry these pairs; the solutions with x even carry the middles at distance two of Lemma I.1.4. A triple of consecutive powerful numbers is a pair $a, a+1$ for which $a-1$ or $a+2$ is powerful as well, and one of the two is excluded at once: the even member of the pair is powerful, hence divisible by 4, so its neighbor at distance two is 2 more than a multiple of 4 and not powerful. We call the other one the *third number* of the pair.

Proposition II.7.2 (the third number). [computed]

- (a) *For the 12159 squarefree $D \leq 20\,000$, the pairs $a, a+1$ of Proposition II.7.1 with $2a+1 < 10^{2000}$ number 6056. The program ran each kernel one rung further, which adds the next pair above this bound for 7 kernels; together these are the 6063 pairs of (b) and (c). A direct search for consecutive powerful numbers up to 10^{12} , which does not use the equation, finds 18 pairs, and the route with $D \leq 10^5$ finds the same 18.*
- (b) *For 4991 of these pairs the third number has a prime factor $p < 5 \cdot 10^5$ with exponent 1, and for 154 more a prime factor $p < 5 \cdot 10^6$ with exponent 1. For the remaining 918, all prime factors below $5 \cdot 10^6$ occur at least twice. After removing them, the rest of the third number is a prime in 24 cases, 5 of them proved by a deterministic test [75] and the other 19, with up to 1687 digits, by elliptic curve certificates [31, 2] that PARI/GP [80] produced and a separate program of ours checked; in 15 further cases the elliptic curve method finds a prime factor with exponent 1.*
- (c) *So for all but 879 of the 6063 pairs a prime with exponent 1 is known for the third number, and it is proved prime. For the 879 others no witness is known; the smallest of their third numbers has 81 digits.*

The computation. (a) For each D we solved $x^2 - Dy^2 = 1$, took the powers $T_k + U_k\sqrt{D}$ with T_k odd below 10^{2000} , and kept those with $D \mid U_k$ and $T_k \equiv \pm 1 \pmod{8}$. For odd D the

second condition follows from the first; for $D = 2D_0$ the two together say $2D_0 \mid U_k/2$. In both cases they are equivalent to $D \mid U_k/2 = R$, so by Proposition II.7.1 these are exactly the pairs. The comparison in (a) was made up to 10^{12} with a separate program that lists all powerful numbers. (b) The third number is $a - 1 = (T - 3)/2$ or $a + 2 = (T + 3)/2$. For an odd prime p , the exponent of p in it is the exponent of p in $T \mp 3$, which we computed from T_k modulo p^2 . For the remaining pairs we computed the third number exactly, divided out all primes below $5 \cdot 10^6$ and tested the rest; a prime rest divides the number exactly once. As controls, the program finds for 5 pairs of the first group the prime found there, and for the pair with $(D, k) = (3, 24)$ it finds that the third number $a - 1$ is itself a prime. $\square$

In words. For every pair we tried to show that its third neighbor is not powerful, by finding a prime that divides it only once. This worked for all but 879 pairs. For those, we could not decide it; that does not make them triples. A triple among them with $2a + 1 < 10^{2000}$ would also have to avoid Part I: its middle is below 10^{2000} , so by Theorem I.3.1 its kernel in the sense of Part I would exceed 10^8 .

8 Open questions

The problems left open in this part are the following. Each is stated where it arises; we collect them here.

- (1) Erdős asked whether the number $N_1(x)$ of pairs $n, n + 1$ of powerful numbers with $n + 1 \leq x$ is bounded by a power of $\log x$ [12]. Proposition II.4.1 gives $N_1(x) \gg \log x$ with an explicit constant; nothing in this part bounds $N_1(x)$ from above.
- (2) Question II.5.3: does the sum of $1/\log F$ over all families converge?
- (3) Question II.6.7: does the rate c_M of the towers at distance two stay bounded?
- (4) The question below, which Corollary II.6.5 and Remark II.6.6 single out.

Question II.8.1 (the kernels that divide U_1). [open] Is there a $\theta < 61/180$ with $E_3(N) \ll N^\theta$? Since $E_3(N) \leq P_2(N)$, Blomer and Schöbel give $E_3(N) \ll N^\gamma$ for every $\gamma > 61/180$ [10, Theorem 3]; the question is whether the squarefree odd m with $m \mid U_1(m)$, $T_1(m)$ even and $T_1(m) \leq N$ are rarer than that.

By Corollary II.6.5(b), a positive answer gives $P_2(N) \ll N^{\max(2/9, \theta)}$, and for $\theta \geq 2/9$ it is equivalent to that bound. Either way it would improve the best exponent we know for distance two. Reuss's exponent $29/100$ is for distance one [63, Theorem 4], and for distance two the best exponent we know is the $61/180$ of Blomer and Schöbel. Remark II.6.6 shows that the case in which Tao's argument is weakest lies in E_3 when n_2 and m_2 are squarefree. What is known is small: the known kernels counted by E_3 are the members of Reinhart's list with $T_1(m)$ even. Their number below $1.5 \cdot 10^{12}$ is 1 [61, Remark 5.5], and below $5.325 \cdot 10^{13}$ it is 3 on his extended list [62], if that list is complete (Remark I.2.5); each of them has $T_1(m) > 10^{94}$ (Proposition I.6.3 for the class 7, and 99 134 digits for the kernel of class 3), and every other kernel has $T_1(m) > m^{3/2}$ by Corollary I.2.3. So $E_3(N) = 0$ for $N < 1.83 \cdot 10^{18}$, and for $N < 3.88 \cdot 10^{20}$ under that assumption. A bound for E_3 is a statement about fundamental units, of the kind that Part I also needs in its wing (ii).

In words. Four questions remain. Two ask whether rates that we computed add up to a finite total. One is Erdős’s question itself. The last asks how many kernels of the kind that divides U_1 there are; a positive answer would improve the best bound we know for pairs at distance two, and it is the same gap that Part I runs into.

The type of every statement

Every statement of this part carries one of six labels, printed after its number. The table lists them; it is generated from the text by the build and checked against a second, independent reading of the source.

statement	title	type
Lemma II.2.1	powerful numbers	proved
Proposition II.2.2	Walker [85, Sections 2 and 3]	literature
Proposition II.2.3	Walker [85, Theorems 2.2, 2.5, 3.2 and 3.5]	literature
Proposition II.3.1	the known pairs	computed
Proposition II.3.2	completeness inside the families	computed
Proposition II.4.1	explicit lower bound	computed
Definition II.5.1	comparison function; a heuristic	heuristic
Proposition II.5.2	the families with a square overtake the comparison function	computed
Question II.5.3	convergence of the rates	open
Proposition II.6.1	the middles that are not divisible by four	computed
Proposition II.6.2	the count is a sum over towers	proved
Corollary II.6.3	lower bound at every height	computed
Proposition II.6.4	localization of the pair count	proved
Corollary II.6.5	what the localization says, and what it does not say	proved
Remark II.6.6	where Tao’s argument is weakest	proved
Question II.6.7	growth of the tower rate	open
Proposition II.7.1	pairs on one Pell equation	proved
Proposition II.7.2	the third number	computed
Question II.8.1	the kernels that divide U_1	open

The types: *proved*, proved in this part; the proof may use published results (6); *literature*, due to the cited source, sometimes with a proof given here for completeness (2); *computed*, the result of an exact computation, or a proof that uses one of our computations (7); *heuristic*, a heuristic, used as a yardstick only (1); *open*, an open question (3).

Part III: further observations

Introduction

Parts I and II study triples and pairs of consecutive powerful numbers. Along the way we measured and proved a number of things that sit at the edge of these two questions and belong to neither. This part collects them. None of them is an attack on the questions of Erdős, and several are consequences or tests of published results; where that is so, the source is named at the statement.

What this part contains. Section 1 looks at the gaps between consecutive powerful numbers: the most frequent gap values up to 10^{14} are powerful, although almost no gap value is, and an elementary lemma gives a mechanism that favors powerful gaps (Proposition III.1.2, Remark III.1.3). Section 2 treats powerful values of cyclotomic polynomials, which arise when a member of a triple is a proper power: two explicit infinite chains, each giving powerful values of both Φ_3 and Φ_6 (Proposition III.2.2), a table up to 10^{12} , and the case $x^4 + 1 = q^3 a^2$ (Proposition III.2.5). Section 3 tests the independence assumption of Part I along the divisors of the index, and states what the test can and cannot see. Section 4 measures the Wieferich fractions of Pell units, the quantity behind the rare events of Part I, against the equidistribution that a conjecture of Katz predicts. Section 5 states two questions.

How to read this part. Every statement is followed by a paragraph *In words*. Every numbered statement carries a label that says whether it is proved, taken from the literature, computed, measured, a heuristic, or open (see the table at the end); in this part most statements are computations or measurements on a finite range, and they are stated as such. A statement that combines several kinds carries the label of the part that needs the most trust: a proof that uses our own computation is labeled computed, and a proof that uses a published result is labeled proved. Every number we computed is generated from the data files of the deposit.

1 The gap spectrum of powerful numbers

Write the powerful numbers in increasing order and call the difference g of two consecutive members a *gap*; the sequence of gaps is OEIS A076446 [57]. For an integer $g \geq 1$ let $\text{sqfull}(g)$ be the product of the prime powers $p^{v_p(g)}$ with $v_p(g) \geq 2$. Then $\text{sqfull}(g) = g$ holds exactly when g is powerful. Up to 10^{14} there are 21 663 503 powerful numbers and 8 235 300 distinct gap values, and only 0.10 % of these values are powerful; most of them are large and occur once. Among the 62 118 values that occur at least 20 times, 4.81 % are powerful. This section shows that the most frequent gap values are nevertheless powerful, and gives a mechanism that favors them.

Lemma III.1.1 (Bajpai, Bennett and Chan [3, proof of Theorem 1.1]). [literature] *Let n and $n+g$ be powerful. Then $\gcd(n, g)$ is powerful and divides $\text{sqfull}(g)$. In particular $\gcd(n, g) = 1$ if g is squarefree, and $g \mid n$ is possible only if g is powerful.*

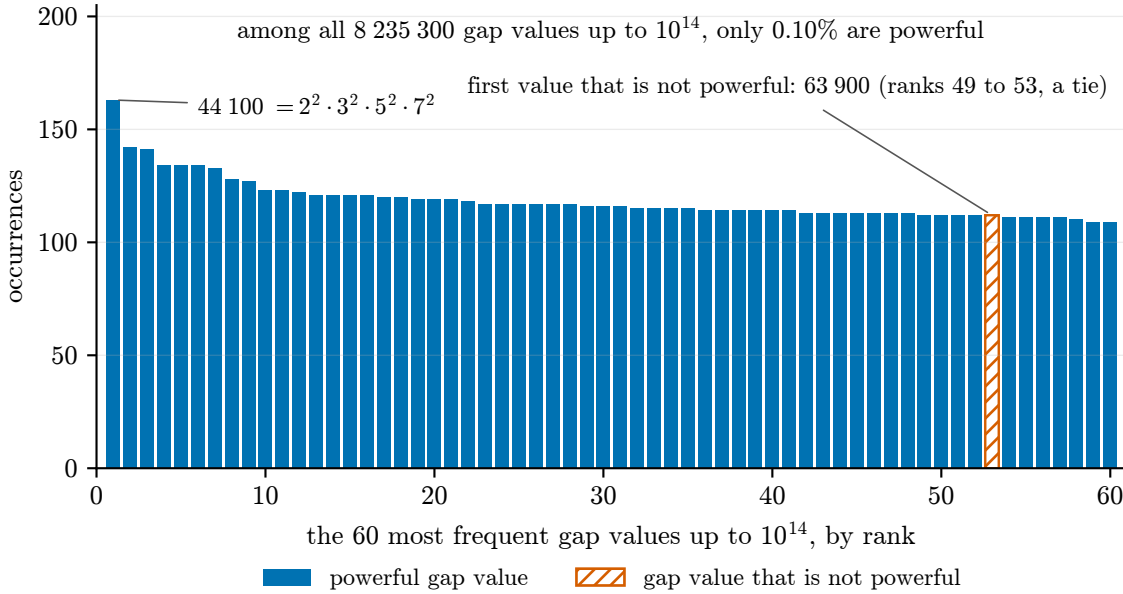


Figure 8: **Up to 10^{14} , the 48 most frequent gap values are all powerful, although only 0.10 % of all gap values are.** The sixty most frequent values of the gap between consecutive powerful numbers up to 10^{14} , by their number of occurrences; values with the same number share their rank and are ordered by size. The first 48 values are all powerful; the first value that is not powerful, 63 900, shares its number of occurrences with powerful values (Proposition III.1.2). Among all 8 235 300 gap values up to 10^{14} , only 0.10 % are powerful. The all-powerful head is shorter at 10^{14} than at 10^{12} (Table 18), so the picture says nothing about larger bounds (Question III.5.1).

Proof. If a prime p divides n and g , it divides $n + g$, so p^2 divides n and $n + g$ and hence their difference g . So every prime of $\gcd(n, g)$ occurs in it at least twice, and with an exponent at most $v_p(g)$, where $v_p(g) \geq 2$. If $g \mid n$, then $\gcd(n, g) = g$ is powerful. $\square$

That $\gcd(n, g)$ is powerful is used, for k -full numbers and without a separate proof, by Bajpai, Bennett and Chan immediately before their equation (4.2); the consequence for $\text{sqfull}(g)$ is immediate, and it is what we use.

In words. If two powerful numbers are a distance g apart, every prime they share with g appears in g at least twice. So the part of g that such a pair can share is its powerful part; for a gap without square factors there is nothing to share.

Proposition III.1.2 (the head of the gap spectrum). [computed] *Up to 10^{14} the 48 most frequent gap values are all powerful; they are the values that occur more than 112 times. The first value that is not powerful is $63\,900 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 71$, with 112 occurrences, and it is the only one among the 103 values that occur at least as often as the hundredth most frequent. Among the 201 values that occur at least as often as the two hundredth, 191 are powerful. Table 18 gives the same count at 10^{12} and 10^{13} .*

bound	powerful numbers	gap values	% of values powerful	most frequent	all powerful to rank	first value not powerful	powerful near rank 200
10^{12}	2 158 391	832 161	0.31	900	69	1656	183 of 215
10^{13}	6 840 384	2 593 517	0.18	44 100	57	63 900	189 of 211
10^{14}	21 663 503	8 235 300	0.10	44 100	48	63 900	191 of 201

Table 18: The gap values between consecutive powerful numbers up to three bounds (Proposition III.1.2). Values of equal frequency share a rank; the column “all powerful to rank” counts the values that occur more often than the first value that is not powerful, and the last column counts the powerful values among those that occur at least as often as the two hundredth most frequent.

If the 50 most frequent values were drawn at random from all gap values up to 10^{14} , about 0.05 of them would be powerful, and about 2.4 if they were drawn from the values that occur at least 20 times. The most frequent value at 10^{13} and 10^{14} is $44\,100 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2$.

In words. Almost no gap value is powerful, but nearly all of the frequent ones are. From 10^{12} to 10^{14} the share of powerful values near the top rises, while the first exception moves up the list; the effect does not simply grow or fade with the bound.

Remark III.1.3 (a ceiling for common divisors). [computed] By Lemma III.1.1, a pair at gap g shares at most $\text{sqfull}(g)$ with g , and only a powerful g can share all of itself. Table 19 lists the 10 values that are not powerful among the 201 values of Proposition III.1.2. Each has a small powerful part that is itself a frequent gap value; values of equal frequency share a rank, so the rank is given as a range. The share of occurrences in which the pair shares the full ceiling is 24.2–44.9% for the 21 values that occur at least as often as the twentieth most frequent, and 20.8–61.8% for these 10. So both sides are consistent with one mechanism, a pair sharing the largest divisor that the lemma allows; only the size of that divisor differs.

g	$\text{sqfull}(g)$	rank of $\text{sqfull}(g)$	occurrences of g	% with $\text{gcd} = \text{sqfull}(g)$
63 900	900	9	112	50.9
82 908	1764	36–41	98	49.0
17 100	900	9	97	43.3
23 324	1372	170–175	96	20.8
151 900	4900	10–11	90	45.6
171 900	900	9	90	52.2
7100	100	357–381	89	61.8
26 100	900	9	89	37.1
127 800	1800	29–31	89	53.9
215 100	900	9	89	51.7

Table 19: The frequent gap values up to 10^{14} that are not powerful (Remark III.1.3): all values that are not powerful and occur at least as often as the two hundredth most frequent.

In words. About half of the pairs at gap 63 900 share the factor 900 with the gap, and 63 900 occurs nearly as often as 900 itself. A powerful gap can share all of itself with a pair; this is one reason why the head of the spectrum is powerful, not a proof that it must be.

Remark III.1.4 (a counting model, measured). [measured] Write $g = \text{sqfull}(g) \cdot c$ with c squarefree, and let $\text{occ}(g)$ be the number of occurrences of g up to 10^{14} . Over the 62 118 gap

values with at least 20 occurrences, which have 2986 distinct parts $\text{sqfull}(g)$ and 9752 distinct cofactors c , we fitted $\log \text{occ}(g) = u(\text{sqfull}(g)) + v(c)$ with one free value per part and per cofactor. Evaluated on the values held back from the fit, in five random splits of about four fifths to one fifth over the gap values, with the mean effect used for a part or cofactor that the fit has not seen and an affine recalibration fitted on the training part, the model accounts for 0.545 ± 0.025 of the variance of $\text{occ}(g)$ and 0.514 ± 0.016 of the variance of $\log \text{occ}(g)$, against 0.306 ± 0.005 for a regression of $\text{occ}(g)$ on plain arithmetic features of g on the same values; the margin is positive in each split. The part alone accounts for 0.012, the cofactor alone for 0.272, and on the fitted values the product $u \cdot v$ is uncorrelated with the residuals ($r = 0.001$). The values of v are reproducible up to a reliability of 0.864, which bounds what any explanation of v can reach; on the 3061 cofactors with at least 4 gap values, the size and the prime factors of c account for 0.425 of v on held-back cofactors in one split, against a ceiling of 0.898 measured on the same cofactors.

The lemma therefore explains why the most frequent values can be powerful, not how often a value occurs: in this model the powerful part alone accounts for 0.012 of the variance and the squarefree rest for 0.272. This is a measurement on a finite range and not a statement about all gaps. A version that splits g over all its powerful divisors instead of the canonical one fits the counts of the observed pairs (g, gcd) closely, but does not predict new gap values significantly better than the plain arithmetic features (0.319 ± 0.012 against 0.307 ± 0.004), because about half of its cofactors occur at a single gap; the record of this negative result is part of the deposit.

In words. For the gap values that occur at least 20 times, knowing the powerful part and the squarefree rest of a gap predicts about half of the variation in how often it occurs, and the rest matters more than the powerful part. The prediction was tested on gaps that the fit had not seen. It describes the data up to 10^{14} ; it is not a law.

Remark III.1.5 (common factors of pairs). [measured] In a sample of every 54th consecutive pair up to 10^{14} , 401 176 pairs in total, the two members are coprime in 47.69% of the cases. The most frequent common divisors are 4, 9, 8, 16, 25, 36, 27, 32, all powerful, as Lemma III.1.1 requires.

In words. About half of the neighboring pairs of powerful numbers share a factor, and the shared factor is always powerful.

We also computed the number variance of the powerful numbers in windows of up to 50 mean spacings. It lies within about 1.2% of a sum of independent lattice variances, and the large-window form of that sum grows like $L^{2/3}$ with the constant of Chan's theorem [20, Theorem 2]; in our windows the measured growth, a slope of 0.63 on a logarithmic scale, has not yet reached the exponent $2/3$. This is a check of Chan's theorem and not a new result.

Prior art and its reach. Apart from the lemma, we found no work on the distribution of the gap values between powerful numbers. We checked the work of Bajpai, Bennett and Chan in full, the OEIS entries A076446 and A001694, and records on Zenodo; the citing works of Bajpai, Bennett and Chan were queried in two databases, which returned three and none. The reach of this search is set by these databases.

2 Powerful values of cyclotomic polynomials

Remark I.4.13 explains where these values come from. If a member of a triple is a proper power x^r with $r \geq 2$, its neighbors $x^r - 1$ and $x^r + 1$ are products of values $\Phi_e(x)$ and $\Phi_e(-x)$ of cyclotomic polynomials. Two such values share only primes that divide $2r$, so a value that is coprime to the others must itself be powerful. This section records what we know about powerful values $\Phi_d(x)$ at integers $x \geq 2$. For $d = 3, 4$ and 6 the polynomial is quadratic, and the question belongs to the study of powerful values of quadratic polynomials by De Koninck, Doyon and Luca [26].

Remark III.2.1 (the values $x^2 + 1$). [computed] A powerful number that is not a square can be written as a^2b^3 with $b > 1$ squarefree, so $x^2 + 1 = \Phi_4(x)$ is powerful exactly when $x^2 - b^3y^2 = -1$ has a solution for some squarefree $b > 1$; this is the form (1)–(2) of De Koninck, Doyon and Luca [26, p. 2], and their Table 1 gives the smallest case, $682^2 + 1 = 5^3 \cdot 61^2$ [26, p. 8]. For each such b the solutions form an infinite family, the odd powers of the smallest one. We solved the equation for all 182 squarefree b with $1 < b \leq 300$: it has a solution for 34 of them. The list of OEIS A175155 [58] of the values x with $x^2 + 1 = a^2b^3$ begins 682, 1 268 860 318, 1 459 639 851 109 444.

In words. The number $x^2 + 1$ is powerful for infinitely many x , but such x are very sparse: after 682 the next value in the list of OEIS A175155 is above a billion. Each of them comes from a Pell-type equation, and only some of these equations have solutions.

Proposition III.2.2 (two explicit chains of powerful values of Φ_3 and Φ_6). [proved] *Let the map $(y, m) \mapsto (127y + 672m, 24y + 127m)$ act on the starting points $(37, 7)$ (the first chain) and $(5, 1)$ (the second chain), and let (y_j, m_j) be the point after j steps. In both chains $y_j^2 - 28m_j^2 = -3$ and y_j is odd for every j ; $7 \mid m_j$ exactly when $j \equiv 0 \pmod{7}$ in the first chain and exactly when $j \equiv 6 \pmod{7}$ in the second; and the two chains contain every solution of $y^2 - 28m^2 = -3$ with $y, m > 0$. For every j with $7 \mid m_j$ the number $x = (y_j - 1)/2$ satisfies $\Phi_3(x) = 343(m_j/7)^2$, which is powerful, and $\Phi_6(x + 1) = \Phi_3(x)$. In particular $\Phi_3(x)$ and $\Phi_6(x)$ are powerful for infinitely many x .*

Proof. Since $127^2 - 28 \cdot 24^2 = 1$, the map preserves $y^2 - 28m^2$, and $37^2 - 28 \cdot 7^2 = 5^2 - 28 \cdot 1^2 = -3$. It preserves the parity of y , because 672 is even and 127 is odd. Modulo 7 it fixes y , because $672 \equiv 0$ and $127 \equiv 1$, and it adds $24y \equiv 3y$ to m . In the first chain $y_0 \equiv 2$ and $m_0 \equiv 0 \pmod{7}$, so $m_j \equiv 6j \pmod{7}$, which vanishes exactly when $j \equiv 0 \pmod{7}$; in the second $y_0 \equiv 5$ and $m_0 = 1$, so $m_j \equiv 1 + 15j \equiv 1 + j \pmod{7}$, which vanishes exactly when $j \equiv 6 \pmod{7}$. For the completeness, let $\alpha = y + z\sqrt{7}$ with $y, z > 0$ and $y^2 - 7z^2 = -3$. Its conjugate is $-3/\alpha$, so $2y = \alpha - 3/\alpha > 0$ gives $\alpha > \sqrt{3}$. Dividing by $\varepsilon = 8 + 3\sqrt{7}$ keeps the norm, and keeps the quotient above $\sqrt{3}$ as long as $\alpha > \sqrt{3}\varepsilon$; so $\alpha = \beta\varepsilon^n$ with $n \geq 0$ and $\sqrt{3} < \beta \leq \sqrt{3}\varepsilon$. For $\beta = y' + z'\sqrt{7}$ this gives $z' = (\beta + 3/\beta)/(2\sqrt{7}) < \sqrt{3}(\varepsilon + 1)/(2\sqrt{7}) < 6$, and among $1 \leq z' \leq 5$ only $z' = 1$ and $z' = 2$ make $7z'^2 - 3$ a square, so $\beta = 2 + \sqrt{7}$ or $\beta = 5 + 2\sqrt{7}$. Multiplying by ε replaces (y, z) by $(8y + 21z, 3y + 8z)$, which swaps the parities of y and z ; so $z = 2m$ is even exactly for $(2 + \sqrt{7})\varepsilon^n$ with n odd and for $(5 + 2\sqrt{7})\varepsilon^n$ with n even. Since $(2 + \sqrt{7})\varepsilon = 37 + 14\sqrt{7}$ and $\varepsilon^2 = 127 + 48\sqrt{7}$, these are the two chains, and multiplying by ε^2 is the map. Finally $4\Phi_3(x) = (2x + 1)^2 + 3 = y_j^2 + 3 = 28m_j^2$, so $\Phi_3(x) = 7m_j^2$, and $\Phi_6(x + 1) = (x + 1)^2 - (x + 1) + 1 = \Phi_3(x)$. $\square$

In the first chain the first member is $x = 18$ with $\Phi_3(18) = 7^3$; the next three have 19, 35 and 52 digits. The value $x = 18$ is the entry $k = 3$, $n = 37$ of Table 1 of De Koninck, Doyon and Luca, since $37^2 + 3 = 4\Phi_3(18)$ [26, p. 8], and they observe that for a quadratic polynomial one powerful value of this kind gives infinitely many [26, p. 7]. Their reduction on that page, applied to Φ_3 with squarefree part 7, leads in our normalization to the equation $y^2 - 28m^2 = -3$ with $7 \mid m$. We did not find there the explicit recurrence, the two chains and their completeness, the parity of y_j , the period 7, or the transfer to Φ_6 . A direct search over all $m \leq 10^6$ finds no solution outside the two chains, as the proposition says. The two chains are disjoint: going forward from $(5, 1)$ the values of y are $5, 1307, \dots$, going backward they are negative from $(-37, 7)$ on, so $(37, 7)$ is not on the second chain. The prime x with $\Phi_3(x) = 7^3 a^2$ reported by Lavi [40], which we know from the description of his record, is the member $j = 13$ of the second chain. Other squarefree parts give other powerful values: the second value for $d = 3$ in Table 20 comes from the equation $y^2 - 52m^2 = -3$ with $13 \mid m$, that is, with 13 in place of 7.

In words. The polynomials $x^2 + x + 1$ and $x^2 - x + 1$ take powerful values infinitely often. A Pell equation produces them one after another, in two chains, and every seventh step of each chain gives one. The idea that one solution gives infinitely many is in the literature; we did not find the explicit chains there.

Proposition III.2.3 (powerful values up to a bound). [computed] *Among the 3 004 199 pairs (d, x) with $3 \leq d \leq 12$, $x \geq 2$ and $\Phi_d(x) \leq 10^{12}$, exactly 6 give a powerful value $\Phi_d(x)$. They are listed in Table 20.*

d	x	$\Phi_d(x)$	factorization
3	18	343	7^3
3	88 916	7 906 143 973	$7^2 \cdot 13^3 \cdot 271^2$
4	682	465 125	$5^3 \cdot 61^2$
5	3	121	11^2
6	19	343	7^3
6	88 917	7 906 143 973	$7^2 \cdot 13^3 \cdot 271^2$

Table 20: The powerful values $\Phi_d(x)$ with $3 \leq d \leq 12$ and $\Phi_d(x) \leq 10^{12}$ (Proposition III.2.3). The pairs for $d = 3$ and $d = 6$ are related by $\Phi_6(x + 1) = \Phi_3(x)$.

In words. Up to the bound, powerful cyclotomic values occur only for $d = 3, 4, 5$ and 6, and only 6 times. The case $d = 5$ is $3^4 + 3^3 + 3^2 + 3 + 1 = 121 = 11^2$, a square.

Remark III.2.4 (a class without squares). [computed] The numbers $x_n = (x^n - 1)/(x - 1)$ satisfy $x_{n+2} = (x + 1)x_{n+1} - xx_n$, a recurrence with $b = x + 1$ and $c = -x$ coprime and $b^2 + 4c = (x - 1)^2 > 0$. If $x \equiv 2 \pmod{4}$, then $c \equiv 2 \pmod{4}$, and Granville's theorem [34, Theorem 3] gives, for every $d \notin \{1, 2, 3, 6\}$, a primitive prime factor of x_d that divides it to an odd power. Such a prime divides $\Phi_d(x)$ and no $\Phi_e(x)$ with $e \mid d$, $e < d$, so $\Phi_d(x)$ is not a square. It can still be powerful, since an odd exponent can be 3. We checked the consequence: for $x \equiv 2 \pmod{4}$, $d \in \{4, 5, 7, \dots, 12\}$ and $\Phi_d(x) \leq 10^{14}$ no value is a square, while outside the class the square $\Phi_5(3) = 11^2$ appears at once, and for these degrees it is the only square outside the class up to 10^{14} . Inside the class, $\Phi_4(682) = 5^3 \cdot 61^2$ is powerful without being a square.

In words. For arguments of the form $4j+2$, a theorem of Granville shows that the cyclotomic values are never squares, apart from four small degrees. Our check up to 10^{14} agrees with it, and the example $\Phi_5(3) = 121$ shows that the condition on the argument cannot simply be dropped. The theorem says nothing about cubes, so it does not decide whether such a value can be powerful.

Proposition III.2.5 (the simplest case of Φ_8). [proved] *There are no positive integers x and a and no prime $q < 2 \cdot 10^{11}$ with $x^4 + 1 = q^3 a^2$.*

Proof. For $q = 2$ there is nothing to prove, since $x^4 + 1 \equiv 1$ or $2 \pmod{16}$ while $8a^2 \equiv 0$ or $8 \pmod{16}$. Let q be odd and write $x^4 + 1 = qy^2$ with $y = qa$. By Cohn [24, p. 1347 and Theorem 1], a solution requires $q \equiv 1 \pmod{8}$ and a solution of $v^2 - qu^2 = -1$; if $\alpha = c + b\sqrt{q}$ is the smallest one, $u_n = (\alpha^n - \bar{\alpha}^n)/(2\sqrt{q})$ and $v_n = (\alpha^n + \bar{\alpha}^n)/2$, then $x^2 = v_A$ and $y = u_A$, where A is the squarefree part of c . Expanding α^n gives $u_n \equiv n c^{n-1} b \pmod{q}$. Since $c^2 \equiv -1 \pmod{q}$, the prime q divides neither c nor A , so $q \mid y = u_A$ forces $q \mid b$. For $q \equiv 1 \pmod{8}$ the fundamental unit $(t + w\sqrt{q})/2$ of $\mathbb{Q}(\sqrt{q})$ has t and w even, because $t^2 - qw^2 = \pm 4$ is impossible with t and w odd; so it lies in $\mathbb{Z}[\sqrt{q}]$, equals α , and $b = w/2$. Hence a solution requires $q \mid w$. The conjecture of Ankeny, Artin and Chowla says that this never happens, and it has been verified for all primes $q \equiv 1 \pmod{4}$ below 10^{11} [81] and between 10^{11} and $2 \cdot 10^{11}$ [82]. $\square$

We checked the last step independently for the 2384 primes $q \equiv 1 \pmod{8}$ below 10^5 : in none of them does q divide b . The proposition covers only the case in which the squarefree part of $x^4 + 1$ is a single prime. For small squarefree parts the literature settles the question completely: Cohn solved $x^4 + 1 = Dy^2$ for all squarefree $D < 10^5$, and apart from $D = x^4 + 1$ with $y = 1$ the only solution is $261^4 + 1 = 70\,258 \cdot 257^2$ [24, abstract]. If $x^4 + 1$ is powerful and D is its squarefree part, then $D \mid y$; neither case allows this, so the squarefree part of a powerful $x^4 + 1$ is at least 10^5 .

In words. The number $x^4 + 1$ is the eighth cyclotomic value. If it were a cube of a prime times a square, a theorem of Cohn would force a very special divisibility in the smallest solution of a Pell equation, and that divisibility is exactly what a well-tested conjecture forbids; computer searches by others confirm it far enough to exclude all primes below $2 \cdot 10^{11}$.

3 An independence test along the divisors of the index

The heuristic of Part I treats the event that a prime of rank d divides $T_d(m)$ at least twice as a rare event of probability about $1/p$, independent of other such events (Proposition I.6.1). Part I tests independence across the primes of one kernel. A lattice point (m, k) with many divisors carries one such event for each divisor d of k , and the divisors are linked: qd is a multiple of d . This section tests whether the events at d and at qd are coupled.

Remark III.3.1 (no joint event along the divisors, and the power of the test). [measured] For the 25 kernels with the most lattice points in the box of Theorem I.3.1, we took each rank d and its smallest prime p of rank d below $2 \cdot 10^6$, and set the status of d to 1 if $p^2 \mid T_d(m)$ and to 0 otherwise. We then formed all pairs (d, qd) with $q \in \{3, 5, 7, 11, 13\}$ for which both ranks have such a prime: 294 920 pairs. The status is 1 at d alone in 38 pairs, at qd alone in 8, and at both in 0; independence predicts 0.001 pairs with both. To see what this test

could detect, we simulated a coupling in which the status at qd copies the status at d with probability ρ , independently for each pair: for $\rho = 0.1$ one expects about 3.8 copied pairs, and in at least 98 % of 400 simulations at least one joint event appeared, against less than 1 % for $\rho = 0$.

In words. If the rare event at one rank made the same event at a multiple rank more likely, pairs with the event at both ranks would appear. There are none. The test is not blind: a coupling in which one time in ten the second event copies the first would almost always have shown up, and one of half that strength would still have shown up in most runs. A much weaker coupling, or one that makes the second event less likely, could not be seen with so few events, and the test uses only the smallest prime of each rank.

The test is formally a χ^2 test on the table, but with fewer than one expected joint event the statistic only records whether a joint event occurs; the simulation measures exactly that. One event at a rank d enters a pair for each q with qd occupied, so the 38 pairs come from fewer distinct events; the simulation copies pair by pair and therefore overstates the power against a coupling that acts once per event. In the prior-art search for this part (arXiv, Zenodo, Google Scholar and OEIS, at the level of titles and abstracts) we found no earlier test of independence along the divisors of the index.

4 Wieferich fractions of Pell units

For a squarefree m and a prime $p \nmid 2m$ let $e = \left(\frac{m}{p}\right)$ and write $\varepsilon_m^{p-e} = A + B\sqrt{m}$ modulo p^2 . Then $A \equiv 1$ and $B \equiv 0 \pmod{p}$, and we call $y_p = ((B/p) \bmod p)/p \in [0, 1)$ the *Wieferich fraction* of ε_m at p . It vanishes exactly when the order of ε_m modulo p^2 equals its order modulo p ; for a prime with a rank this is the condition $p^2 \mid T_{\alpha_m(p)}(m)$ of Part I. In Katz's language [36], ε_m is a point on the torus $x^2 - my^2 = 1$, and y_p is its Wieferich fraction for the framing whose Lie lattice is $\mathbb{Z}\sqrt{m}$, which Katz describes [36, Section 6]. His Framed Wieferich Conjecture [36, Conjecture 6.1] predicts that such fractions are equidistributed.

Remark III.4.1 (no deviation from equidistribution, memory or coupling). [measured] For the 25 kernels of Section 3 and all primes $p < 3 \cdot 10^6$ with $p \nmid 2mU_1$, that is, for 5 420 294 pairs (m, p) , we computed y_p .

- (a) *Near zero.* For six thresholds t from 10^{-2} to 10^{-7} the number of $y_p < t$ agrees with the exact expectation for uniform y_p ; the largest deviation is 1.37 standard deviations. For example, $y_p < 10^{-4}$ holds 593 times against 583.9 expected, and $y_p = 0$ holds 56 times against $\sum 1/p = 50.71$. Only 16 of these zeros lie at primes that divide some $T_k(m)$, which happens exactly when the order of ε_m modulo p is divisible by 4; they are the Wieferich events of Part I on these kernels (Remark I.6.11), and the other zeros do not appear there. Split by the 2-adic valuation of the order of ε_m modulo p (odd, 2 mod 4, 0 mod 4) and by e into 6 classes, the smallest p -value of a χ^2 test of uniformity is 0.014, or 0.08 after the Bonferroni correction.
- (b) *Along the primes.* For each kernel, the correlation of y_p with the fraction j primes later, for $j = 1, \dots, 20$, gives 500 tests; 7 exceed $|z| = 2.576$ against about 5 expected, the largest is 3.30, with Bonferroni $p = 0.48$. A memory planted in test data, with 5 percent of the values replaced by their predecessor, is found with $z = 21.4$.

- (c) *Between kernels.* For the 300 pairs of kernels, the correlation of their fractions at common primes has largest $|z| = 3.61$, with Bonferroni $p = 0.09$; a planted correlation of strength 0.02 is found with $z = 10.0$.

In words. For every unit and every prime there is a number between 0 and 1 that measures how close the prime comes to being a Wieferich prime for that unit. Over more than five million such numbers, on 25 units and the primes below $3 \cdot 10^6$, our tests find no deviation from independent uniform random numbers: near zero, from one prime to the next, and from one unit to another. Katz's conjecture predicts (a); applied to the product of two such tori it also predicts (c); (b) goes beyond it. The only computation of Wieferich fractions that his paper mentions is one for elliptic curves, by Bombieri, reported as compatible with the conjecture. Censuses of the zeros $y_p = 0$ exist for single units, for example Ryan's for $1 + \sqrt{2}$ up to 10^9 [67, Sections 2.1 and 5.1]; we did not find a test of the distribution of the fractions themselves, near zero, along the primes or between units. A finite test cannot prove equidistribution; it shows no deviation at the sizes the controls can detect.

5 Questions

Question III.5.1 (the most frequent gap). [open] Let $g(x)$ be the most frequent gap between consecutive powerful numbers up to x . Is $g(x)$ powerful for all large x , and does the share of powerful values among the N most frequent gaps up to x tend to 1 when N is fixed and $x \rightarrow \infty$?

Proposition III.1.2 and Table 18 are consistent with the first part at three bounds up to 10^{14} . For the second part they point both ways: the share of powerful values among the two hundred most frequent rises, while the leading block in which all values are powerful shrinks. Lemma III.1.1 gives powerful gaps an advantage; it does not show that the advantage wins in the limit.

In words. Is the most common distance between neighboring powerful numbers always itself powerful, however far one looks? Up to 10^{14} it is, but the data do not show a clear trend for the values just below the top.

Question III.5.2 (powerful values of $x^4 + 1$). [open] Is $x^4 + 1$ powerful for some integer $x \geq 1$?

By Proposition III.2.5 and the paragraph after it, the squarefree part of such a value is at least 10^5 , and it is not a prime below $2 \cdot 10^{11}$. For $x^2 + 1$ the answer is yes (Remark III.2.1).

In words. Can the eighth cyclotomic value $x^4 + 1$ ever be powerful? The fourth one, $x^2 + 1$, can. For $x^4 + 1$ only special shapes are excluded so far.

Remark III.5.3 (a conjecture on the most frequent gap). [heuristic] We state a weak and a strong form for the most frequent gap $g(x)$ of Question III.5.1. *Weak form:* $g(x)$ tends to infinity with x , and $g(x)$ is powerful for all large x . *Strong form:* for all large x , $g(x)$ is the square of a product of the first primes, $(2 \cdot 3 \cdots p_r)^2$, with r growing with x . The strong form implies the weak one; the weak one claims only what the data and Lemma III.1.1 support directly, and the strong one also names the shape. Up to 10^{12} the most frequent value is $900 = (2 \cdot 3 \cdot 5)^2$, and up to 10^{13} and 10^{14} it is $44100 = (2 \cdot 3 \cdot 5 \cdot 7)^2$ (Table 18). The reason we expect it is Lemma III.1.1: if $g = P^2$ with P squarefree and $P^2 \mid n$, then $n = P^2m$ and

$n + g = P^2(m + 1)$, so the primes of P already occur at least twice in both numbers, and only the other primes of m and $m + 1$ must occur squared. A larger P frees more primes but leaves fewer $m \leq x/P^2$; the smallest primes free the most for the smallest cost, and as x grows the balance moves to a larger P . A balance of this kind is behind the conjecture of Odlyzko, Rubinstein and Wolf that the most frequent gap between consecutive primes eventually runs through the products of the first primes, 30, 210, 2310 [53]; we found no such statement for powerful numbers. Two values do not make a trend. The weak form fails if $g(x)$ stays bounded, or if a value that is not powerful is the most frequent gap for arbitrarily large x . The strong form fails in addition if a powerful value of another shape, such as 3600 or 3528, which follow 900 and 44 100 closely in the counts, is the most frequent gap for arbitrarily large x . Which of the two forms is right is open.

In words. We expect the most common distance between neighboring powerful numbers to keep growing and to stay powerful (weak form), and we suspect that it is always the square of $2 \cdot 3 \cdot 5 \cdots$, as 900 and 44 100 are (strong form). For the primes a similar pattern is conjectured, with 30, 210, 2310. This is an expectation supported by two values and a mechanism, not a result.

The type of every statement

Every statement of this part carries one of six labels, printed after its number. The table lists them; it is generated from the text by the build and checked against a second, independent reading of the source.

statement	title	type
Lemma III.1.1	Bajpai, Bennett and Chan [3, proof of Theorem 1.1]	literature
Proposition III.1.2	the head of the gap spectrum	computed
Remark III.1.3	a ceiling for common divisors	computed
Remark III.1.4	a counting model, measured	measured
Remark III.1.5	common factors of pairs	measured
Remark III.2.1	the values $x^2 + 1$	computed
Proposition III.2.2	two explicit chains of powerful values of Φ_3 and Φ_6	proved
Proposition III.2.3	powerful values up to a bound	computed
Remark III.2.4	a class without squares	computed
Proposition III.2.5	the simplest case of Φ_8	proved
Remark III.3.1	no joint event along the divisors, and the power of the test	measured
Remark III.4.1	no deviation from equidistribution, memory or coupling	measured
Question III.5.1	the most frequent gap	open
Question III.5.2	powerful values of $x^4 + 1$	open
Remark III.5.3	a conjecture on the most frequent gap	heuristic

The types: *proved*, proved in this part; the proof may use published results (2); *literature*, due to the cited source, sometimes with a proof given here for completeness (1); *computed*, the result of an exact computation, or a proof that uses one of our computations (5); *measured*, a

measurement, with a model that is not proved (4); *heuristic*, a heuristic, used as a yardstick only (1); *open*, an open question (2).

Data and code availability

Every number of this volume that we computed is generated from a result file of a data deposit or from a cited constant; no computed number is typed by hand. The data deposit is archived with the collected volume, DOI 10.5281/zenodo.23127547.

What the deposit contains. The deposit has two parts. The core holds what the text cites: (i) the result files, in JSON and plain text, behind every computed number, together with the raw output of the scripts that wrote them; (ii) the scripts, written in Python, each stating before it runs what it expects to find and the positive and negative controls that it runs; (iii) a manifest with the SHA-256 hash and the size of every file, which the build checks before it reads a result file; (iv) the decimal expansions of the building blocks of Part I that the searches left open (Proposition I.6.8), including those that the algebraic split of Remark I.6.13 settles; (v) the certificates and the second routes for the computations that settle a building block; and (vi) the file `OPEN_PROBLEMS.md`, which lists the open problems of the volume. The archive holds the remaining scripts and results of the investigation, including negative results that the text does not print.

Where the files are. The collected volume and its three parts are published as four linked records on Zenodo, and the same files are in one repository on GitHub. The record of the collected volume (DOI 10.5281/zenodo.23127547) holds the volume and the whole deposit, the core and the archive described above. The record of each part holds that part and the result files that its text cites, and links to the record of the volume. The repository (github.com/berkay-yueksel-sayim/erdos-364-365-powerful-numbers-pell) holds the four PDFs and the deposit, and its README lists the four records. The records of Parts I, II and III have the DOIs 10.5281/zenodo.23127703, 10.5281/zenodo.23127705 and 10.5281/zenodo.23127707.

How the numbers get into the text. The source of the text refers to each computed number by a key. A build tool reads the result file named for the key, verifies it against the manifest, computes the number in its printed form, and writes it into the text; it stops if a key has no source or if a file differs from the manifest. The same tool generates the table of the types of the statements and the list of open problems from the source of the text, and it is part of the core.

Software. The scripts use Python 3.12 with gmpy2 2.3, SymPy 1.14 and NumPy 2.4. The elliptic curve method and its relatives are run with GMP-ECM 7.0.6 [90], used as an installed program inside a container image built from the Debian package and not part of the deposit; the recipe of the image is in the core. The elliptic curve certificates for primality were produced in the same way with PARI/GP 2.17.2 [80] and are part of the deposit; a separate program of ours checks each of them, so the result does not depend on that program. Further

work is planned to support these certificates with a second prover written independently by us. The text is set with pdfTeX.

Licenses. The text and the data are licensed under the Creative Commons Attribution 4.0 International License (CC BY 4.0), and the code under the Apache License, Version 2.0. Programs of others that the scripts use as tools, such as GMP-ECM, are not part of the deposit and keep their own licenses.

Use of AI tools

Generative AI tools were used throughout this work: large language models of the Claude family by Anthropic, mainly Claude Opus 5 and 5.5, and also Claude Fable 5.1 and Claude Sonnet 5.5 and 5, between August and October 2026, through the Claude Code environment. They were used to draft the text and the mathematical arguments, to write the computer programs, to run and check the computations, and to search for and read the literature. The author works independently, is not trained as a number theorist, and learned the subject actively during the work; the tools made it possible to carry out and check computations on this scale. The author led the work in stages: each stage was set by a prompt that stated what it should achieve, within a stage the tools followed a plan agreed with the author, and the author considered the results before giving the next prompt, revising the plan when the results required it. The author contributed ideas where they helped and framed how a problem should be attacked, decided which statements to keep, and is responsible for the content. The programs of the project, including its own checking and build tools, were developed with the AI tools, and the author started the long computations, such as the factor searches, on the author's own machine.

Several parts of the checking do not rest on these tools. The elliptic curve primality certificates were produced by PARI/GP 2.17.2 and are verified by a separate program, and the factor searches use GMP-ECM 7.0.6. Every computed number is generated by a program from deposited data or from a cited constant, and the programs carry positive and negative controls. Proofs, numbers and citations were also checked in independent passes by further instances of the AI tools that were given no access to the drafting context. These tools are not authors.

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